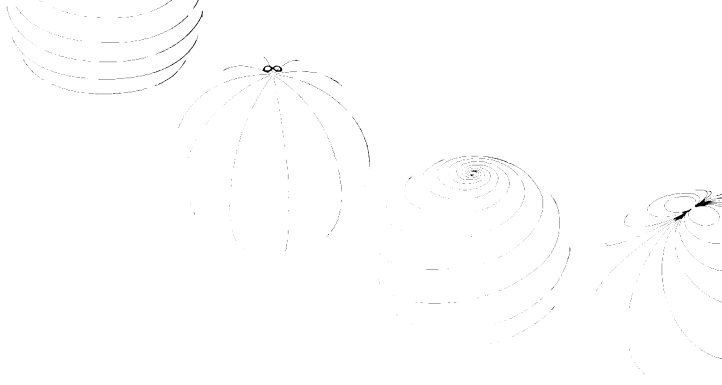




CHAPTER 2

Complex Functions as Transformations

2.1 Introduction

A complex function f is a rule that assigns to a complex number z an image complex number $w = f(z)$. In order to investigate such functions it is essential that we be able to visualize them. Several methods exist for doing this, but (until Chapter 10) we shall focus almost exclusively on the method introduced in the previous chapter. That is, we shall view z and its image w as points in the complex plane, so that f becomes a *transformation of the plane*.

Conventionally, the image points w are drawn on a fresh copy of $\mathbb{C}$, called the *image plane* or the *w-plane*. This convention is illustrated in [2.1], which depicts the transformation $z \mapsto w = f(z) = (1 + i\sqrt{3})z = 2e^{i\pi/3}z$ (cf. figure [1.5], p. 9).

Usually, the real and imaginary parts of z are denoted x and y , and those of the image point w are denoted u and v , so that $w = f(z) = u(z) + iv(z)$, where $u(z)$ and $v(z)$ are real functions of z . The precise forms of these functions will depend on whether we describe z with Cartesian or polar coordinates. For instance, writing $z = x + iy$ in the above example yields

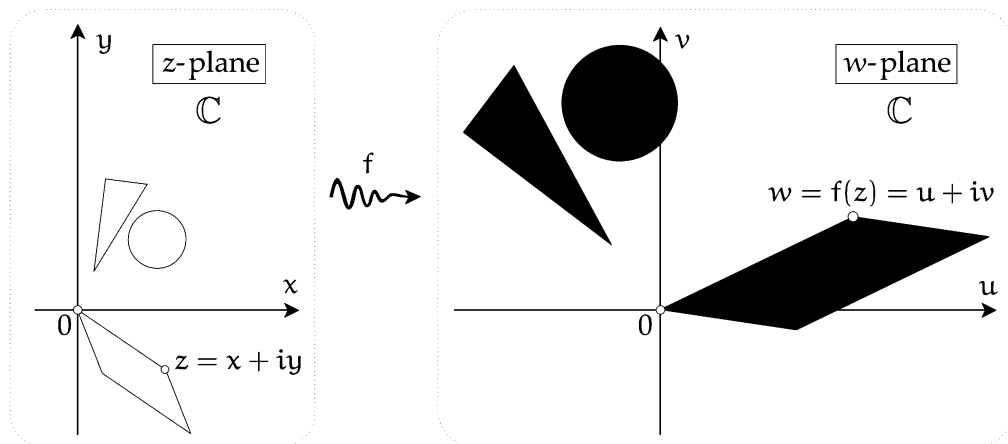
$$u(x + iy) = x - \sqrt{3}y \quad \text{and} \quad v(x + iy) = \sqrt{3}x + y,$$

while writing $z = re^{i\theta}$ and $(1 + i\sqrt{3}) = 2e^{i\pi/3}$ yields

$$u(re^{i\theta}) = 2r \cos \left[\theta + \frac{\pi}{3} \right] \quad \text{and} \quad v(re^{i\theta}) = 2r \sin \left[\theta + \frac{\pi}{3} \right].$$

Of course we may also describe the w -plane with polar coordinates so that $w = f(z) = Re^{i\phi}$, where $R(z)$ and $\phi(z)$ are real functions of z . With the same example as before, the transformation becomes

$$R(re^{i\theta}) = 2r \quad \text{and} \quad \phi(re^{i\theta}) = \theta + \frac{\pi}{3}.$$



[2.1] A complex function $f(z)$ is most commonly viewed as a mapping of points in one copy of $\mathbb{C}$ (the “z-plane”) to points in another copy of $\mathbb{C}$ (the “w-plane”): $z \mapsto w = f(z)$. Here, $f(z) = (1 + i\sqrt{3})z = 2e^{i\pi/3}z$ is the combined effect of an expansion by factor 2 and a rotation by angle $(\pi/3)$. Most commonly, we write $z = x + iy$ and $w = u + iv$.

We shall find that we can gain considerable insight into a given f by drawing pictures showing its effect on points, curves, and shapes. However, it would be nice if we could simultaneously grasp the behaviour of f for *all* values of z . One such method is to instead represent f as a *vector field*, whereby $f(z)$ is depicted as a vector emanating from the point z ; for more detail, the reader is invited to read the beginning of Chapter 10.

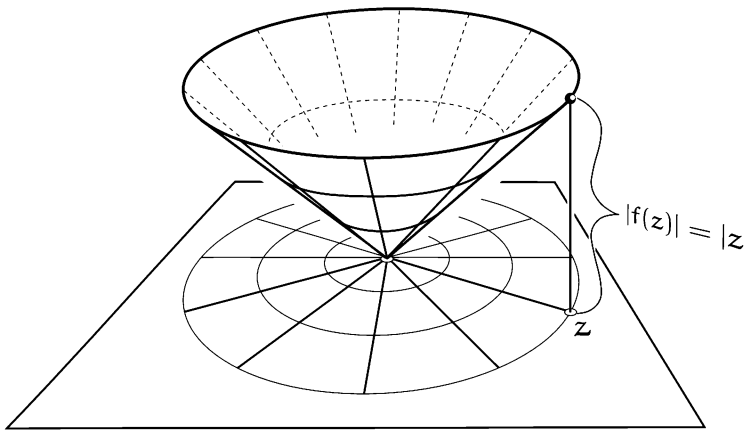
Yet other methods are based on the idea of a graph. In the case of a real function $f(x)$ of a real variable x we are accustomed to the convenience of visualizing the overall behaviour of f by means of its graph, i.e., the curve in the two-dimensional xy -plane made up of the points $(x, f(x))$. In the case of a complex function this approach does not seem viable because to depict the pair of complex numbers $(z, f(z))$ we would need *four* dimensions: two for $z = x + iy$ and two for $f(z) = u + iv$.

Actually, the situation is not quite as hopeless as it seems. First, note that although two-dimensional space is needed to draw the graph of a real function f , the graph itself [the set of points $(x, f(x))$] is only a one-dimensional curve, meaning that only one real number (namely x) is needed to identify each point within it. Likewise, although four-dimensional space is needed to draw the set of points with coordinates $(x, y, u, v) = (z, f(z))$, the graph itself is *two-dimensional*, meaning that only two real numbers (namely x and y) are needed to identify each point within it. Thus, intrinsically, the graph of a complex function is merely a two-dimensional surface (a so-called *Riemann surface*), and it is thus susceptible to visualization in ordinary three-dimensional space. This approach will not be explored in this book, though the last three chapters in particular should prove helpful in understanding Riemann’s original physical insights, as expounded by [Klein \(1881\)](#). See also

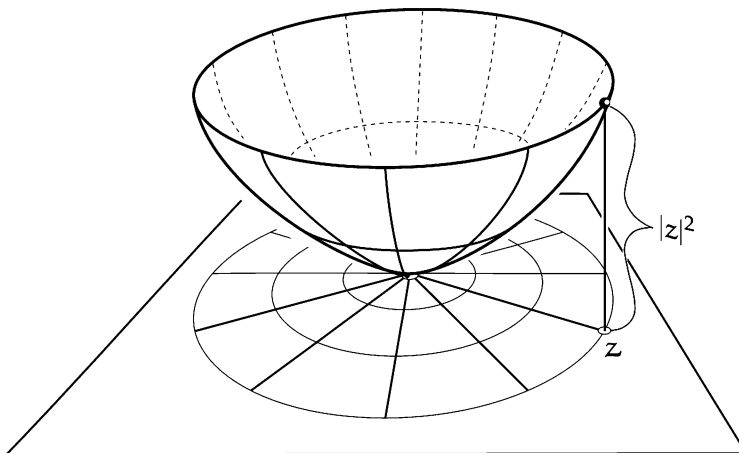
Springer (1981, Ch. 1), which essentially reproduces Klein's monograph, but with additional helpful commentary.

There is another type of graph of a complex function that is sometimes useful. The image $f(z)$ of a point z may be described by its distance $|f(z)|$ from the origin, and the angle $\arg[f(z)]$ it makes with the real axis. Let us discard half of this information (the angle) and try to depict how the modulus $|f(z)|$ varies with z . To do so, imagine the complex z -plane lying horizontally in space, and construct a point at height $|f(z)|$ vertically above each point z in the plane, thereby producing a surface called the *modular surface* of f . Figure [2.2] illustrates the conical modular surface of $f(z) = z$, while [2.3] illustrates the paraboloid modular surface of $f(z) = z^2$.

A note on computers. Beginning in this chapter, we will often suggest that you use a computer to expand your understanding of the mathematical phenomenon



[2.2] The height above $\mathbb{C}$ of the **modular surface** of a complex function $f(z)$ is its modulus, $|f(z)|$. Here, $f(z) = z$, producing a cone.



[2.3] The paraboloid modular surface of $f(z) = z^2$.

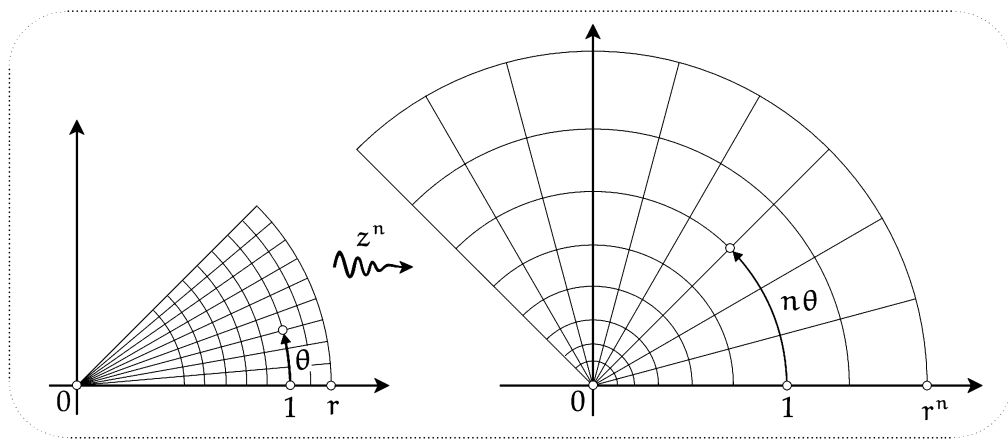
under discussion. However, we wish to stress that the *specific* uses of the computer that we have suggested in the text are only a beginning. Think of the computer as a physicist would his laboratory—you may use it to check your existing ideas about the construction of the world, or as a tool for discovering new phenomena which then demand new ideas for their explanation. In the Preface we make concrete suggestions (probably of only fleeting relevance) as to how your laboratory should be equipped.

2.2 Polynomials

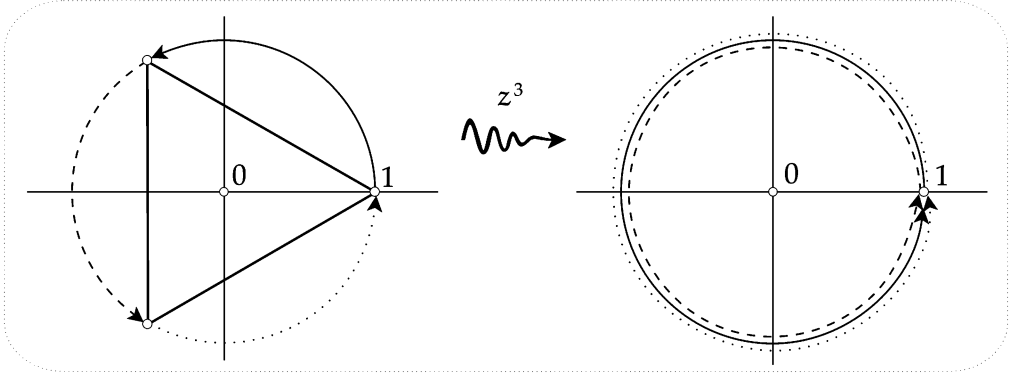
2.2.1 Positive Integer Powers

Consider the mapping $z \mapsto w = z^n$, where n is a positive integer. Writing $z = r e^{i\theta}$ this becomes $w = r^n e^{in\theta}$, i.e., the distance is raised to the n^{th} power and the angle is multiplied by n . Figure [2.4] is intended to make this a little more vivid by showing the effect of the mapping on some rays and arcs of origin-centred circles. As you can see, here $n = 3$.

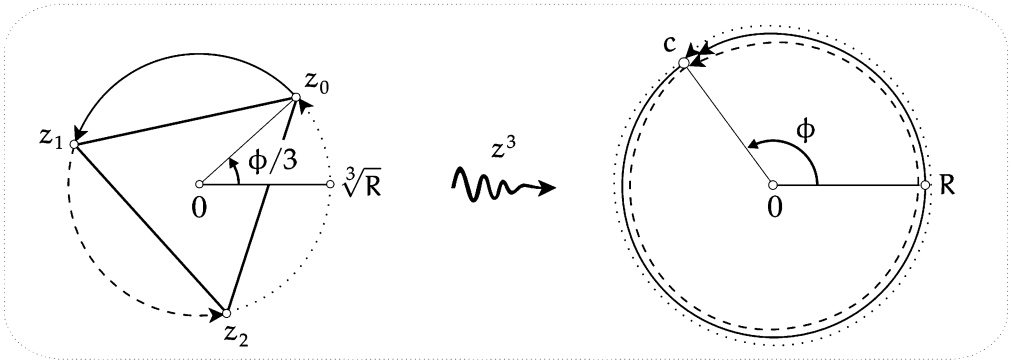
On page 29 we saw that the n solutions of $z^n = 1$ are the vertices of the regular n -gon inscribed in the unit circle, with one vertex at 1. This can be understood more vividly from our new transformational point of view. If $w = f(z) = z^n$ then the solutions of $z^n = 1$ are the points in the z -plane that are mapped by f to the point $w = 1$ in the w -plane. Now imagine a particle in orbit round the unit circle in the z -plane. Since $1^n = 1$, the image particle $w = f(z)$ will also orbit round the unit circle (in the w -plane), but *with n times the angular speed of the original particle*. Thus each time z executes $(1/n)$ of a revolution, w will execute a complete revolution and return to the same image point. The preimages of any given w on the unit



[2.4] The geometric effect of the power mapping, $z \mapsto z^n$.



[2.5] As z rotates around the unit circle, starting at 1, its image under $z \mapsto z^3$ rotates three times as fast, executing a complete revolution every time z executes one third of a revolution. Thus the cube roots of 1 form an equilateral triangle. More generally, the same reasoning explains why the n^{th} roots of unity form a regular n -gon.



[2.6] Extending the reasoning of the previous figure, the cube roots of an arbitrary complex number c will form an equilateral triangle, and the n^{th} roots will form a regular n -gon.

circle will therefore be successive positions of z as it repeatedly executes $(1/n)$ of a revolution, i.e., they will be the vertices of a regular n -gon. With $w = 1$, figure [2.5] illustrates this idea for the mapping $w = f(z) = z^3$.

More generally, [2.6] shows how to solve $z^3 = c = R e^{i\phi}$ by inscribing an equilateral triangle in the circle $|z| = \sqrt[3]{R}$. By the same reasoning, it is clear that the solutions of $z^n = c$ are the vertices of the regular n -gon inscribed in the circle $|z| = \sqrt[n]{R}$, with one vertex at angle (ϕ/n) .

To arrive at the same result symbolically, first note that if ϕ is one value of $\arg c$, then the complete set of possible angles is $(\phi + 2m\pi)$, where m is an arbitrary integer. Setting $z = r e^{i\theta}$,

$$r^n e^{in\theta} = z^n = c = R e^{i(\phi+2m\pi)} \implies r^n = R \quad \text{and} \quad n\theta = \phi + 2m\pi,$$

so the solutions are $z_m = \sqrt[n]{R} e^{i(\phi+2m\pi)/n}$. Each time we increase m by 1, z_m is rotated by $(1/n)$ of a revolution (because $z_{m+1} = e^{\frac{2\pi i}{n}} z_m$), producing the vertices

of a regular n -gon. Thus the complete set of solutions will be obtained if we let m take any n consecutive values, say $m = 0, 1, 2, \dots, (n - 1)$.

2.2.2 Cubics Revisited*

As an instructive application of these ideas, let us reconsider the problem of solving a cubic equation in x . For simplicity, we shall assume in the following that the coefficients of the cubic are all real.

In the previous chapter we saw [Ex. 1] that the general cubic could always be reduced to the form $x^3 = 3px + 2q$. We then found [Ex. 2] that this could be solved using Cardano's formula,

$$x = s + t, \quad \text{where} \quad s^3 = q + \sqrt{q^2 - p^3}, \quad t^3 = q - \sqrt{q^2 - p^3}, \quad \text{and} \quad st = p.$$

Once again, observe that if $q^2 < p^3$ then this formula involves complex numbers.

On the other hand, we also saw [Ex. 3] that the cubic could be solved using Viète's formula:

$$\text{if } q^2 \leq p^3, \text{ then } x = 2\sqrt{p} \cos \left[\frac{1}{3}(\phi + 2m\pi) \right],$$

where m is an integer and $\phi = \cos^{-1}(q/p\sqrt{p})$. At the time of its discovery, Viète's "angle trisection" method was a breakthrough, because it solved the cubic (using only real numbers) precisely when Cardano's formula involved "impossible", complex numbers. For a long time thereafter, Viète's method was thought to be entirely different from Cardano's, and it is sometimes presented in this way even today. We shall now take a closer look at these two methods and see that they are really the same.

If $q^2 \leq p^3$, then in Cardano's formula s^3 and t^3 are *complex conjugates*:

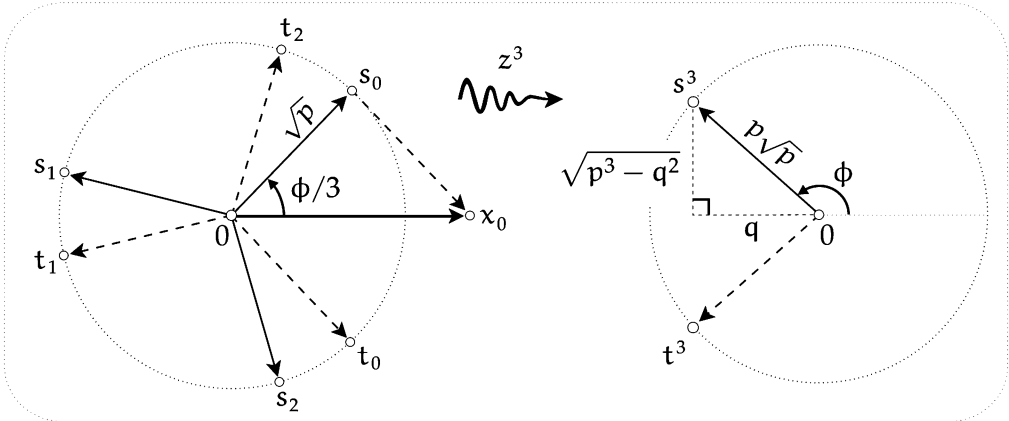
$$s^3 = q + i\sqrt{p^3 - q^2} \quad \text{and} \quad t^3 = \overline{s^3} = q - i\sqrt{p^3 - q^2}.$$

These complex numbers are illustrated on the RHS of [2.7]. By Pythagoras' Theorem, they both have length $|s^3| = p\sqrt{p}$, and so the angle ϕ occurring in Viète's formula is simply the angle of s^3 .

Since s^3 and t^3 lie on the circle of radius $(\sqrt{p})^3$, their preimages under the mapping $z \mapsto z^3$ will lie on the circle of radius $\sqrt{p}$. The LHS of [2.7] shows these preimages; note that the three values of t are the complex conjugates of the three values of s .

According to the Fundamental Theorem of Algebra, the original cubic should have three solutions. However, by combining each of the three values of s with each of the three values of t , it would seem that Cardano's formula $x = s + t$ yields *nine* solutions.

The resolution lies in the fact that we also require $st = p$. Since p is real, this means s and t must have equal and opposite angles. In the formula $x = s + t$, each



[2.7] The solution of the cubic $x^3 = 3px + 2q$ was long thought to have two completely different solutions. If $q^2 \leq p^3$, Cardano's 1545 formula says that $x = s + \bar{s}$, where $s^3 = q + i\sqrt{p^3 - q^2}$. On the other hand, Viète's 1591 "angle trisection method" yields $x = 2\sqrt{p} \cos \left[\frac{1}{3}(\phi + 2m\pi) \right]$, where $\phi = \cos^{-1}(q/p\sqrt{p})$. The figure reveals that the two methods are in fact the same.

of the three values of s must therefore be paired with the conjugate value of t . We can now see how Cardano's formula becomes Viète's formula:

$$x_m = s_m + t_m = s_m + \bar{s}_m = 2\sqrt{p} \cos \left[\frac{1}{3}(\phi + 2m\pi) \right].$$

In Ex. 4 the reader is invited to consider the case $q^2 > p^3$.

2.2.3 Cassinian Curves*

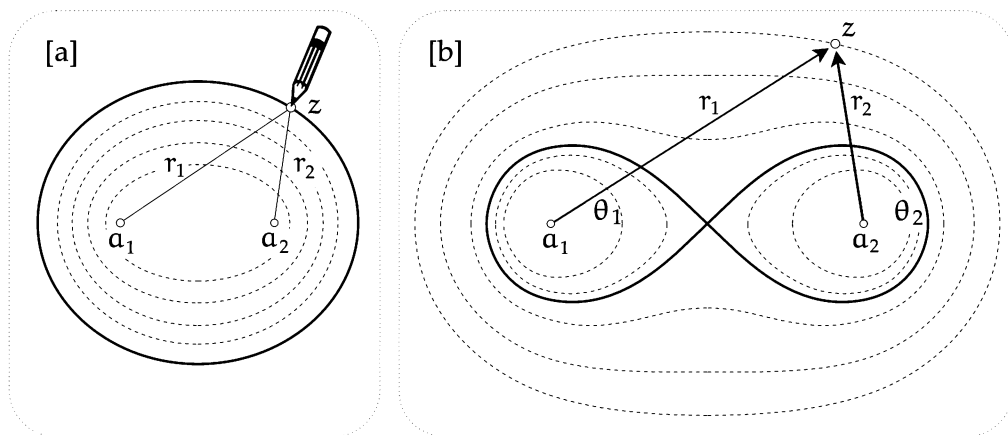
Consider [2.8a]. The ends of a piece of string of length l are attached to two fixed points a_1 and a_2 in $\mathbb{C}$, and, with its tip at z , a pencil holds the string taut. The figure illustrates the well known fact that if we move the pencil (continuing to keep the string taut) it traces out an *ellipse*, with foci a_1 and a_2 . Writing $r_{1,2} = |z - a_{1,2}|$, the equation of the ellipse is thus

$$r_1 + r_2 = l.$$

By choosing different values of l we obtain the illustrated family of confocal ellipses.

In 1687 Newton published his great *Principia*, in which he demonstrated that the planets orbit in such ellipses, with the sun at one of the foci. Seven years earlier, however, Giovanni Cassini had instead proposed that the orbits were curves for which the *product* of the distances is constant:

$$r_1 \cdot r_2 = \text{const.} = k^2. \quad (2.1)$$



[2.8] **[a]** Ellipses with foci a_1 and a_2 have equation $r_1 + r_2 = \text{const}$. In 1687 Newton proved that the planets orbit the sun in such ellipses, with the sun at one focus. **[b]** Seven years earlier, however, Giovanni Cassini had instead proposed that they orbit the sun in what are now called Cassinian curves, with equation $r_1 r_2 = \text{const}$. The figure eight curve is called the *lemniscate*, and it played a pivotal role in understanding elliptic integrals and elliptic functions.

These curves are illustrated in [2.8b]; they are called *Cassinian curves*, and the points a_1 and a_2 are again called *foci*.

The following facts will become clearer in a moment, but you might like to think about them for yourself. If k is small then the curve consists of two separate pieces, resembling small circles centred at a_1 and a_2 . As k increases, these two components of the curve become more egg shaped. When k reaches a value equal to half the distance between the foci then the pointed ends of the egg shapes meet at the midpoint of the foci, producing a figure eight [shown solid]. Increasing the value of k still further, the curve first resembles an hourglass, then an ellipse, and finally a circle.

Although Cassinian curves turned out to be useless as a description of planetary motion, the figure eight curve proved extremely valuable in quite another context. In 1694 it was rediscovered by James Bernoulli and christened the *lemniscate*—it then became the catalyst in unravelling the behaviour of the so-called *elliptic integrals* and *elliptic functions*. See [Stillwell \(2010\)](#) and [Siegel \(1969\)](#) for more on this fascinating story.

Cassinian curves arise naturally in the context of complex polynomials. A general quadratic $Q(z) = z^2 + pz + q$ will have two roots (say, a_1 and a_2) and so can be factorized as $Q(z) = (z - a_1)(z - a_2)$. In terms of [2.8b], this becomes

$$Q(z) = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

Therefore, by virtue of (2.1), $z \mapsto w = Q(z)$ will map each curve in [2.8b] to an origin-centred circle, $|w| = k^2$, and it will map the foci to the origin.

If we follow this transformation by a translation of c , i.e., if we change $z \mapsto Q(z)$ to $z \mapsto Q(z) + c$, then the images will instead be concentric circles centred at $c = (\text{image of foci})$. Conversely, given any quadratic mapping $z \mapsto w = Q(z)$, the preimages of a family of concentric circles in the w -plane centred at c will be the Cassinian curves whose foci are the preimages of c .

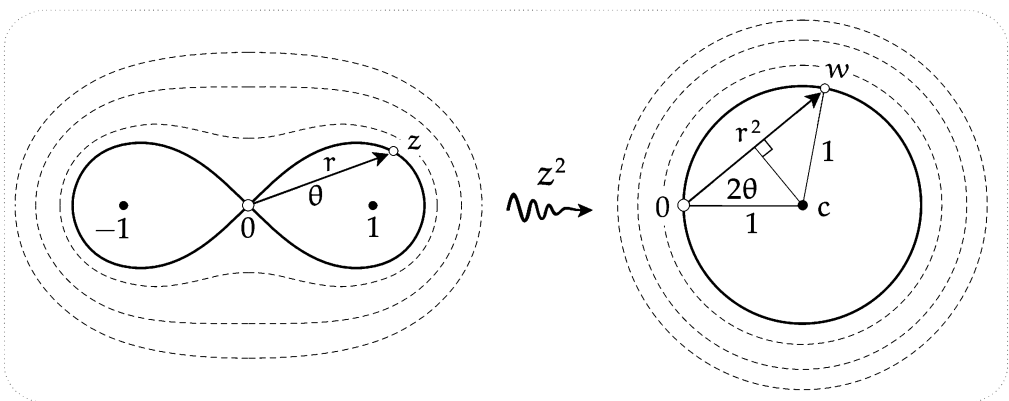
In particular, consider the case $c = 1$ and $w = Q(z) = z^2$. The preimages of $w = 1$ are $z = \pm 1$, so these are the foci, and the Cassinian curves are thus centred at the origin. See [2.9]. Since Q leaves the origin fixed, the lemniscate must be mapped (as illustrated) to the circle of radius 1 passing through the origin. Writing $z = r e^{i\theta}$, $w = r^2 e^{i2\theta}$, and so we see from the figure that the polar equation of the lemniscate is

$$r^2 = 2 \cos 2\theta. \quad (2.2)$$

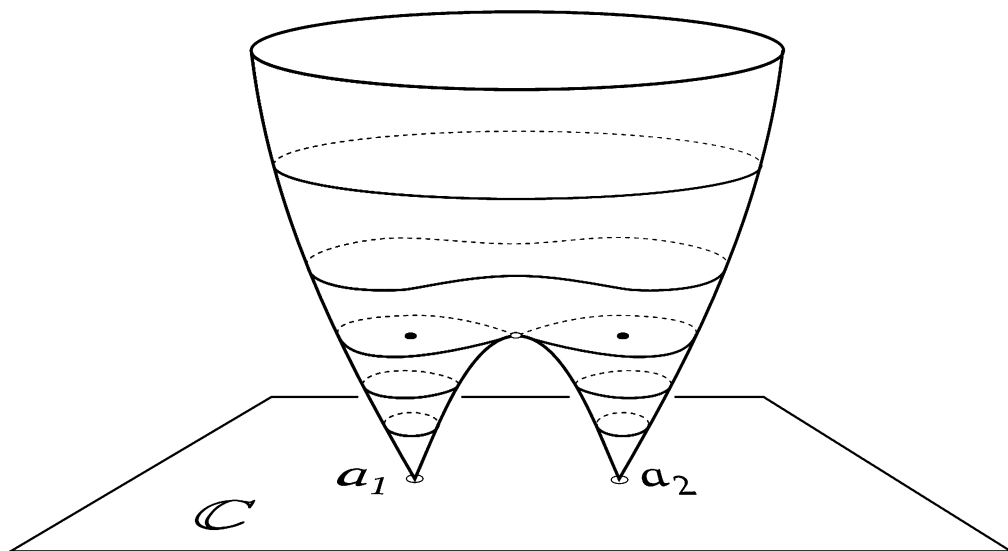
Returning to [2.8b], the form of the Cassinian curves may be grasped more intuitively by sketching the modular surface of $Q(z) = (z - a_1)(z - a_2)$. First observe that as z moves further and further away from the origin, $Q(z)$ behaves more and more like z^2 . Indeed, since the ratio $[Q(z)/z^2]$ is easily seen [exercise] to tend to unity as $|z|$ tends to infinity, we may say that $Q(z)$ is ultimately equal to z^2 in this limit. Thus, for large values of $|z|$, the modular surface of Q will look like the paraboloid in [2.3].

Next, consider the behaviour of the surface near a_1 . Writing $D = |a_1 - a_2|$ for the distance between the foci, we see [exercise] that $|Q(z)|$ is ultimately equal to Dr_1 as z tends to a_1 . Thus the surface meets the plane at a_1 in a cone like that shown in [2.2]. Of course the same thing happens at a_2 .

Combining these facts, we obtain the surface shown in [2.10]. Since a Cassinian curve satisfies $|Q(z)| = r_1 r_2 = k^2$, it is the intersection of this surface with a plane



[2.9] On the left, the Cassinian curves with foci ± 1 have equation $|(z-1)(z+1)| = \text{const.}$ It follows that their images under $z \mapsto z^2$ are the circles illustrated on the right, centred at 1. It then follows immediately that the equation of the lemniscate is $r^2 = 2 \cos 2\theta$.



[2.10] The form of the Cassinian curves in [2.8b] may be grasped intuitively as horizontal sections of the modular surface of $Q(z) = (z - a_1)(z - a_2)$.

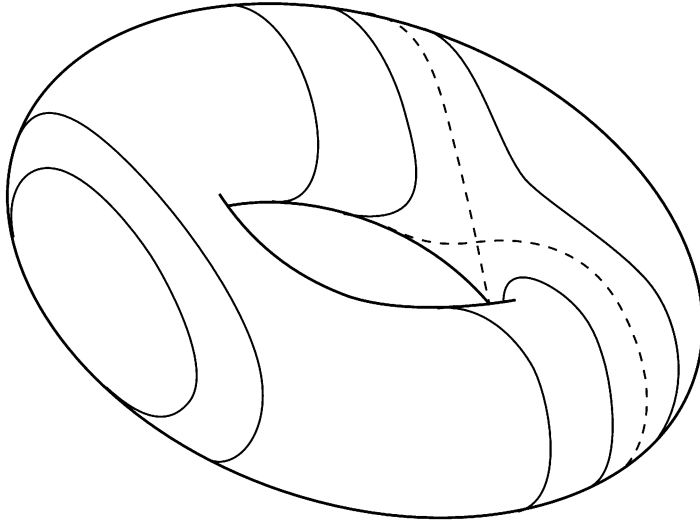
parallel to C , and at height k^2 above it. As k increases from 0 to a large value, it is now easy to follow the evolution of the curves in [2.8b] by looking at how this intersection varies as the plane moves upward in [2.10]. Thus the Cassinian curves may be viewed as a geographical contour map of the modular surface of the quadratic.

Interestingly, Cassinian curves were already known to the ancient Greeks. Around 150 BCE, Perseus considered the intersection curves of a *torus* [obtained by rotating a circle C about an exterior line l in its plane] with planes parallel to l . It turns out that if the distance of the plane from l equals the radius of C then the resulting *spiral section of Perseus* is a Cassinian curve. See [2.11]; in particular, note how the lemniscate [dashed] makes its surprise appearance when the plane touches the inner rim of the torus. We have adapted this figure from [Brieskorn and Knörrer \(2012\)](#), to which the reader is referred for more details.

Returning to the complex plane, there is a natural way to define Cassinian curves with more than two foci: A Cassinian curve with n foci, $a_1, a_2, \dots, a_n$, is the locus of a point for which the product of the distances to the foci remains constant. A straightforward extension of the above ideas shows that these curves are the preimages of origin-centred circles $|w| = \text{const.}$ under the mapping given by the n^{th} degree polynomial whose roots are the foci:

$$z \mapsto w = P_n(z) = (z - a_1)(z - a_2) \cdots (z - a_n).$$

Equivalently, the Cassinian curves are the cross-sections of the modular surface of $P_n(z)$. This surface has n cone-like legs resting on C at $a_1, a_2, \dots, a_n$, and for large values of $|z|$ it resembles the axially symmetric modular surface of z^n .



[2.11] The Spiric Sections of Perseus (150 BCE). *The intersection of a torus with a plane whose distance from the axis of symmetry equals the radius of the generating circle turns out to be a Cassinian curve. If the plane touches the inner rim, the **lemniscate** makes a surprise appearance!*

2.3 Power Series

2.3.1 The Mystery of Real Power Series

Many real functions $F(x)$ can be expressed (e.g., via Taylor's Theorem) as power series:

$$F(x) = \sum_{j=0}^{\infty} c_j x^j = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots,$$

where the c_j 's are real constants. Of course, this infinite series will normally only converge to $F(x)$ in some origin-centred *interval of convergence* $-R < x < R$. But how is R (the *radius of convergence*) determined by $F(x)$?

It turns out that this question has a beautifully simple answer, *but only if we investigate it in the complex plane*. If we instead restrict ourselves to the real line—as mathematicians were forced to in the era in which such series were first employed—then the relationship between R and $F(x)$ is utterly mysterious. Historically, it was precisely this mystery¹ that led Cauchy to several of his breakthroughs in complex analysis.

¹ Cauchy was investigating the convergence of series solutions to *Kepler's equation*, which describes where a planet is in its orbit at any given time.

To see that there is a mystery, consider the power series representations of the functions

$$G(x) = \frac{1}{1-x^2} \quad \text{and} \quad H(x) = \frac{1}{1+x^2}.$$

The familiar infinite geometric series,

$$\frac{1}{1-x} = \sum_{j=0}^{\infty} x^j = 1 + x + x^2 + x^3 + \cdots \quad \text{if and only if } -1 < x < 1, \quad (2.3)$$

immediately yields

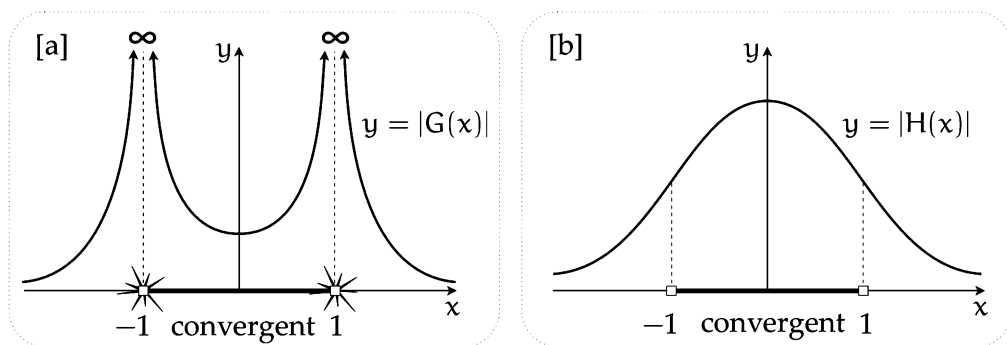
$$G(x) = \sum_{j=0}^{\infty} x^{2j} \quad \text{and} \quad H(x) = \sum_{j=0}^{\infty} (-1)^j x^{2j},$$

where both series have the same interval of convergence, $-1 < x < 1$.

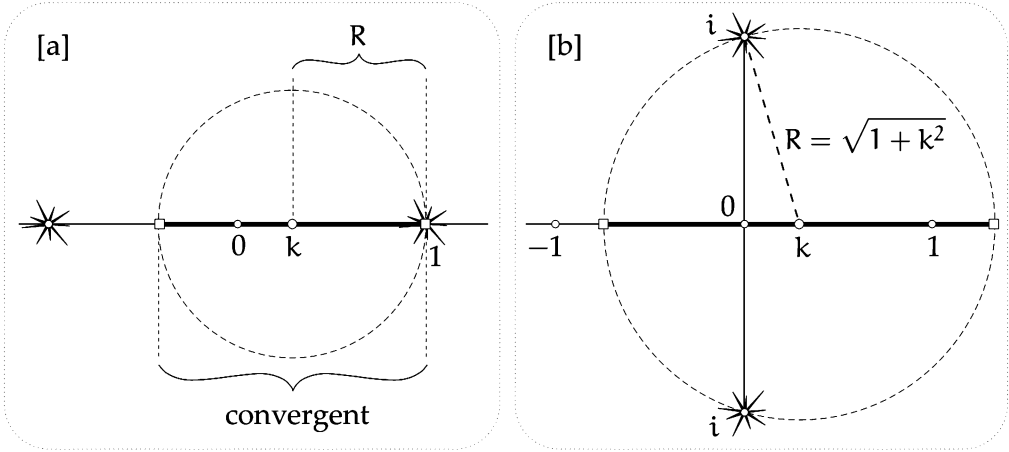
It is easy to understand the interval of convergence of the series for $G(x)$ if we look at the graph [2.12a]. The series becomes divergent at $x = \pm 1$ because these points are *singularities* of the function itself, i.e., they are places where $|G(x)|$ becomes infinite. But if we look at $y = |H(x)|$ in [2.12b], there seems to be no reason for the series to break down at $x = \pm 1$. Yet break down it does.

To begin to understand this, let us expand these functions into power series centred at $x = k$ (instead of $x = 0$), i.e., into series of the form $\sum_{j=0}^{\infty} c_j X^j$, where $X = (x - k)$ measures the displacement of x from the centre k . To expand G we first generalize (2.3) by expanding $1/(a - x)$ about k :

$$\frac{1}{a-x} = \frac{1}{a-(X+k)} = \frac{1}{(a-k)} \frac{1}{\left[1 - \left(\frac{X}{a-k}\right)\right]},$$



[2.12] [a] The convergence of the power series for $G(x) = \frac{1}{1-x^2}$ is readily understood: clearly it must end when we arrive at the singularities $x = \pm 1$. **[b]** In contrast to this, the fact that the power series for $H(x) = \frac{1}{1+x^2}$ also stops converging at ± 1 is utterly mysterious!



[2.13] The radius of convergence is the distance to the nearest singularity. [a] It is intuitively clear that the radius of convergence R of the power series for $G(x) = \frac{1}{1-x^2}$ centred at k is the distance from k to the nearest singularity. [b] The previously mysterious formula $R = \sqrt{1^2 + k^2}$ for the convergence of the real power series for $H(x) = \frac{1}{1+x^2}$ is explained by its imaginary singularities at $\pm i$.

and so

$$\frac{1}{a-x} = \sum_{j=0}^{\infty} \frac{X^j}{(a-k)^{j+1}}, \quad \text{if and only if } |X| < |a-k|. \quad (2.4)$$

To apply this result to G , we factorize $(1-x^2) = (1-x)(1+x)$ and then decompose G into partial fractions:

$$\frac{1}{1-x^2} = \frac{1}{2} \left[\frac{1}{1-x} - \frac{1}{-1-x} \right] = \frac{1}{2} \sum_{j=0}^{\infty} \left[\frac{1}{(1-k)^{j+1}} - \frac{1}{(-1-k)^{j+1}} \right] X^j,$$

where $|X| < |1-k|$ and $|X| < |1+k|$. Thus the interval of convergence $|X| < R$ is given by

$$R = \min \{|1-k|, |1+k|\} = (\text{distance from } k \text{ to the nearest singularity of } G).$$

This readily comprehensible result is illustrated in [2.13a]; ignore the shaded disc for the time being.

In the case of $H(x)$, I cannot think of an elegant method of finding the expansion using only real numbers, but see Ex. 9 for an attempt. Be that as it may, it can be shown that the radius of convergence of the series in X is given by the strange formula $R = \sqrt{1+k^2}$. As with Cotes' work in the previous chapter, we have here a result about real functions that is trying to tell us about the existence of the complex plane.

If we picture the real line as embedded in a plane then Pythagoras' Theorem tells us that $R = \sqrt{1^2 + k^2}$ should be interpreted as the distance from the centre k

of the expansion to either of the fixed points that lie *off the line*, one unit from 0 in a direction at right angles to the line. See [2.13b]. If the plane is thought of as $\mathbb{C}$, then these points are $\pm i$, and

$$R = (\text{distance from } k \text{ to } \pm i).$$

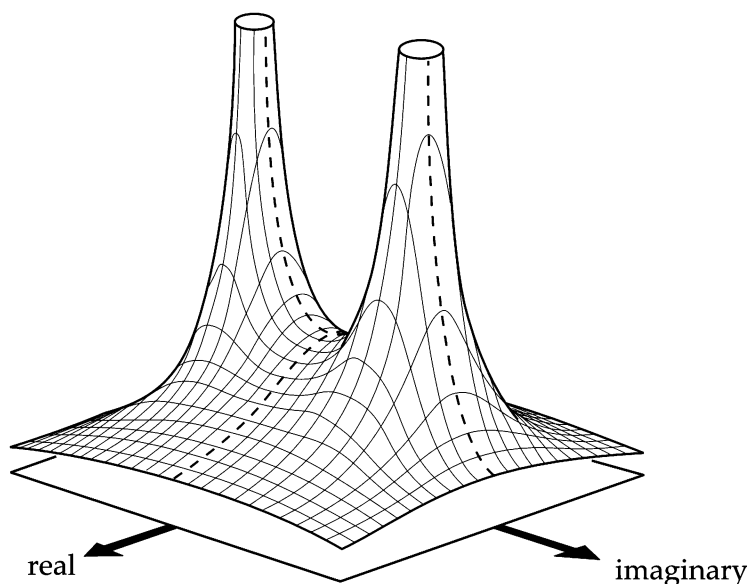
The mystery begins to unravel when we turn to the *complex* function

$$h(z) = \frac{1}{1+z^2},$$

which is identical to $H(x)$ when z is restricted to the real axis of the complex plane. In fact there is a sense—we cannot be explicit yet—in which $h(z)$ is the *only* complex function that agrees with H on this line.

While [2.12b] shows that $h(z)$ is well-behaved for real values of z , it is clear that $h(z)$ has *two singularities in the complex plane*, one at $z = i$ and the other at $z = -i$; these are shown as little explosions in [2.13b]. Figure [2.14] tries to make this more vivid by showing the modular surface of $h(z)$, the singularities at $\pm i$ appearing as “volcanoes” erupting above these points. We will sort through the details in a moment, but the mystery has all but disappeared: *in both [2.13a] and [2.13b], the radius of convergence is the distance to the nearest singularity.*

If we intersect the surface in [2.14] with a vertical plane through the real axis then we recover the deceptively tranquil graph in [2.12b], but if we instead slice



[2.14] Twin volcanoes erupt in the modular surface of $h(z) = \frac{1}{1+z^2}$, directly above its singularities at $z = \pm i$. The graphs in [2.12] can now be understood in a unified way, as the vertical sections of this surface in the imaginary and real directions. The mysterious radius of convergence of the real power series for $h(x)$ is thereby explained!

the surface along the imaginary axis then we obtain the graph in [2.12a]. That this is no accident may be seen by first noting that $G(x)$ is just the restriction to the real axis of the complex function $g(z) = 1/(1 - z^2)$. Since $g(z) = h(iz)$, h and g are essentially the same: if we rotate the plane by $(\pi/2)$ and then do h , we obtain g . In particular the modular surface of g is simply [2.14] rotated by $(\pi/2)$, the volcanoes at $\pm i$ being rotated to ± 1 .

2.3.2 The Disc of Convergence

Let us consider the convergence of complex power series, leaving aside for the moment the question of whether a given complex function can be expressed as such a series.

A complex power series $P(z)$ (centred at the origin) is an expression of the form

$$P(z) = \sum_{j=0}^{\infty} c_j z^j = c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad (2.5)$$

where the c_j 's are complex constants, and z is a complex variable. The partial sums of this infinite series are just the ordinary polynomials,

$$P_n(z) = \sum_{j=0}^n c_j z^j = c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots + c_n z^n.$$

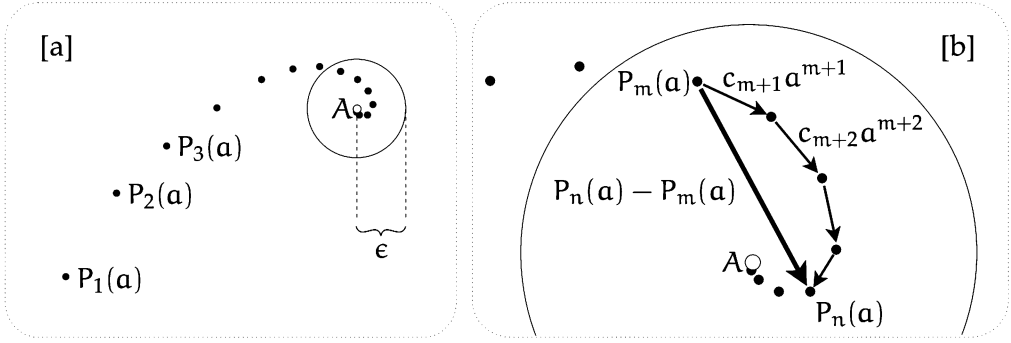
For a given value of $z = a$, the sequence of points $P_1(a), P_2(a), P_3(a), \dots$ is said to *converge to the point A* if for any given positive number ϵ , no matter how small, there exists a positive integer N such that $|A - P_n(a)| < \epsilon$ for every value of n greater than N . Figure [2.15a] illustrates that this is much simpler than it sounds: all it says is that once we reach a certain point $P_N(a)$ in the sequence $P_1(a), P_2(a), P_3(a), \dots$, all of the subsequent points lie within an arbitrarily small disc of radius ϵ centred at A .

In this case we say that the power series $P(z)$ *converges* to A at $z = a$, and we write $P(a) = A$. If the sequence $P_1(a), P_2(a), P_3(a), \dots$ does not converge to a particular point, then the power series $P(z)$ is said to *diverge* at $z = a$. Thus for each point z , $P(z)$ will either converge or diverge.

Figure [2.15b] shows a magnified view of the disc in [2.15a]. If $n > m > N$ then $P_m(a)$ and $P_n(a)$ both lie within this disc, and consequently the distance between them must be less than the diameter of the disc:

$$|c_{m+1}a^{m+1} + c_{m+2}a^{m+2} + \cdots + c_n a^n| = |P_n(a) - P_m(a)| < 2\epsilon. \quad (2.6)$$

Conversely, it can be shown [exercise] that if this condition is met then $P(a)$ converges. Thus we have a new way of phrasing the definition of convergence: $P(a)$ *converges if and only if there exists an N such that inequality (2.6) holds (for arbitrarily small ϵ) whenever m and n are both greater than N .*



[2.15] [a] The sequence $P_n(a)$ is said to converge to A if it inevitably enters every disc centred at A , no matter how small we make its radius ϵ , and the disc then acts like a black hole: having entered the disc, the sequence can never escape it! **[b]** The direct route from $P_m(a)$ to $P_n(a)$ has length $|P_n(a) - P_m(a)|$ and is shorter than the indirect route that goes via $P_{m+1}(a)$, $P_{m+2}(a)$, etc., which has length $\tilde{P}_n(a) - \tilde{P}_m(a)$. But if $P(a)$ is absolutely convergent then the indirect route must have a length that tends to zero for sufficiently large m and n , and therefore the direct route must, too. Therefore, absolute convergence implies convergence.

The complex power series $P(z)$ is said to be *absolutely convergent* at $z = a$ if the real series

$$\tilde{P}(z) \equiv \sum_{j=0}^{\infty} |c_j z^j| = |c_0| + |c_1 z| + |c_2 z^2| + |c_3 z^3| + \cdots,$$

converges there. Absolute convergence is certainly different from ordinary convergence. For example, [exercise] $P(z) = \sum z^j/j$ is convergent at $z = -1$, but it is not absolutely convergent there. On the other hand,

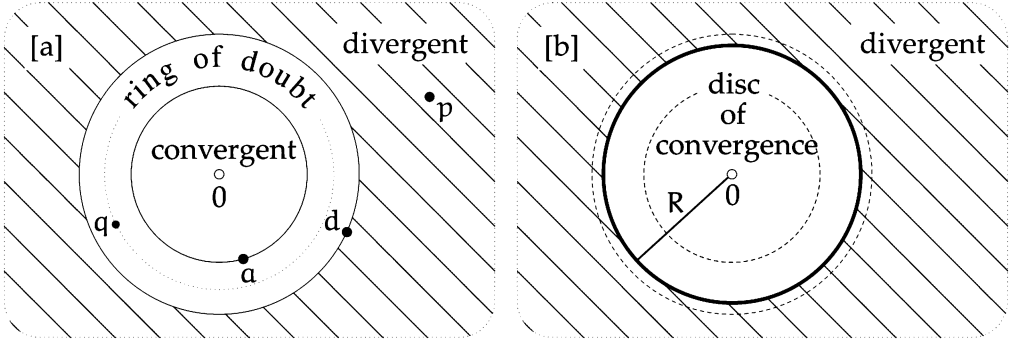
If $P(z)$ is absolutely convergent at some point, then it will also be convergent at that point. (2.7)

Thus absolute convergence is a stronger requirement than convergence.

To establish (2.7), suppose that $P(z)$ is absolutely convergent at $z = a$, so that (by definition) $\tilde{P}(a)$ is convergent. In terms of the partial sums $\tilde{P}_n(z) = \sum_{j=0}^n |c_j z^j|$ of the real series $\tilde{P}(z)$, this says that for sufficiently large values of m and n we can make $[\tilde{P}_n(a) - \tilde{P}_m(a)]$ as small as we please. But, referring to [2.15b], we see that

$$\tilde{P}_n(a) - \tilde{P}_m(a) = |c_{m+1} a^{m+1}| + |c_{m+2} a^{m+2}| + \cdots + |c_n a^n|$$

is the total length of the roundabout journey from $P_m(a)$ to $P_n(a)$ that goes via $P_{m+1}(a)$, $P_{m+2}(a)$, etc. Since $|P_n(a) - P_m(a)|$ is the length of the shortest journey from $P_m(a)$ to $P_n(a)$,



[2.16] [a] If $P(z)$ converges at a then it also converges everywhere inside the shaded disc $|z| < |a|$. And if it diverges at d then it also diverges in the striped region $|z| > |d|$. Its behaviour is therefore unknown only in “the ring of doubt”, $|a| \leq |z| \leq |d|$. Testing a point q half way across this ring then halves the width of the ring. **[b]** Iterating this q -algorithm, the ring of doubt shrinks and converges to a definite circle, the circle of convergence, the interior of which is the disc of convergence, and the radius of which, R , is the radius of convergence.

$$|P_n(a) - P_m(a)| \leq \tilde{P}_n(a) - \tilde{P}_m(a).$$

Thus $|P_n(a) - P_m(a)|$ must also become arbitrarily small for sufficiently large m and n . Done.

We can now establish the following fundamental fact:

If $P(z)$ converges at $z = a$, then it will also converge everywhere inside the disc $|z| < |a|$. (2.8)

See [2.16a]. In fact we will show that $P(z)$ is *absolutely* convergent in this disc; the result then follows directly from (2.7).

If $P(a)$ converges then the length $|c_n a^n|$ of each term must die away to zero as n goes to infinity [why?]. In particular, there must be a number M such that $|c_n a^n| < M$ for all n . If $|z| < |a|$ then $\rho = |z|/|a| < 1$ and so $|c_n z^n| < M\rho^n$. Thus,

$$\tilde{P}_n(z) - \tilde{P}_m(z) \leq M(\rho^{m+1} + \rho^{m+2} + \dots + \rho^n) = \frac{M}{1-\rho} [\rho^{m+1} - \rho^{n+1}], \quad (2.9)$$

where the RHS is as small as we please for sufficiently large m and n . Done.

If $P(z)$ does not converge everywhere in the plane then there must be at least one point d where it diverges. Now suppose that $P(z)$ were to converge at some point p further away from the origin than d . See [2.16a]. By (2.8) it would then converge everywhere inside the disc $|z| < |p|$, and in particular it would converge at d , contradicting our initial hypothesis. Thus,

If $P(z)$ diverges at $z = d$, then it will also diverge everywhere outside the circle $|z| = |d|$. (2.10)

At this stage we have settled the question of convergence everywhere except in the “ring of doubt”, $|a| \leq |z| \leq |d|$, shown in [2.16a]. Suppose we take a point q half way across the ring of doubt (i.e., on the circle $|z| = \frac{|a|+|d|}{2}$), then check whether $P(q)$ converges or not. Regardless of the outcome, (2.8) and (2.10) enable us to obtain a new ring of doubt that is half as wide as before. For example, if $P(q)$ is convergent then $P(z)$ is convergent for $|z| < |q|$, and the new ring of doubt is $|q| \leq |z| \leq |d|$. Repeating this test procedure in the new ring will again halve its width. Continuing in this manner, the ring of doubt will narrow to a definite circle $|z| = R$ (called the *circle of convergence*) such that $P(z)$ converges everywhere inside the circle, and diverges everywhere outside the circle. See [2.16b]. The radius R is called the *radius of convergence*—at last we see where this name comes from!—and the interior of the circle is called the *disc of convergence*.

Note that this argument tells us nothing about the convergence of $P(z)$ on the circle of convergence. In principle, we can imagine convergence at all, some, or none of the points on this circle, and one can actually find examples of power series that realize each of these three possibilities. See Ex. 11.

All of the above results immediately generalize to a power series centred at an arbitrary point k , that is to a series of the form $P(z) = \sum c_j Z^j$, where $Z = (z - k)$ is the complex number from the centre k to the point z . Thus, restating our main conclusion (due to Niels Abel) in general form,

Given a complex power series $P(z)$ centred at k , there exists a circle $|z - k| = R$ centred at k such that $P(z)$ converges everywhere inside the circle, and $P(z)$ diverges everywhere outside the circle. (2.11)

Of course one can also have a series that converges everywhere, but this may be thought of as the limiting case in which the circle of convergence is infinitely large.

Returning to figures [2.13a] and [2.13b], we now recognize the illustrated discs as the discs of convergence of the series for $1/(1 \mp z^2)$.

2.3.3 Approximating a Power Series with a Polynomial

Implicit in the definition of convergence is a simple but very important fact: if $P(a)$ converges, then its value can be *approximated* by the partial sum $P_m(a)$, and by choosing a sufficiently large value of m we can make the approximation as accurate as we wish. Combining this observation with (2.11),

At each point z in the disc of convergence, $P(z)$ can be approximated with arbitrarily high precision by a polynomial $P_m(z)$ of sufficiently high degree.

For simplicity's sake, let us investigate this further in the case that $P(z)$ is centred at the origin. The error $E_m(z)$ at z associated with the approximation $P_m(z)$ can be defined as the distance $E_m(z) = |P(z) - P_m(z)|$ between the exact answer and the approximation. For a fixed value of m , the error $E_m(z)$ will vary as z moves

around in the disc of convergence. Clearly, since $E_m(0) = 0$, the error will be extremely small if z is close to the origin, but what if z approaches the circle of convergence? The answer depends on the particular power series, but *it can happen that the error becomes enormous!* [See Ex. 12.] This does not contradict the above result: for any fixed z , no matter how close to the circle of convergence, the error $E_m(z)$ will become arbitrarily small as m tends to infinity.

This problem is avoided if we restrict z to the disc $|z| \leq r$, where $r < R$, because this prevents z from getting *arbitrarily* close to the circle of convergence, $|z| = R$. In attempting to approximate $P(z)$ within this disc, it turns out that we can do the following. We first decide on the maximum error (say ϵ) that we are willing to put up with, then choose (once and for all) an approximating polynomial $P_m(z)$ of sufficiently high degree that the error is smaller than ϵ throughout the disc. That is, throughout the disc, the approximating point $P_m(z)$ lies less than ϵ away from the true point, $P(z)$. One describes this by saying that $P(z)$ is *uniformly convergent* on this disc:

If $P(z)$ has disc of convergence $|z| < R$, then $P(z)$ is uniformly convergent
on the closed disc $|z| \leq r$, where $r < R$. (2.12)

Although we may not have uniform convergence on the whole disc of convergence, the above result shows that this is really a technicality: we do have uniform convergence on a disc that *almost* fills the complete disc of convergence, say $r = (0.999999999) R$.

To verify (2.12), first do Ex. 12, then have a good look at (2.9).

2.3.4 Uniqueness

If a complex function can be expressed as a power series, then it can only be done so in one way—the power series is *unique*. This is an immediate consequence of the *Identity Theorem*:

If

$$c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots = d_0 + d_1 z + d_2 z^2 + d_3 z^3 + \cdots$$

for all z in a neighbourhood (no matter how small) of 0, then the power
series are identical: $c_j = d_j$.

Putting $z = 0$ yields $c_0 = d_0$, so they may be cancelled from both sides. Dividing by z and again putting $z = 0$ then yields $c_1 = d_1$, and so on. [Although this was easy, Ex. 13 shows that it is actually rather remarkable.] The result can be strengthened considerably: *If the power series merely agree along a segment of curve (no matter how small) through 0, or if they agree at every point of an infinite sequence of points that converges to 0, then the series are identical.* The verification is essentially the same, only instead of putting $z = 0$, we now take the limit as z approaches 0, either along the segment of curve or through the sequence of points.

We can perhaps make greater intuitive sense of these results if we first recall that a power series can be approximated with arbitrarily high precision by a *polynomial* of sufficiently high degree. Given two points in the plane (no matter how close together) there is a unique line passing through them. Thinking in terms of a graph $y = f(x)$, this says that a polynomial of degree 1, say $f(x) = c_0 + c_1x$, is uniquely determined by the images of any two points, no matter how close together. Likewise, in the case of degree 2, if we are given three points (no matter how close together), there is only one parabolic graph $y = f(x) = c_0 + c_1x + c_2x^2$ that can be threaded through them. This idea easily extends to complex functions: *there is one, and only one, complex polynomial of degree n that maps a given set of $(n + 1)$ points to a given set of $(n + 1)$ image points.* The above result may therefore be thought of as the limiting case in which the number of known points (together with their known image points) tends to infinity.

Earlier we alluded to a sense in which $h(z) = 1/(1 + z^2)$ is the *only* complex function that agrees with the real function $H(x) = 1/(1 + x^2)$ on the real line. Yet clearly we can easily write down infinitely many complex functions that agree with $H(x)$ in this way. For example,

$$g(z) = g(x + iy) = \frac{\cos[x^2y] + i \sin[y^2]}{e^y + x^2 \ln(e + y^4)}.$$

Then in what sense can $h(z)$ be considered the unique generalization of $H(x)$?

We already know that $h(z)$ can be expressed as the power series $\sum_{j=0}^{\infty} (-1)^j z^{2j}$, and this fact yields [exercise] a provisional answer: $h(z)$ is the only complex function that (i) agrees with $H(x)$ on the real axis, and (ii) can be expressed as a power series in z . This still does not completely capture the sense in which $h(z)$ is unique, but it's a start.

More generally, suppose we are given a real function $F(x)$ that can be expressed as a power series in x on a (necessarily origin-centred) segment of the real line: $F(x) = \sum_{j=0}^{\infty} c_j x^j$. Then the complex power series $f(z) = \sum_{j=0}^{\infty} c_j z^j$ with the same coefficients can be used to define the unique complex function $f(z)$ that (i) agrees with F on the given segment of the real axis, and (ii) can be expressed as a power series in z .

For example, consider the *complex exponential function*, written e^z , the geometry of which we will discuss in the next section. Since $e^x = \sum_{j=0}^{\infty} x^j / j!$,

$$e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \frac{1}{4!}z^4 + \cdots.$$

Note that our heuristic, power-series approach to Euler's formula [Chapter 1] is starting to look more respectable!

2.3.5 Manipulating Power Series

The fact that power series can be approximated with arbitrarily high precision by polynomials implies [see Ex. 14] that

Two power series with the same centre can be added, multiplied, and divided in the same way as polynomials. (2.13)

If the two series $P(z)$ and $Q(z)$ have discs of convergence D_1 and D_2 , then the resulting series for $[P+Q]$ and PQ will both converge in the smaller of D_1 and D_2 , though they may in fact converge within a still larger disc. No such general statement is possible in the case of $(P/Q) = P(1/Q)$, because the convergence of the series for $(1/Q)$ is limited not only by the boundary circle of D_2 , but also by any points inside D_2 where $Q(z) = 0$.

Let us illustrate (2.13) with a few examples. Earlier we actually assumed this result in order to find the series for $1/(1-z^2)$ centred at k . Using the partial fraction decomposition

$$\frac{1}{1-z^2} = \frac{(1/2)}{1-z} + \frac{(1/2)}{1+z},$$

we obtained two power series for the functions on the RHS, and then *assumed* that these power series could be added like two polynomials, by adding the coefficients.

In the special case $k = 0$ we can check that this procedure works, because we already *know* the correct answer for the series centred at the origin:

$$\frac{1}{1-z^2} = 1 + z^2 + z^4 + z^6 + \dots$$

Since

$$\begin{aligned} \frac{1}{1-z} &= 1 + z + z^2 + z^3 + z^4 + z^5 + \dots \\ \text{and} \quad \frac{1}{1+z} &= 1 - z + z^2 - z^3 + z^4 - z^5 + \dots, \end{aligned}$$

we see that adding the coefficients of these series does indeed yield the correct series for $1/(1-z^2)$.

Since

$$\frac{1}{1-z^2} = \left[\frac{1}{1-z} \right] \left[\frac{1}{1+z} \right],$$

we can recycle this example to illustrate the correctness of multiplying power series as if they were polynomials:

$$\begin{aligned} &[1 + z + z^2 + z^3 + z^4 + z^5 + \dots][1 - z + z^2 - z^3 + z^4 - z^5 + \dots] \\ &= 1 + (1-1)z + (1-1+1)z^2 + (1-1+1-1)z^3 + (1-1+1-1+1)z^4 + \dots, \end{aligned}$$

which is again the correct series for $1/(1-z^2)$.

Next, let's use (2.13) to find the series for $1/(1-z)^2$:

$$\begin{aligned} & [1 + z + z^2 + z^3 + z^4 + z^5 + \cdots][1 + z + z^2 + z^3 + z^4 + z^5 + \cdots] \\ &= 1 + (1+1)z + (1+1+1)z^2 + (1+1+1+1)z^3 + (1+1+1+1+1)z^4 + \cdots, \end{aligned}$$

and so $(1-z)^{-2} = \sum_{j=0}^{\infty} (j+1)z^j$.

You may check for yourself that the above series for $(1-z)^{-1}$ and $(1-z)^{-2}$ are both special cases of the general *Binomial Theorem*, which states that if n is any real number (not just a positive integer), then within the unit disc,

$$(1+z)^n = 1 + nz + \frac{n(n-1)}{2!}z^2 + \frac{n(n-1)(n-2)}{3!}z^3 + \frac{n(n-1)(n-2)(n-3)}{4!}z^4 + \cdots \quad (2.14)$$

Historically, this result was one of Newton's key weapons in developing calculus, and later it played an equally central role in the work of Euler.

In Exs. 16, 17, 18, we show how manipulation of power series may be used to demonstrate the Binomial Theorem, first for all negative integers, then for all rational powers. Although we shall not discuss it further, the case of an irrational power ρ may be treated by taking an infinite sequence of rational numbers that converges to ρ . Later we shall use other methods to establish a still more general version of (2.14) in which the power n is allowed to be a *complex* number!

Next we describe how to divide two power series $P(z)$ and $Q(z)$. In order to find the series $P(z)/Q(z) = \sum c_j z^j$, one multiplies both sides by $Q(z)$ to obtain $P(z) = Q(z) \sum c_j z^j$, and then multiplies the two power series on the right. By the uniqueness result, the coefficients of this series must equal the known coefficients of $P(z)$, and this enables one to calculate the c_j 's. An example will make this process much clearer.

In order to find the coefficients c_j in the series $1/e^z = \sum c_j z^j$, we multiply both sides by e^z to obtain

$$\begin{aligned} 1 &= [1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \cdots][c_0 + c_1 z + c_2 z^2 + c_3 z^3 + c_4 z^4 + \cdots] \\ &= c_0 + (c_0 + c_1)z + \left(\frac{c_0}{2!} + \frac{c_1}{1!} + \frac{c_2}{0!}\right)z^2 + \left(\frac{c_0}{3!} + \frac{c_1}{2!} + \frac{c_2}{1!} + \frac{c_3}{0!}\right)z^3 \cdots \end{aligned}$$

By the uniqueness result, we may equate coefficients on both sides to obtain an infinite set of linear equations:

$$\begin{aligned} 1 &= c_0, \\ 0 &= c_0 + c_1, \\ 0 &= c_0/2! + c_1/1! + c_2/0!, \\ 0 &= c_0/3! + c_1/2! + c_2/1! + c_3/0!, \text{ etc.} \end{aligned}$$

Successively solving the first few of these equations [exercise] quickly leads to the guess $c_n = (-1)^n/n!$, which is then easily verified [exercise] by considering

the binomial expansion of $(1 - 1)^m$, where m is a positive integer. Thus we find that

$$1/e^z = 1 - z + \frac{1}{2!}z^2 - \frac{1}{3!}z^3 + \frac{1}{4!}z^4 - \frac{1}{5!}z^5 + \cdots = e^{-z},$$

just as with the real function e^x .

2.3.6 Finding the Radius of Convergence

Given a complex power series $P(z) = \sum c_j z^j$, there are several ways of determining its radius of convergence directly from its coefficients. Since they are formally identical to the methods used on real series, we merely state them, leaving it to you to generalize the standard real proofs.

The *ratio test* says that

$$R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|,$$

provided this limit exists. For example, if

$$P(z) = 1 + z + \frac{z^2}{2^2} + \frac{z^3}{3^2} + \frac{z^4}{4^2} + \cdots,$$

then

$$R = \lim_{n \rightarrow \infty} \frac{1/n^2}{1/(n+1)^2} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^2 = 1.$$

If $|c_n/c_{n+1}|$ tends to infinity then (formally) $R = \infty$, corresponding to convergence everywhere in the plane. For example, $e^z = \sum_{j=0}^{\infty} z^j/j!$ converges everywhere, because

$$R = \lim_{n \rightarrow \infty} \frac{1/n!}{1/(n+1)!} = \lim_{n \rightarrow \infty} (n+1) = \infty.$$

When the ratio test fails, or becomes difficult to apply, we can often use the *root test*, which says that

$$R = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{|c_n|}},$$

provided this limit exists. For example, if we first recall [we will discuss this later] that the real function e^x may be written as

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n,$$

then applying the root test to the series

$$P(z) = \sum_{j=1}^{\infty} \left(\frac{j-3}{j}\right)^{j^2} z^j,$$

yields [exercise] $R = e^3$.

On occasion both the ratio and root tests will fail, but there exists a slightly refined version of the latter which can be shown to work in *all* cases. It is called the *Cauchy-Hadamard Theorem*, and it says that

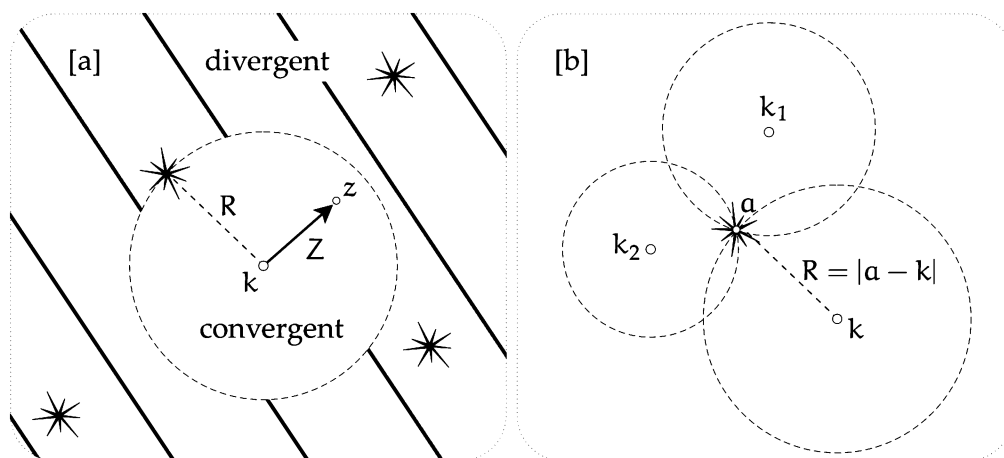
$$R = \frac{1}{\limsup \sqrt[n]{|c_n|}}.$$

We will not discuss this further since it is not needed in this book.

The above examples of power series were plucked out of thin air, but often our starting point is a known complex function $f(z)$ which is then *expressed* as a power series. The problem of determining R then has a conceptually much more satisfying answer. Roughly²,

If $f(z)$ can be expressed as a power series centred at k , then the radius of convergence is the distance from k to the nearest singularity of $f(z)$. (2.15)

Figure [2.17a] illustrates this, the singularities of $f(z)$ being represented as explosions. To understand which functions can be expanded into power series we need deep results from later in the book, but we are already in a position to verify that a rational function [the ratio of two polynomials] can be, and that the radius of convergence for its expansion is given by (2.15).



[2.17] [a] Here, explosions mark the locations of singularities of a function $f(z)$ that we wish to expand into a power series centred at k , i.e., a series in powers of $Z \equiv (z - k)$. Then the radius of convergence $R =$ (the distance from k to the nearest singularity.)
[b] Here we see the three discs of convergence that arise from three different choices of the centre k , assuming that a is the nearest singularity for all three centres.

² Later [p. 107] we shall have to modify the statement in the case that $f(z)$ is a “multifunction”, having more than one value for a given value of z .

To begin with, reconsider [2.13a] and [2.13b], both of which are examples of (2.15). Recall that in [2.13b] we merely *claimed* that $R = \sqrt{1 + k^2}$ for the series expansion of $h(z) = 1/(1 + z^2)$ centred at the real point k . We now verify this and explicitly find the series.

To do so, first note that (2.4) easily generalizes to

$$\frac{1}{a - z} = \sum_{j=0}^{\infty} \frac{Z^j}{(a - k)^{j+1}}, \quad \text{if and only if } |Z| < |a - k|, \quad (2.16)$$

where a and k are now arbitrary *complex* numbers, and $Z = (z - k)$ is the complex number connecting the centre of the expansion to z . The condition $|z - k| < |a - k|$ for convergence is that z lie in the interior of the circle centred at k and passing through a . See [2.17b], which also shows the discs of convergence when we instead choose to expand $1/(a - z)$ about k_1 or k_2 . Since the function $1/(a - z)$ has just one singularity at $z = a$, we have verified (2.15) for this particular function.

Earlier we found the expansion of $1/(1 - x^2)$ by factorizing the denominator and using partial fractions. We are now in a position to use exactly the same approach to find the expansion of $h(z) = 1/(1 + z^2)$ centred at an arbitrary complex number k :

$$\frac{1}{1 + z^2} = \frac{1}{(z - i)(z + i)} = \frac{1}{2i} \left[\frac{1}{-i - z} - \frac{1}{i - z} \right].$$

Applying (2.16) to both terms then yields

$$\frac{1}{1 + z^2} = \sum_{j=0}^{\infty} \frac{1}{2i} \left[\frac{1}{(-i - k)^{j+1}} - \frac{1}{(i - k)^{j+1}} \right] Z^j. \quad (2.17)$$

The series for $1/(\pm i - z)$ converge inside the concentric circles $|z - k| = |\pm i - k|$ centred at k and passing through the points $\mp i$, which are the singularities of $h(z)$. But (2.17) will only converge when *both* these series converge, i.e., in the disc $|z - k| < R$ where R is the distance from the centre k to the nearest singularity of $h(z)$. Thus we have confirmed (2.15) for $h(z)$.

In particular, if k is real then (2.17) converges in the disc shown in [2.13b]. If z is restricted to the real axis then $h(z)$ reduces to the real function $1/(1 + x^2)$, and the expansion of this function into powers of $X = (x - k)$ can be deduced easily from (2.17). Since k is now real, $|i - k| = \sqrt{1 + k^2}$, and we may write $(i - k) = \sqrt{1 + k^2} e^{i\phi}$, where $\phi = \arg(i - k)$ is the appropriate value of $\tan^{-1}(-1/k)$. Thus [exercise]

$$\frac{1}{1 + x^2} = \sum_{j=0}^{\infty} \left[\frac{\sin(j + 1)\phi}{(\sqrt{1 + k^2})^{j+1}} \right] X^j. \quad (2.18)$$

Again, we have here a result concerning real functions that would be very difficult to obtain using only real numbers.

The above analysis of $1/(1+z^2)$ can easily be generalized [exercise] to show that any rational function can be expressed as a power series, with radius of convergence given by (2.15).

2.3.7 Fourier Series*

On the 21st of December 1807, Joseph Fourier announced to the French Academy a discovery so remarkable that his distinguished audience found it *literally* incredible. His claim was that any³ real periodic function $F(\theta)$, no matter how capricious its graph, may be decomposed into a sum of sinusoidal waves of higher and higher frequency. For simplicity's sake, let the period be 2π ; then the *Fourier series* is

$$F(\theta) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} [a_n \cos n\theta + b_n \sin n\theta],$$

where [see Ex. 20]

$$a_n = \frac{1}{\pi} \int_0^{2\pi} F(\theta) \cos n\theta \, d\theta \quad \text{and} \quad b_n = \frac{1}{\pi} \int_0^{2\pi} F(\theta) \sin n\theta \, d\theta. \quad (2.19)$$

This optional section is addressed primarily to readers who have already encountered such series. For those who have not, we hope that this brief discussion (together with the exercises at the end of the chapter) may whet your appetite for more on this fascinating subject⁴.

In the world of the real numbers there appears to be no possible connection between the concepts of Fourier series and Taylor series, but when we pass into the complex realm a beautiful and remarkable fact emerges:

Taylor series and Fourier series of real functions are merely two different ways of viewing complex power series.

We will explain this cryptic pronouncement by means of an example.

Consider the complex function $f(z) = 1/(1-z)$. Writing $z = r e^{i\theta}$, one finds [exercise] that the real and imaginary parts of $f(r e^{i\theta})$ are given by

$$f(r e^{i\theta}) = u(r e^{i\theta}) + i v(r e^{i\theta}) = \left[\frac{1 - r \cos \theta}{1 + r^2 - 2r \cos \theta} \right] + i \left[\frac{r \sin \theta}{1 + r^2 - 2r \cos \theta} \right].$$

Let's concentrate on just one of these real functions, say v .

³ Later it was found that *some* restrictions must be placed on F , but they are astonishingly weak.

⁴ In many areas of mathematics it is hard to find even one really enlightening book, but Fourier analysis has been blessed with at least two: [Lanczos \(1966\)](#), and [Körner \(1988\)](#).

If z moves outward from the origin along a ray $\theta = \text{const.}$ then $v(r e^{i\theta})$ becomes a function of r alone, say $V_\theta(r)$. For example,

$$V_{\frac{\pi}{4}}(r) = \frac{r}{\sqrt{2}(1+r^2) - 2r}.$$

If z instead travels round and round a circle $r = \text{const.}$ then v becomes a function of θ alone, say $\tilde{V}_r(\theta)$. For example,

$$\tilde{V}_{\frac{1}{2}}(\theta) = \frac{2 \sin \theta}{5 - 4 \cos \theta}.$$

Note that this is a periodic function of θ , with period 2π . The reason is simple and applies to any $\tilde{V}_r(\theta)$ arising from a (single-valued) function $f(z)$: each time z makes a complete revolution and returns to its original position, $f(z)$ travels along a closed loop and returns to its original position.

Now, to see the unity of Taylor and Fourier series, recall that (within the unit disc) $f(z) = 1/(1-z)$ can be expressed as a convergent complex power series:

$$\begin{aligned} f(r e^{i\theta}) &= 1 + (r e^{i\theta}) + (r e^{i\theta})^2 + (r e^{i\theta})^3 + (r e^{i\theta})^4 + \dots \\ &= 1 + r(\cos \theta + i \sin \theta) + r^2(\cos 2\theta + i \sin 2\theta) + r^3(\cos 3\theta + i \sin 3\theta) + \dots \end{aligned}$$

In particular,

$$v(r e^{i\theta}) = r \sin \theta + r^2 \sin 2\theta + r^3 \sin 3\theta + r^4 \sin 4\theta + r^5 \sin 5\theta + \dots$$

If we put $\theta = (\pi/4)$, we immediately obtain the Taylor series for $V_{\frac{\pi}{4}}(r)$:

$$\frac{r}{\sqrt{2}(1+r^2) - 2r} = V_{\frac{\pi}{4}}(r) = \frac{1}{\sqrt{2}} r + r^2 + \frac{1}{\sqrt{2}} r^3 - \frac{1}{\sqrt{2}} r^5 - r^6 - \frac{1}{\sqrt{2}} r^7 + \frac{1}{\sqrt{2}} r^9 + \dots$$

Once again, consider how difficult this would be to obtain using only real numbers. From this we find, for example, that

$$\left. \frac{d^{98}}{dr^{98}} \left[\frac{r}{\sqrt{2}(1+r^2) - 2r} \right] \right|_{r=0} = 98!$$

If we instead put $r = (1/2)$, we immediately obtain the Fourier series for $\tilde{V}_{\frac{1}{2}}(\theta)$:

$$\frac{2 \sin \theta}{5 - 4 \cos \theta} = \tilde{V}_{\frac{1}{2}}(\theta) = \frac{1}{2} \sin \theta + \frac{1}{2^2} \sin 2\theta + \frac{1}{2^3} \sin 3\theta + \frac{1}{2^4} \sin 4\theta + \dots$$

The absence of cosine waves in this series correctly reflects the fact that $\tilde{V}_{\frac{1}{2}}(\theta)$ is an *odd* function of θ .

This connection between complex power series and Fourier series is not merely aesthetically satisfying, it can also be very practical. The conventional derivation of the Fourier series of $\tilde{V}_{\frac{1}{2}}(\theta)$ requires that we evaluate the tricky integrals in (2.19),

whereas we have obtained the result using only simple algebra! Indeed, we can now use our Fourier series to do integration:

$$\int_0^{2\pi} \left[\frac{2 \sin \theta \sin n\theta}{5 - 4 \cos \theta} \right] d\theta = \frac{\pi}{2^n}.$$

Further examples may be found in Exs. 21, 37, 38.

We end with a premonition of things to come. The coefficients in a Taylor series may be calculated by differentiation, while those in a Fourier series may be calculated by integration. Since these two types of series are really the same in the complex plane, this suggests that there exists some hidden connection between differentiation and integration that only complex numbers can reveal. Later we shall see how Cauchy confirmed this idea in spectacular fashion.

2.4 The Exponential Function

2.4.1 Power Series Approach

We have seen that the only complex function expressible as a power series that generalizes the real function e^x to complex values is

$$e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \frac{1}{4!}z^4 + \cdots,$$

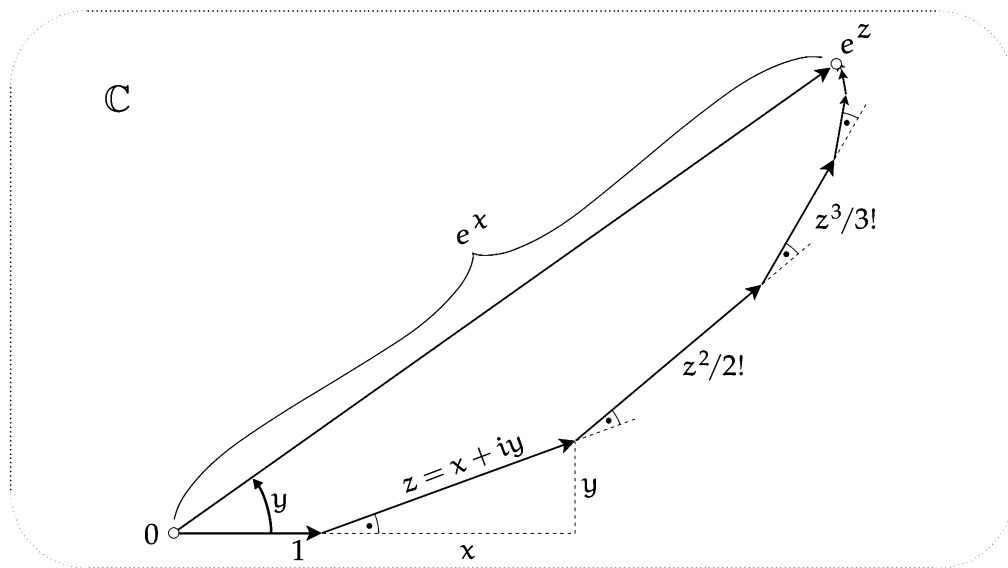
which converges everywhere in $\mathbb{C}$. We now investigate the geometric nature of this function.

Figure [2.18] visualizes the above series as a spiral journey, the angle between successive legs of the journey being fixed and equal to $\bullet = \arg(z) = \arg(x + iy)$. In the special case where this angle is a right angle, we saw in Chapter 1 that the spiral converges to a point on the unit circle given by Euler's formula, $e^{iy} = \cos y + i \sin y$. In fact this special spiral enables us to figure out what happens in the case of the general spiral in [2.18]: for an arbitrary value of $z = x + iy$, *the spiral converges to the illustrated point at distance e^x and at angle y* . In other words,

$$e^{x+iy} = e^x e^{iy}.$$

This is a consequence of the fact that if a and b are arbitrary complex numbers, then $e^a e^b = e^{a+b}$. To verify this we simply multiply the two series:

$$\begin{aligned} e^a e^b &= \left[1 + a + \frac{1}{2!}a^2 + \frac{1}{3!}a^3 + \cdots \right] \left[1 + b + \frac{1}{2!}b^2 + \frac{1}{3!}b^3 + \cdots \right] \\ &= 1 + (a + b) + \left[\frac{a^2 + 2ab + b^2}{2!} \right] + \left[\frac{a^3 + 3a^2b + 3ab^2 + b^3}{3!} \right] + \cdots \\ &= 1 + (a + b) + \frac{1}{2!}(a + b)^2 + \frac{1}{3!}(a + b)^3 + \cdots \\ &= e^{a+b}. \end{aligned}$$



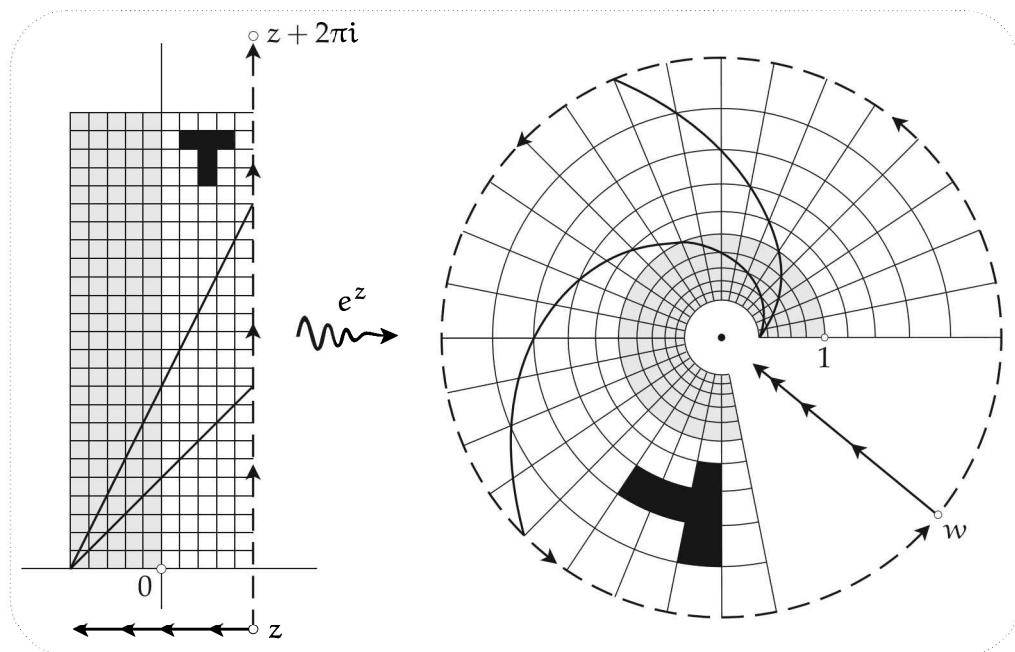
[2.18] The power series for e^z can be visualized as a spiral journey, the angle between successive legs of the journey being fixed and equal to $\bullet = \arg(z) = \arg(x+iy)$. In the text we show that the power series has the property that $e^a e^b = e^{a+b}$, so that $e^{x+iy} = e^x e^{iy}$. Thus the spiral ends at the point that has length e^x and angle y , as illustrated.

Here we have left it to you to show that the *general* term in the penultimate line is indeed $(a+b)^n/n!$.

2.4.2 The Geometry of the Mapping

Figure [2.19] illustrates the essential features of the mapping $z \mapsto w = e^z$. Study it carefully, noting the following facts:

- If z travels upward at a steady speed s , then w rotates about the origin at angular speed s . After z has travelled a distance of 2π , w returns to its starting position. Thus the mapping is *periodic*, with period $2\pi i$.
- If z travels westward at a steady speed, w travels towards the origin, with ever decreasing speed. Conversely, if z travels eastward at a steady speed, w travels away from the origin with ever increasing speed.
- Combining the previous two facts, the entire w -plane (with the exception of $w = 0$) will be filled by the image of any horizontal strip in the z -plane of height 2π .
- A line in general position is mapped to a spiral of the type discussed in the previous chapter.
- Euler's formula $e^{iy} = \cos y + i \sin y$ can be interpreted as saying that e^z wraps the imaginary axis round and round the unit circle like a piece of string.



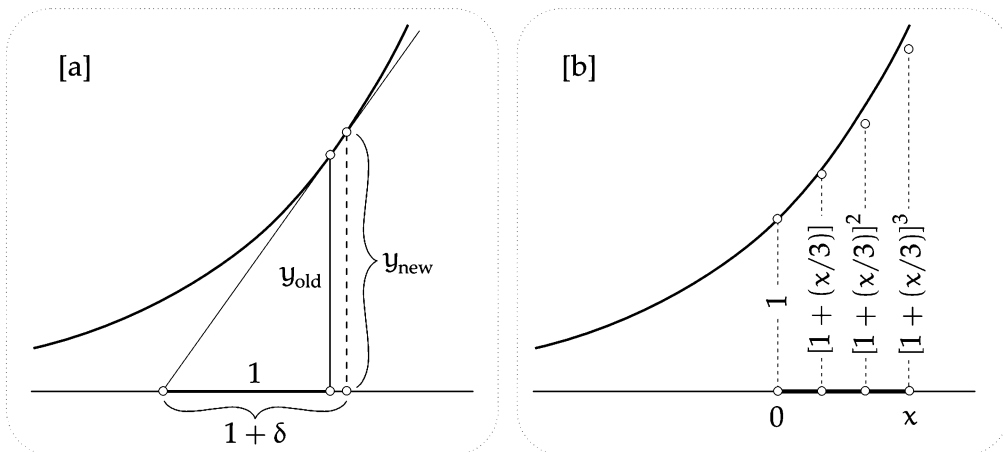
[2.19] The geometry of the complex exponential follows from the fact that e^{x+iy} has length e^x and angle y . Thus horizontal lines $y = \text{const.}$ map to rays, and vertical lines $x = \text{const.}$ map to circles. Mysteriously, for now, a computer can be used to empirically confirm that small squares are ultimately mapped to small squares, in the limit that they shrink and vanish. This is a central mystery of complex analysis, and it will begin to unravel in Chapter 4.

- The half-plane to the left of the imaginary axis is mapped to the interior of the unit circle, and the half-plane to the right of the imaginary axis is mapped to the exterior of the unit circle.
- The images of the small squares closely resemble squares, and (related to this) any two intersecting lines map to curves that intersect at the same angle as the lines themselves.

The last of these observations is not intended to be self-evident—in Chapter 4 we will begin to explore this fundamental property and to see that it is shared by many other important complex mappings.

2.4.3 Another Approach

The advantage of the power series approach to e^z is that it suggests that there is something unique about this generalization of e^x to complex values. The disadvantage is the amount of unilluminating algebra needed to decipher the geometric



[2.20] [a] The fact that $(e^x)' = e^x$ implies that the shaded triangle has unit base. It follows that $y_{\text{new}} \approx (1 + \delta) y_{\text{old}}$ as δ vanishes. **[b]** Now start with $x = 0$ and $y_{\text{old}} = 1$. Taking $\delta = (x/n)$ and repeatedly moving δ to the right, the height is ultimately multiplied by $[1 + \frac{x}{n}]$ each time, so $e^x \approx [1 + \frac{x}{n}]^n$.

meaning of the series. We now describe a different approach in which the geometry lies much closer to the surface. The idea is to generalize the real result,

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n. \quad (2.20)$$

Here is one way of understanding (2.20). As we discussed in Chapter 1, $f(x) = e^x$ may be defined by the property $f'(x) = f(x)$. Figure [2.20a] interprets this in terms of the graph of $y = f(x)$. Drawing a tangent at an arbitrary point, the base of the shaded triangle is always equal to 1. As you see from the figure, it follows that if the height is y_{old} at some point x , then moving x an infinitesimal distance δ to the right yields a new height given by

$$y_{\text{new}} = (1 + \delta) y_{\text{old}}.$$

To find the height e^x at x , we divide the interval $[0, x]$ into a large number n of very short intervals of length (x/n) . Since the height at $x = 0$ is 1, the height at (x/n) will be approximately $[1 + (x/n)] \cdot 1$, and so the height at $2(x/n)$ will be approximately $[1 + (x/n)] \cdot [1 + (x/n)] \cdot 1$, and so....., and so the height at $x = n(x/n)$ will be approximately $[1 + (x/n)]^n$. [For clarity's sake, [2.20b] illustrates this geometric progression with the small (hence inaccurate) value $n = 3$.] It is now plausible that the approximation $[1 + (x/n)]^n$ becomes more and more accurate as n tends to infinity, thereby yielding (2.20). Try using a computer to verify empirically that the accuracy does indeed increase with n .

Generalizing (2.20) to complex values, we may define e^z as

$$e^z = \lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n. \quad (2.21)$$

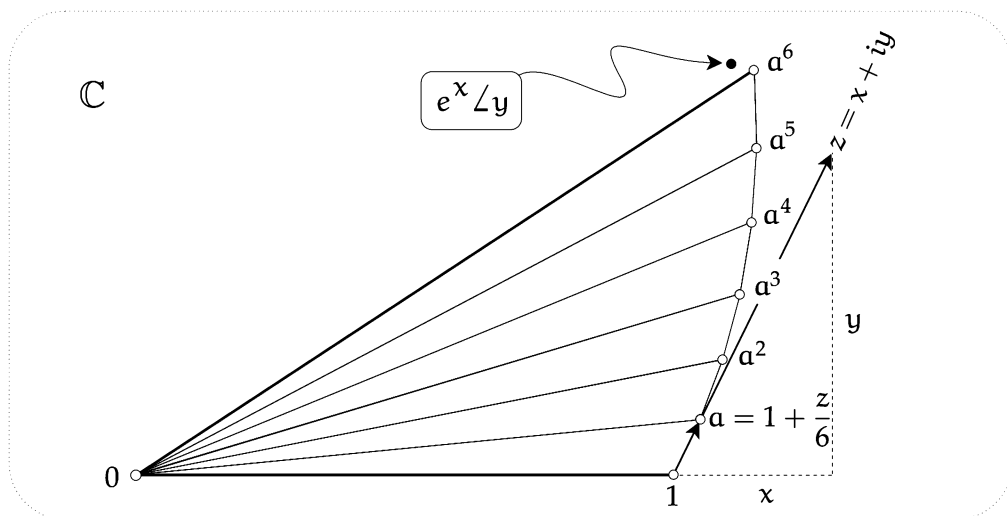
First we should check that this is the *same* generalization of e^x that we obtained using power series. Using the Binomial Theorem to write down the first few terms of the n^{th} degree polynomial $[1 + (z/n)]^n$, we get

$$\begin{aligned} \left(1 + \frac{z}{n}\right)^n &= 1 + n \left[\frac{z}{n}\right] + \frac{n(n-1)}{2!} \left[\frac{z}{n}\right]^2 + \frac{n(n-1)(n-2)}{3!} \left[\frac{z}{n}\right]^3 + \dots \\ &= 1 + z + \frac{\left(1 - \frac{1}{n}\right)}{2!} z^2 + \frac{\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)}{3!} z^3 + \dots, \end{aligned}$$

which makes it clear that we do recover the original power series as n tends to infinity.

Next we turn to the geometry of (2.21). In deciphering the power series for e^z we felt free to assume Euler's formula, because in Chapter 1 we used the power series to derive that result. However, it would smack of circular reasoning if we were to assume Euler's formula while following our new approach to e^z , based on (2.21). Temporarily, we shall therefore revert to our earlier notation and write $r\angle\theta$ instead of $re^{i\theta}$; the fact we wish to understand is therefore written $e^{x+iy} = e^x \angle y$.

With $n = 6$, figure [2.21] uses Ex. 1.5, p. 52, to geometrically construct the successive powers of $a \equiv [1 + (z/n)]$ for a specific value of z . [All six shaded triangles are similar; the two kinds of shading merely help to distinguish one triangle from the next.] Even with this small value of n , we see empirically that in this particular case $[1 + (z/n)]^n$ is close to $e^x \angle y$. To understand this mathematically, we will try to approximate $a = [1 + (z/n)]$ when n is large.

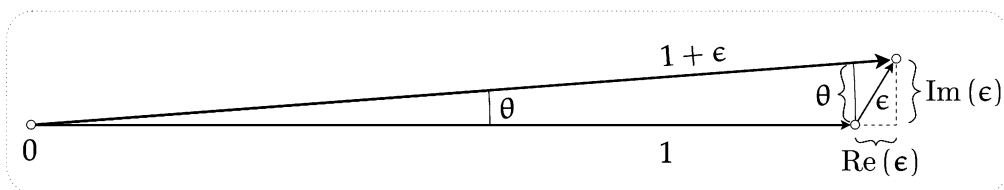


[2.21] Begin with the lightly shaded triangle with vertices 0, 1, and $a \equiv [1 + (z/n)]$, here illustrated with $n = 6$. Now successively construct triangles that are all similar to the original, as illustrated. We have thereby constructed the successive powers of a , culminating in $a^n = [1 + (z/n)]^n$. Our earlier computations demonstrate that this must ultimately be equal to $e^x \angle y$, and this concrete example seems to confirm this. But what is the geometrical explanation?!

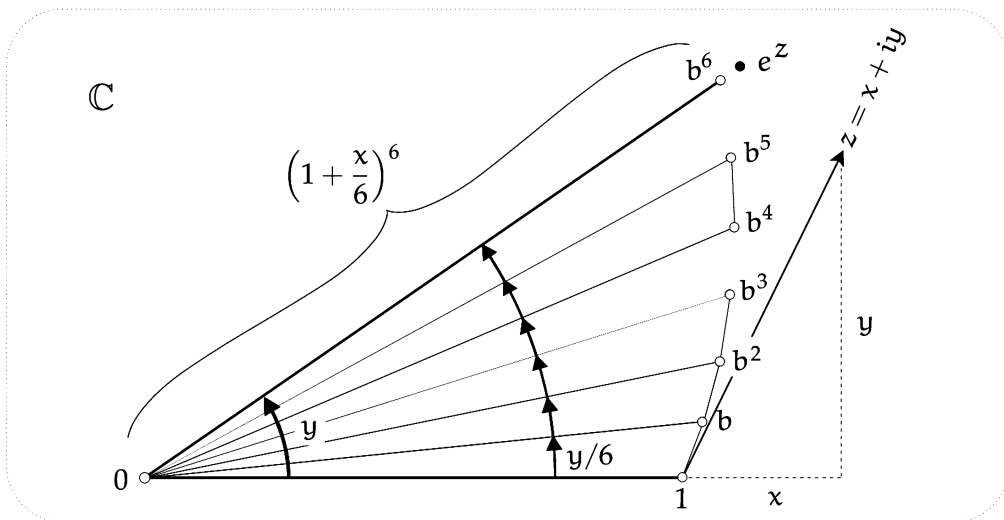
Let ϵ be a small, ultimately vanishing, complex number. Consider the length r and angle θ of the number $(1 + \epsilon) = r\angle\theta$ shown in [2.22]. The origin-centred circular arc [not shown] connecting $(1 + \epsilon)$ to the point r on the real axis almost coincides with the illustrated perpendicular from $(1 + \epsilon)$ to the real axis. Thus r is approximately equal to $[1 + \text{Re}(\epsilon)]$, and is ultimately equal⁵ to it as ϵ tends to zero. Similarly, we see that the angle θ (the illustrated arc of the unit circle) is ultimately equal to $\text{Im}(\epsilon)$. Thus

$$(1 + \epsilon) \asymp [1 + \operatorname{Re}(\epsilon)] \angle \operatorname{Im}(\epsilon), \quad \text{as } \epsilon \text{ vanishes.}$$

Now set $\epsilon = (z/n) = (x + iy)/n$. With the same values of z and n as in [2.21], figure [2.23] shows the approximation $b \equiv (1 + \frac{x}{n}) \angle (\frac{y}{n})$ to a , together with its successive powers.



[2.22] As ϵ vanishes, we see that $(1 + \epsilon) \asymp [1 + \operatorname{Re}(\epsilon)] \angle \operatorname{Im}(\epsilon)$.



[2.23] Geometrical explanation of [2.21]. Taking $\epsilon = (z/n) = (x + iy)/n$ in the previous figure, we may now approximate $\alpha \equiv [1 + (z/n)]$ as $\beta \equiv (1 + \frac{x}{n}) \angle (\frac{y}{n})$, because $\alpha \asymp \beta$ as n goes to infinity. Here we once again illustrate the case $n = 6$. Thus, $[1 + \frac{z}{n}]^n \asymp [(1 + \frac{x}{n}) \angle (\frac{y}{n})]^n = (1 + \frac{x}{n})^n \angle y \asymp e^x \angle y$, and the mystery is solved!

⁵ Once again, " $\asymp$ " denotes Newton's concept of *ultimate equality*; see the new Preface.

Returning to the general case, the geometry of (2.21) should now be clear. As n goes to infinity,

$$\left[1 + \frac{z}{n}\right]^n \asymp \left[\left(1 + \frac{x}{n}\right) \angle \left(\frac{y}{n}\right)\right]^n = \left(1 + \frac{x}{n}\right)^n \angle y.$$

Thus, in this limit as n goes to infinity, and using (2.20), we deduce that

$$e^{x+iy} = e^x \angle y,$$

as was to be shown. In particular, if we put $x = 0$ then we recover Euler's formula, $e^{iy} = 1 \angle y$, and so we are entitled to write $e^{x+iy} = e^x e^{iy}$.

For a slightly different way of looking at (2.21), see Ex. 22.

2.5 Cosine and Sine

2.5.1 Definitions and Identities

In the previous chapter Euler's formula enabled us to express cosine and sine in terms of the exponential function evaluated along the imaginary axis:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \quad \text{and} \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

Now that we understand the effect of e^z on arbitrary points (not merely points on the imaginary axis), it is natural to extend the definitions of cosine and sine to the complex functions

$$\cos z \equiv \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z \equiv \frac{e^{iz} - e^{-iz}}{2i}. \quad (2.22)$$

Of course another way of generalizing $\cos x$ and $\sin x$ would be via their power series, discussed in the previous chapter. This leads to the alternative definitions,

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots, \quad \text{and} \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots.$$

However, by writing down the series for $e^{\pm iz}$ you can easily check that these two approaches both yield the same complex functions.

From the definitions (2.22) we see that $\cos z$ and $\sin z$ have much in common with their real ancestors. For example, $\cos(-z) = \cos z$, and $\sin(-z) = -\sin z$. Also, since e^z is periodic with period $2\pi i$, it follows that $\cos z$ and $\sin z$ are also periodic, but with period 2π . The meaning of this periodicity will become clearer when we examine the geometry of the mappings.

Other immediate consequences of (2.22) are the following important generalizations of Euler's formula:

$$e^{iz} = \cos z + i \sin z \quad \text{and} \quad e^{-iz} = \cos z - i \sin z.$$

WARNING: $\cos z$ and $\sin z$ are now complex numbers—they are *not* the real and imaginary parts of e^{iz} .

It is not hard to show that all the familiar identities for $\cos x$ and $\sin x$ continue to hold for our new complex functions. For example, we still have

$$\cos^2 z + \sin^2 z = (\cos z + i \sin z)(\cos z - i \sin z) = e^{iz} e^{-iz} = e^0 = 1,$$

despite the fact that this identity no longer expresses Pythagoras' Theorem. Similarly, we will show that if a and b are arbitrary complex numbers then

$$\cos(a + b) = \cos a \cos b - \sin a \sin b \quad (2.23)$$

$$\sin(a + b) = \sin a \cos b + \cos a \sin b, \quad (2.24)$$

despite the fact that these identities no longer express the geometric rule for multiplying points on the unit circle. First,

$$\begin{aligned} \cos(a + b) + i \sin(a + b) &= e^{i(a+b)} \\ &= e^{ia} e^{ib} \\ &= (\cos a + i \sin a)(\cos b + i \sin b) \\ &= (\cos a \cos b - \sin a \sin b) + i(\sin a \cos b + \cos a \sin b), \end{aligned}$$

exactly as in the previous chapter. However, in view of the warning above, we *do not* obtain (2.23) and (2.24) simply by equating real and imaginary parts. Instead [exercise] one first finds the analogous identity for $\cos(a + b) - i \sin(a + b)$, then adds it to (or subtracts it from) the one above.

2.5.2 Relation to Hyperbolic Functions

Recall the definitions of the hyperbolic cosine and sine functions:

$$\cosh x \equiv \frac{e^x + e^{-x}}{2} \quad \text{and} \quad \sinh x \equiv \frac{e^x - e^{-x}}{2}.$$

By interpreting each of these as the average (i.e., midpoint) of e^x and $\pm e^{-x}$, it is easy to obtain the graphs $y = \cosh x$ and $y = \sinh x$ shown in [2.24a] and [2.24b].

As you probably know, $\cosh x$ and $\sinh x$ satisfy identities that are remarkably similar to those satisfied by $\cos x$ and $\sin x$, respectively. For example, if r_1 and r_2 are arbitrary real numbers, then [exercise]

$$\cosh(r_1 + r_2) = \cosh r_1 \cosh r_2 + \sinh r_1 \sinh r_2 \quad (2.25)$$

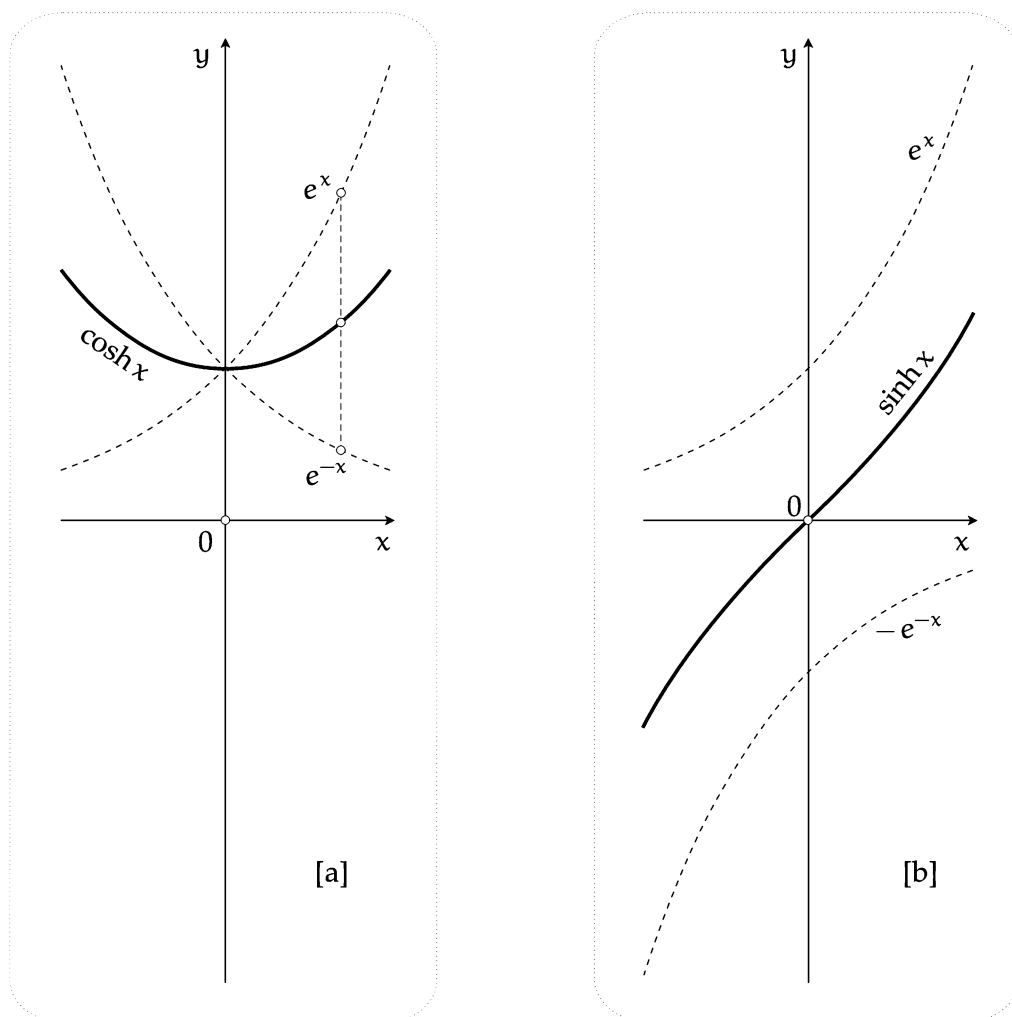
$$\sinh(r_1 + r_2) = \sinh r_1 \cosh r_2 + \cosh r_1 \sinh r_2. \quad (2.26)$$

Nevertheless, [2.24] shows that the actual behaviour of the hyperbolic functions is quite unlike the circular functions: they are not periodic, and they become arbitrarily large as x tends to infinity. It is therefore surprising and pleasing that the introduction of complex numbers brings about a unification of these two types of functions.

We begin to see this if we restrict $z = iy$ to the imaginary axis, for then

$$\cos(iy) = \cosh y \quad \text{and} \quad \sin(iy) = i \sinh y.$$

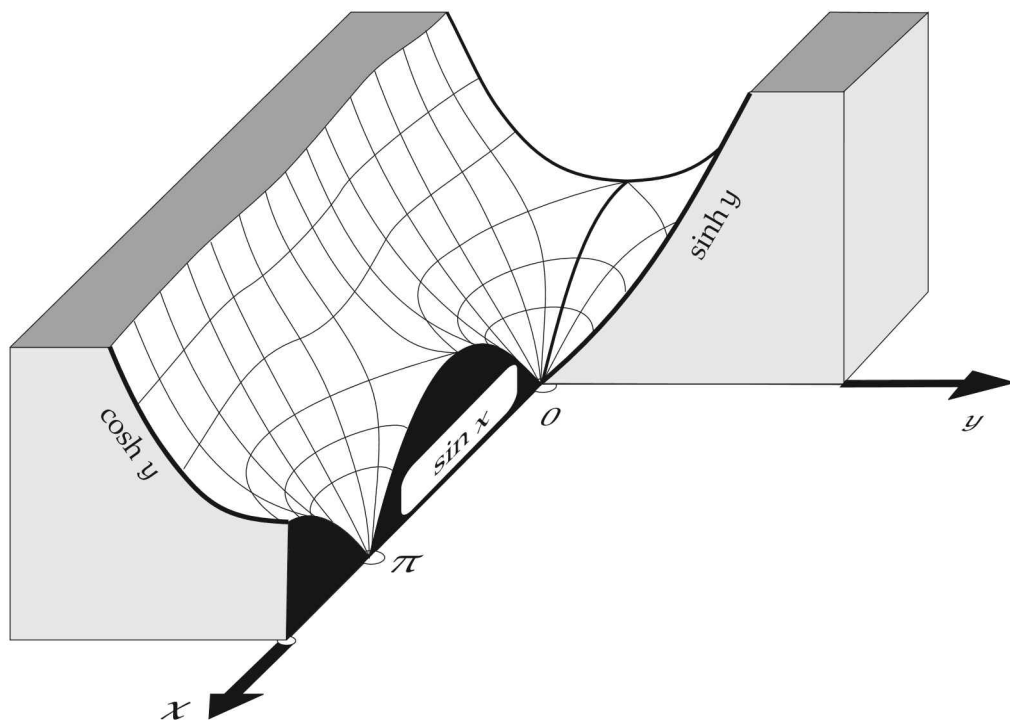
This connection becomes particularly vivid if we consider the modular surface of $\sin z$. Since $|\sin z|$ is ultimately equal to $|z|$ as z approaches the origin, it follows that



[2.24] [a] The graph of $\cosh x$ may be visualized as the average (i.e., midpoint) of the graphs of e^x and e^{-x} . [b] The graph of $\sinh x$ may be visualized in the same way, as the average of e^x and $-e^{-x}$.

the surface rises above the origin in the form of a cone. Also, $|\sin(z + \pi)| = |\sin z|$, so there is an identical cone at each multiple of π along the real axis. These are the only points [exercise] at which the surface hits the plane. Figure [2.25]—which we have adapted from [Markushevich \(2005\)](#)—shows a portion of the surface. Notice that this surface also yields the $\cosh$ graph, for if we restrict $z = (3\pi/2) + iy$ to the line $x = (3\pi/2)$, for example, then $|\sin z| = \cosh y$.

A practical benefit of this unification is that if you can remember (or quickly derive using Euler's formula) a trig identity involving cosine and sine, then you can immediately write down the corresponding identity for the hyperbolic functions. For example, if we substitute $a = ir_1$ and $b = ir_2$ into (2.23) and (2.24), then we obtain (2.25) and (2.26).



[2.25] Cutting open the modular surface of the complex function $\sin z$ reveals the previously hidden unity of the circular and hyperbolic functions.

The connection between the circular and hyperbolic functions becomes stronger still if we generalize the latter to complex functions in the obvious way:

$$\cosh z \equiv \frac{e^z + e^{-z}}{2} \quad \text{and} \quad \sinh z \equiv \frac{e^z - e^{-z}}{2}.$$

Since we now have

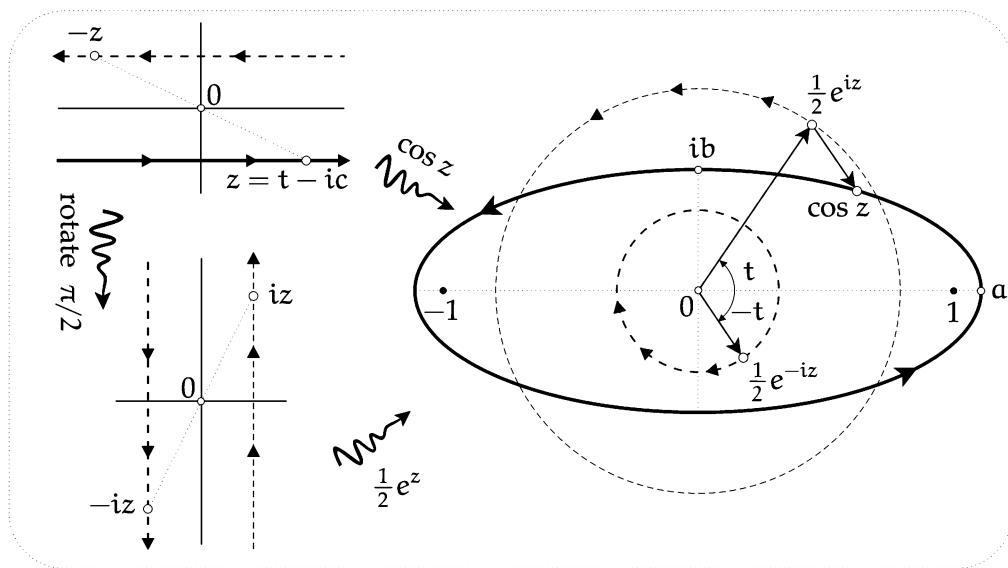
$$\cosh z = \cos(iz) \quad \text{and} \quad \sinh z = -i \sin(iz),$$

the distinction between the two kinds of function has all but evaporated: $\cosh$ is the composition of a rotation through $(\pi/2)$, followed by $\cos$; also, $\sinh$ is the composition of a rotation through $(\pi/2)$, followed by $\sin$, followed by a rotation through $-(\pi/2)$.

2.5.3 The Geometry of the Mapping

Just as in the real case, $\sin z = \cos(z - \frac{\pi}{2})$, which means that we may obtain $\sin$ from $\cos$ by first translating the plane by $-(\pi/2)$. It follows from the preceding remarks that it is sufficient to study just $\cos z$ in order to understand all four functions, $\cos z$, $\sin z$, $\cosh z$, and $\sinh z$. We now consider the geometric nature of the mapping $z \mapsto w = \cos z$.

We begin by finding the image of a horizontal line $y = -c$ lying below the real axis. It is psychologically helpful to picture the line as the orbit of a particle moving



[2.26] Understanding the geometry of e^z allows us to understand the geometry of $\cos z$. Consider the illustrated horizontal line $z(t) = t - ic$ (where t is time) traced from left to right. Then iz travels upward along the illustrated vertical line, and $\frac{1}{2}e^{iz}$ therefore travels counterclockwise along an origin-centred circle. By the same reasoning, $\frac{1}{2}e^{-iz}$ travels clockwise along the smaller circle. Finally, $\cos z$ is the sum of these two counter-rotating circular motions, and it traces an ellipse. It turns out that all horizontal lines are mapped to confocal ellipses, with foci ± 1 !

eastward at unit speed, whose position at time t is $z = t - ic$. See [2.26], in which the line is shown heavy and unbroken. As z traces this line, $-z$ traces the line $y = c$, but in the opposite direction. Applying the mapping $z \mapsto iz$ (which is a rotation of $\frac{\pi}{2}$), the image particles trace the vertical lines $x = \pm c$, again with unit speed and in opposite directions. Finally applying $z \mapsto \frac{1}{2}e^z$, the image particles orbit with equal and opposite angular speeds in origin-centred circles of radii $\frac{1}{2}e^{\pm c}$.

The image orbit under $z \mapsto w = \cos z$ of the original particle travelling on the line $y = -c$ is just the sum of these counter-rotating circular motions. This is clearly some kind of symmetrical oval hitting the real and imaginary axes at $a = \cosh c$ and $ib = i \sinh c$. It is also clear that $\cos z$ executes a complete orbit of this oval with each movement of 2π by z ; this is the geometric meaning of the periodicity of $\cos z$.

I haven't found a simple geometric explanation, but it's easy to show symbolically that the oval traced by $\cos z$ is a perfect ellipse. Writing $w = u + iv$, we find from the figure [exercise] that $u = a \cos t$ and $v = b \sin t$, which is the familiar parametric representation of the ellipse $(u/a)^2 + (v/b)^2 = 1$. Furthermore,

$$\sqrt{a^2 - b^2} = \sqrt{\cosh^2 c - \sinh^2 c} = 1,$$

so the foci are at ± 1 , independent of which particular horizontal line z travels along.

Try mulling this over. How does the shape of the ellipse change as we vary c ? How do we recover the real cosine function as c tends to zero? What is the orbit of $\cos z$ as z travels eastward along the line $y = c$, above the real axis? What is the image of the vertical line $x = c$ under $z \mapsto \cosh z$? What is the orbit of $\sin z$ as z travels eastward along the line $y = c$; how does it differ from the orbit of $\cos z$; and is the resulting variation of $|\sin z|$ consistent with the modular surface shown in [2.25]?

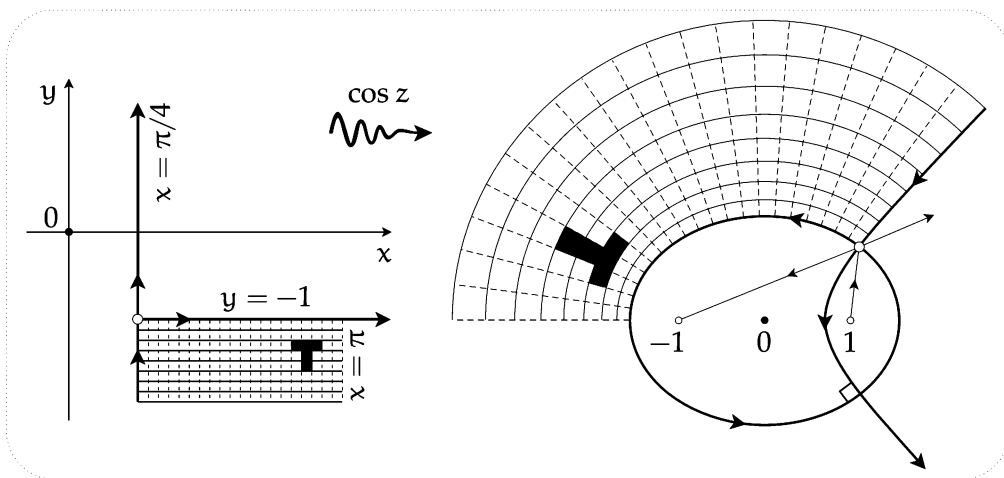
Before reading on, try using the idea in [2.26] to sketch for yourself the image under $z \mapsto \cos z$ of a *vertical* line.

As illustrated in [2.27], the answer is a *hyperbola*. We can show this using the addition rule (2.23), which yields

$$u + iv = \cos(x + iy) = \cos x \cosh y - i \sin x \sinh y.$$

On a horizontal line, y is constant, so $(u/\cosh y)^2 + (v/\sinh y)^2 = 1$, as before. On a vertical line, x is constant, so $(u/\cos x)^2 - (v/\sin x)^2 = 1$, which is the equation of a hyperbola. Furthermore, since $\cos^2 x + \sin^2 x = 1$, it follows that the foci of the hyperbola are always ± 1 , independent of which vertical line is being mapped.

Figure [2.27] tries to make these results more vivid by showing the image of a grid of horizontal and vertical lines. Note the empirical fact that *each small square in*



[2.27] Using the same reasoning as the previous figure, vertical lines are mapped by $\cos z$ to hyperbolas, and calculation reveals that these are also confocal, with the same foci as the ellipses, ± 1 ! Next, recall the reflection property of the conic sections: a ray of light emitted from a focus is reflected directly towards the other focus by the ellipse, and directly away from the other focus by the hyperbola, as illustrated. It follows that these two sets of conic sections are the orthogonal trajectories of each other. Furthermore—and this is an additional, surprising, and presently mysterious fact—as the size of the squares in the grid on the left tends to zero, their images under $\cos z$ are ultimately squares, too, as illustrated!

the grid is mapped by $\cos z$ to an image shape that is again approximately square. This is the same surprising (and visually pleasing) phenomenon that we observed in the case of $z \mapsto e^z$.

We hope your curiosity is piqued—later chapters are devoted to probing this phenomenon in depth. In the present case of $z \mapsto \cos z$ we can at least give a mathematical explanation of part of the result, namely, that the sides of the image “squares” do indeed meet at right angles; in other words, each ellipse cuts each hyperbola at right angles.

This hinges on the fact that these ellipses and hyperbolas are *confocal*. To prove the desired result [exercise], think of each curve as a mirror, then appeal to the familiar *reflection property* of the conic sections: a ray of light emitted from a focus is reflected directly towards the other focus by the ellipse, and directly away from the other focus by the hyperbola, as illustrated in [2.27].

2.6 Multifunctions

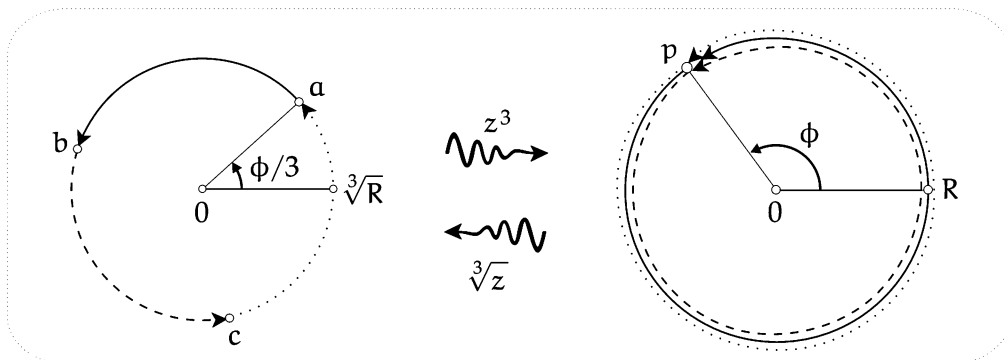
2.6.1 Example: Fractional Powers

Thus far we have considered a complex function f to be a rule that assigns to each point z (perhaps restricted to lie in some region) a single complex number $f(z)$. This familiar conception of a function is unduly restrictive. Using examples, we now discuss how we may broaden the definition of a function to allow $f(z)$ to have many different values for a single value of z . In this case f is called a “many-valued function”, or, as we shall prefer, a *multifunction*.

We have, in effect, already encountered such multifunctions. For example, we know that $\sqrt[3]{z}$ has three different values (if z is not zero), so it is a three-valued multifunction. In greater detail, [2.28] recalls how we can find the three values of $\sqrt[3]{p}$ using the mapping $z \mapsto z^3$. Having found one solution a , we can find the other two (b and c) using the fact that as $z = r e^{i\theta}$ orbits round an origin-centred circle, $z^3 = r^3 e^{i3\theta}$ orbits with three times the angular speed, executing a complete revolution each time z executes one third of a revolution. Put differently, reversing the direction of the mapping divides the angular speed by three. This is an essential ingredient in understanding the mapping $z \mapsto \sqrt[3]{z}$, which we will now study in detail.

Writing $z = r e^{i\theta}$, we have $\sqrt[3]{z} = \sqrt[3]{r} e^{i(\theta/3)}$. Here $\sqrt[3]{r}$ is uniquely defined as the real cube root of the length of z ; the sole source of the three-fold ambiguity in the formula is the fact that there are infinitely many different choices for the angle θ of a given point z .

Think of z as a moving point that is initially at $z = p$. If we arbitrarily choose θ to be the angle ϕ shown in [2.28], then $\sqrt[3]{p} = a$. As z gradually moves away from p , θ gradually changes from its initial value ϕ , and $\sqrt[3]{z} = \sqrt[3]{r} e^{i(\theta/3)}$ gradually moves



[2.28] The cube root is a three-valued multifunction. Since $z \mapsto z^3$ multiplies angles by 3, $z \mapsto \sqrt[3]{z}$ divides angles by 3. Now suppose we start at the illustrated point p at angle ϕ on the circle of radius R . Then it would seem clear that its cube root is the illustrated point $\sqrt[3]{p} = a$ at angle $(\phi/3)$ on the circle of radius $\sqrt[3]{R}$. But now suppose p executes a complete revolution and returns to its starting point. Then $\sqrt[3]{z}$ rotates at one third the angular speed and arrives at b : a second equally good value of $\sqrt[3]{p}$. If p keeps rotating, executing a second revolution, we arrive at $\sqrt[3]{p} = c$, and a third revolution returns us to the original value $\sqrt[3]{p} = a$. Thus the three values of $\sqrt[3]{p}$ are a , b , and c , and they form an equilateral triangle.

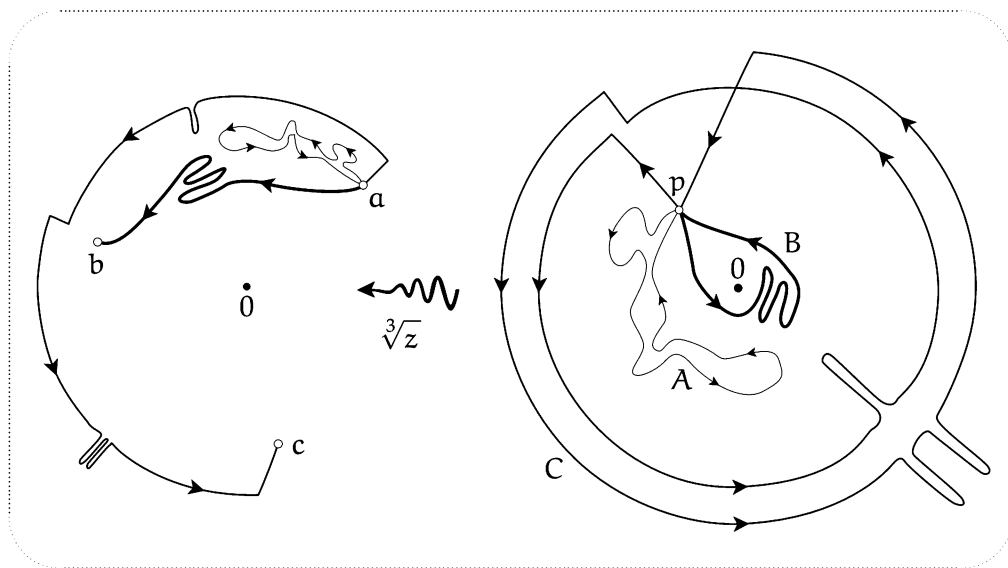
away from its initial position a , but in a completely determined way—its distance from the origin is the cube root of the distance of z , and its angular speed is one third that of z .

Figure [2.29] illustrates this. Usually we draw mappings going from left to right, but here we have reversed this convention to facilitate comparison with [2.28].

As z travels along the closed loop A (finally returning to p), $\sqrt[3]{z}$ travels along the illustrated closed loop and returns to its original value a . However, if z instead travels along the closed loop B , which goes round the origin once, then $\sqrt[3]{z}$ does *not* return to its original value but instead ends up at a different cube root of p , namely b . Note that the detailed shape of B is irrelevant, all that matters is that it encircles the origin once. Similarly, if z travels along C , encircling the origin twice, then $\sqrt[3]{z}$ ends up at c , the third and final cube root of p . Clearly, if z were to travel along a loop [not shown] that encircled the origin three times, then $\sqrt[3]{z}$ would return to its original value a .

The premise for this picture of $z \mapsto \sqrt[3]{z}$ was the arbitrary choice of $\sqrt[3]{p} = a$, rather than b or c . If we instead chose $\sqrt[3]{p} = b$, then the orbits on the left of [2.29] would simply be rotated by $(2\pi/3)$. Similarly, if we chose $\sqrt[3]{p} = c$, then the orbits would be rotated by $(4\pi/3)$.

The point $z = 0$ is called a *branch point* of $\sqrt[3]{z}$. More generally, let $f(z)$ be a multifunction and let $a = f(p)$ be one of its values at some point $z = p$. Arbitrarily choosing the initial position of $f(z)$ to be a , we may follow the movement of $f(z)$ as z travels along a closed loop beginning and ending at p . When z returns to p , $f(z)$



[2.29] Writing $z = r e^{i\theta}$, we have $\sqrt[3]{z} = \sqrt[3]{r} e^{i(\theta/3)}$. Once again, let us arbitrarily start at $\sqrt[3]{p} = a$, and then let p move along the three illustrated paths, A, B, and C. As you see, which cube root we end up at does not depend on the detailed shape of the path, it only depends on the net change in θ —how many complete revolutions have been executed when p returns home. The cube root rotates one third of a revolution each time p loops around the branch point at the origin.

will either return to a or it will not. A branch point $z = q$ of f is a point such that $f(z)$ fails to return to a as z travels along any loop that encircles q once.

Returning to the specific example $f(z) = \sqrt[3]{z}$, we have seen that if z executes three revolutions round the branch point at $z = 0$ then $f(z)$ returns to its original value. If $f(z)$ were an ordinary, single-valued function then it would return to its original value after only one revolution. Thus, relative to an ordinary function, *two extra* revolutions are needed to restore the original value of $f(z)$. We summarize this by saying that 0 is a branch point of $\sqrt[3]{z}$ of order two.

More generally, if q is a branch point of some multifunction $f(z)$, and $f(z)$ first returns to its original value after N revolutions round q , then q is called an *algebraic branch point* of order $(N - 1)$; an algebraic branch point of order 1 is called a *simple branch point*. We should stress that it is perfectly possible that $f(z)$ never returns to its original value, no matter how many times z travels round q . In this case q is called a *logarithmic branch point*—the name will be explained in the next section.

By extending the above discussion of $\sqrt[3]{z}$, check for yourself that if n is an integer then $z^{(1/n)}$ is an n -valued multifunction whose only (finite) branch point is at $z = 0$, the order of this branch point being $(n - 1)$. More generally, the same is true for any fractional power $z^{(m/n)}$, where (m/n) is a fraction reduced to lowest terms.

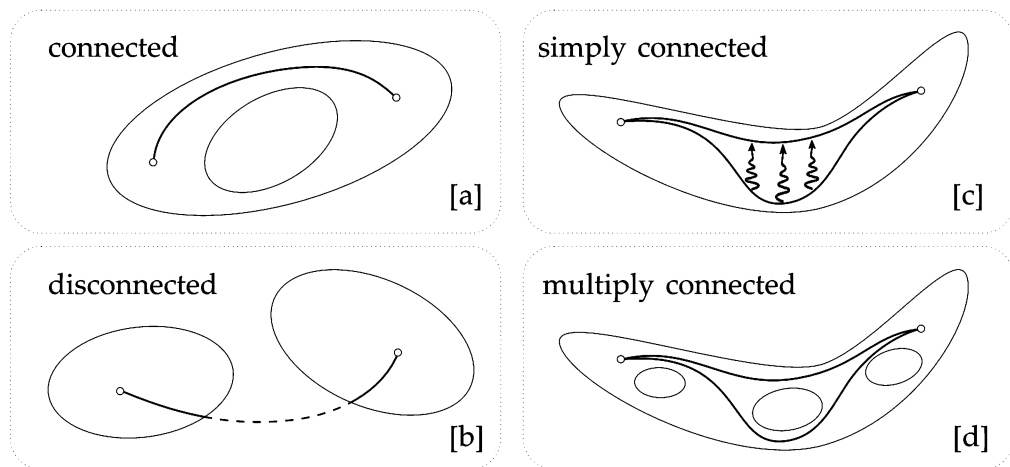
2.6.2 Single-Valued Branches of a Multifunction

Next we will show how we may extract three ordinary, single-valued functions from the three-valued multifunction $\sqrt[3]{z}$. First, [2.30] introduces some terminology which we need for describing sets of points in $\mathbb{C}$.

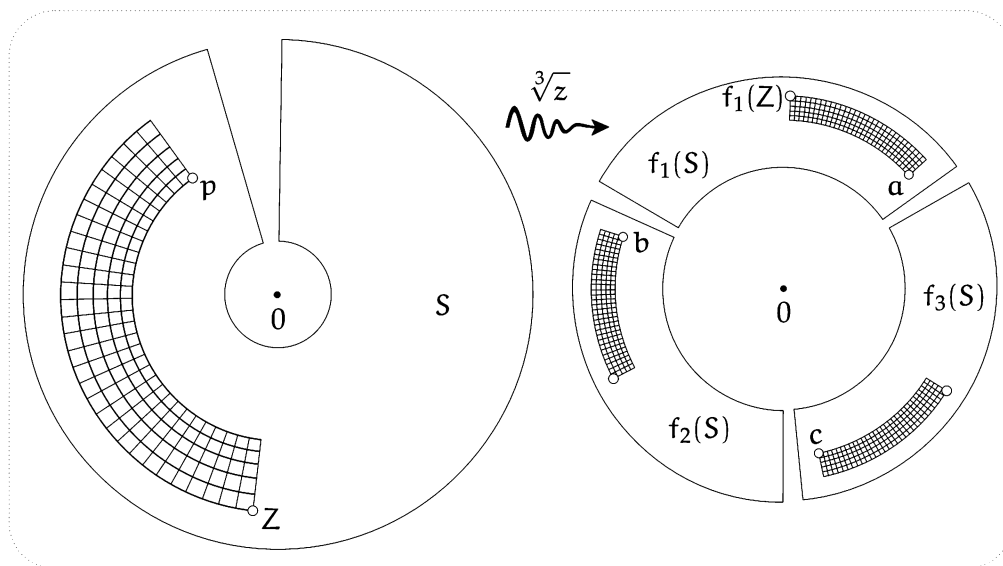
A set S is said to be *connected* (see [2.30a]) if any two points in S can be connected by an unbroken curve lying entirely within S . Conversely, if there exist pairs of points that cannot be connected in this way (see [2.30b]), then the set is *disconnected*. Amongst connected sets we may single out the *simply connected* sets (see [2.30c]) as those that do not have holes in them. More precisely, if we picture the path connecting two points in the set as an elastic string, then this string may be continuously deformed into any other path connecting the points, without any part of the string ever leaving the set. Conversely, if the set does have holes in it then it is *multiply connected* (see [2.30d]) and there exist two paths connecting two points such that one path cannot be deformed into the other.

Now let us return to [2.29]. By arbitrarily picking one of the three values of $\sqrt[3]{p}$ at $z = p$, and then allowing z to move, we see that we obtain a unique value of $\sqrt[3]{Z}$ associated with any particular path from p to Z . However, we are still dealing with a multifunction: by going round the branch point at 0 we can end up at any one of the three possible values of $\sqrt[3]{Z}$.

On the other hand, the value of $\sqrt[3]{Z}$ does not depend on the detailed shape of the path: *if we continuously deform the path without crossing the branch point then we obtain the same value of $\sqrt[3]{Z}$* . This shows us how we may obtain a single-valued function. If we restrict z to any simply connected set S that contains p but does not contain the branch point, then every path in S from p to Z will yield the same value of $\sqrt[3]{Z}$,



[2.30] Here is the terminology that is used to describe different types of regions in the plane.



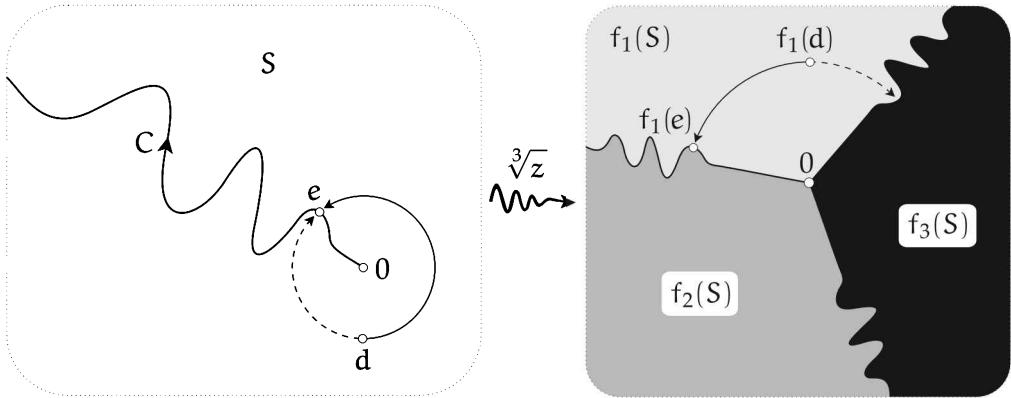
[2.31] If we chose a simply connected region S that does not contain the branch point at 0 , then no path within S can encircle the branch point at 0 , and therefore it is possible to define a single-valued branch of $\sqrt[3]{z}$ within S . For example, if we arbitrarily take $\sqrt[3]{p} = a$, then let $z = p$ move about within S and arrive at $z = Z$, say, then $\sqrt[3]{z}$ arrives at a unique value $f_1(Z) = \sqrt[3]{Z}$ that is independent of the path. But since there are three values of $\sqrt[3]{p}$, there are three distinct branches of $\sqrt[3]{z}$. Incidentally, note that these three branches display the ubiquitous (yet mysterious) preservation of small squares.

which we will call $f_1(Z)$. Since the path is irrelevant, f_1 is an ordinary, single-valued function of position on S ; it is called a *branch* of the original multifunction $\sqrt[3]{z}$.

Figure [2.31] illustrates such a set S , together with its image under the branch f_1 of $\sqrt[3]{z}$. Here we have reverted to our normal practice of depicting the mapping going from left to right. If we instead choose $\sqrt[3]{p} = b$ then we obtain a second branch f_2 of $\sqrt[3]{z}$, while $\sqrt[3]{p} = c$ yields the third and final branch f_3 . Notice, incidentally, that all three branches display the by now ubiquitous (yet mysterious) preservation of small squares.

We now describe how we may enlarge the domain S of the branches so as to obtain the cube roots of any point in the plane. First of all, as illustrated in [2.32], we draw an arbitrary (but not self-intersecting) curve C from the branch point 0 out to infinity; this is called a *branch cut*. Provisionally, we now take S to be the plane with the points of C removed—this prevents any closed path in S from encircling the branch point. We thereby obtain on S the three branches f_1 , f_2 , and f_3 . For example, the figure shows the cube root $f_1(d)$ of d .

What about a point such as e on C ? Imagine that z is travelling round an origin-centred circle through e . The figure illustrates the fact that $f_1(z)$ approaches two different values according as z arrives at e with positive or negative angular speed.



[2.32] In order to define a **single-valued branch** of $f(z) = \sqrt[3]{z}$ throughout the entire complex plane, we begin by drawing an arbitrary curve C that starts at the branch point 0 and extends out to infinity, as illustrated, and define S to be $\mathbb{C}$ with the points of C removed. The curve C is called a **branch cut**. Out of the three possible choices for $\sqrt[3]{a}$, let us arbitrarily choose the illustrated point $f_1(d)$. Then $f_1(Z)$ is single-valued throughout S , because it is impossible for z to encircle the branch point at 0 as it travels from d to Z . But we still need to define $f_1(e)$ at a point such as e that lies on C . By convention, it is defined to be the value of $f_1(z)$ when z arrives at e travelling counterclockwise, i.e., arriving from the right of C . The two other branches are defined in exactly the same way.

If we (arbitrarily) define $f_1(e)$ to be the value of $f(z)$ when z travels *counterclockwise* round the circle, then $f_1(z)$ is well defined on the whole plane. Similarly for the other two branches.

Of course the branch cut C is the work of man—the multifunction $\sqrt[3]{z}$ is oblivious to our desire to dissect it into three single-valued functions. As we have just seen, this shows up in the fact that the resulting branches are *discontinuous* on C , despite the fact that the three values of $\sqrt[3]{z}$ *always* move continuously as z moves continuously. As z crosses C travelling counterclockwise then we must switch from one branch to the next in order to maintain continuous motion of $\sqrt[3]{z}$: for example, f_1 switches to f_2 . If z executes three counterclockwise revolutions round the branch point, then the branches permute cyclically, each finally returning to itself: using an arrow to denote a crossing of C ,

$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} \rightarrow \begin{pmatrix} f_2 \\ f_3 \\ f_1 \end{pmatrix} \rightarrow \begin{pmatrix} f_3 \\ f_1 \\ f_2 \end{pmatrix} \rightarrow \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix}.$$

A common choice for C is the negative real axis. If we do not allow z to cross the cut then we may restrict the angle $\theta = \arg(z)$ to lie in the range $-\pi < \theta \leq \pi$. This is called the *principal value of the argument*, written $\text{Arg}(z)$; note the capital first letter. With this choice of θ , the single-valued function $\sqrt[3]{r} e^{i(\theta/3)}$ is called the *principal branch* of the cube root; let us write it as $[\sqrt[3]{z}]$. Note that the principal branch agrees

with the real cube root function on the positive real axis, but *not* on the negative real axis; for example, $[\sqrt[3]{-8}] = 2e^{i(\pi/3)}$. Also note that the other two branches associated with this choice of C can be expressed in terms of the principal branch as $e^{i(2\pi/3)}[\sqrt[3]{z}]$ and $e^{i(4\pi/3)}[\sqrt[3]{z}]$.

It should be clear how the above discussion extends to a general fractional power.

2.6.3 Relevance to Power Series

Earlier we explained the otherwise mysterious interval of convergence for a real function such as $1/(1+x^2)$ by extending the function off the real line and into the complex plane: the obstruction to convergence was the existence of points at which the complex function became infinite (singularities). We now discuss the more subtle fact that branch points also act as obstacles to the convergence of power series.

The real Binomial Theorem says that if n is any real number (not just a positive integer), then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 + \dots$$

If n is a positive integer then the series terminates at x^n and the issue of convergence does not arise. If n is not a positive integer then the ratio test tells us that the interval of convergence of the power series is $-1 < x < 1$. This interval is easily understood when n is negative, because the function then has a singularity at $x = -1$. But how, for example, are we to explain this interval of convergence in the case $n = (1/3)$?

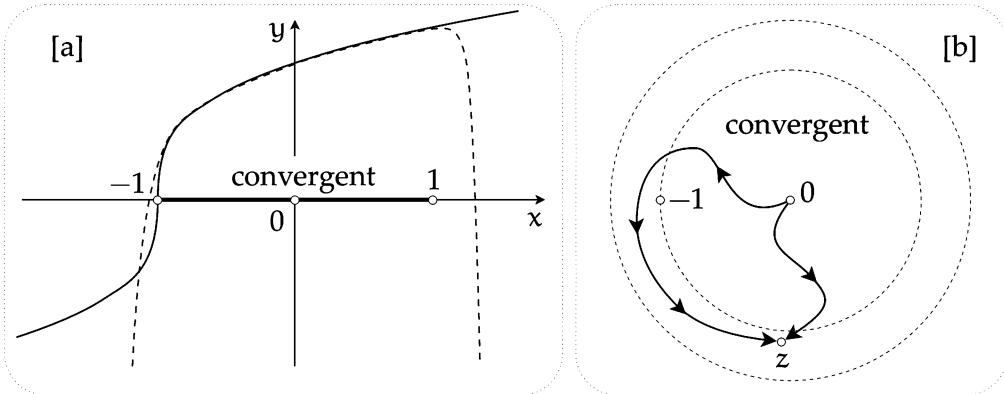
Figure [2.33a] shows the graph $y = (1+x)^{1/3}$ of the real function $f(x) = (1+x)^{1/3}$, which is well defined for all x since every real number has a unique real cube root. Looking at this graph, there seems to be no good reason for the series to break down at ± 1 , yet break down it does. This is illustrated rather vividly by the dashed curve, which is the graph of the 30th degree polynomial obtained by truncating the binomial series at x^{30} . As you can see, this curve follows $y = f(x)$ very closely (actually more closely than illustrated) between ± 1 , but just beyond this interval it suddenly starts to deviate wildly.

Unlike the case of $1/(1+x^2)$, observe that the mystery does not disappear when we extend the real function $f(x)$ to the complex function $f(z) = (1+z)^{1/3}$, because $f(z)$ *does not have any singularities*.

We have already discussed the fact [see (2.14) and Exs. 16, 17, 18] that the Binomial Theorem extends to the complex plane. In the present case it says that

$$f(z) = (1+z)^{1/3} = 1 + \frac{1}{3}z - \frac{1}{9}z^2 + \frac{5}{81}z^3 - \frac{10}{243}z^4 + \frac{22}{729}z^5 - \dots,$$

with convergence inside the unit disc shown in [2.33b]. In common with all power series, the RHS of the above equation is a single-valued function. For example, at



[2.33] [a] The real function $f(x) = (1+x)^{\frac{1}{3}}$ can be expressed as a power series, but only for $-1 < x < 1$; the dashed curve shows the 30th degree polynomial obtained by truncating the binomial series at x^{30} . But why does convergence break down here?! There are no singularities in sight, and, unlike the resolution of the convergence mystery of $\frac{1}{1+x^2}$, the complex generalization $f(z) = (1+z)^{\frac{1}{3}}$ also does not have any singularities! **[b]** But $f(z)$ does have a branch point at -1 . Suppose the series were to converge at the illustrated point z outside the unit disc. Then the two illustrated paths could encircle the branch point, yielding two different values of $f(z)$. But the power series is single-valued, so this is impossible!

$z = 0$ the series equals 1. But while $f(x)$ was an ordinary single-valued function of x , the LHS of the above equation is a three-valued multifunction of z , with a second order branch point at $z = -1$. For example, $f(0)$ takes three values: 1, $e^{i\frac{2\pi}{3}}$, and $e^{-i\frac{2\pi}{3}}$. We now recognize that the power series represents just one branch of $f(z)$, namely the one for which $f(0) = 1$.

This solves the mystery. For suppose that the series were to converge inside the larger circle in [2.33b], and in particular at the illustrated point z . Starting at $z = 0$ with the value $f(0) = 1$, then travelling along the two illustrated paths to z , we clearly end up with two *different* values of $f(z)$, because together the two paths enclose the branch point at -1 . But the power series cannot mimic this behaviour since it is necessarily single-valued—its only way out is to cease converging outside the unit disc. We have demanded the impossible of the power series, and it has responded by committing suicide!

This example shows that a branch point is just as real an obstacle to convergence as a singularity. Quite generally, this argument shows that if a branch of a multifunction can be expressed as a power series, the disc of convergence cannot be large enough to contain any branch points of the multifunction. This strongly suggests a further generalization of the (unproven) statement (2.15):

If a complex function or a branch of a multifunction can be expressed as a power series, the radius of convergence is the distance to the nearest singularity or branch point. (2.27)

Much later in the book we will develop the tools necessary to confirm this conjecture.

2.6.4 An Example with Two Branch Points

Choosing the positive value of the square root, [2.34a] illustrates the graph $y = f(x) = \sqrt{1+x^2}$, which is a hyperbola. Again, the Binomial Theorem yields a power series that mysteriously only converges between ± 1 , namely,

$$f(x) = (1+x^2)^{\frac{1}{2}} = 1 + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \frac{1}{16}x^6 - \frac{5}{128}x^8 + \cdots.$$

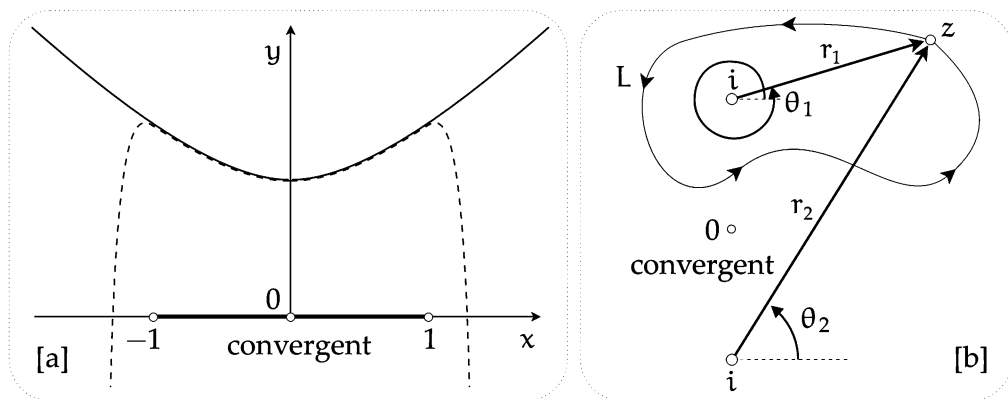
The divergence of the series beyond this interval is vividly conveyed by the dashed curve, which is the graph of the 20th degree polynomial obtained by truncating the binomial series at x^{20} .

As before, the explanation lies in $\mathbb{C}$, where $f(x)$ becomes the two-valued multifunction $f(z) = \sqrt{z^2+1}$. This can be rewritten as $f(z) = \sqrt{(z-i)(z+i)}$, which makes it clear that $f(z)$ has two simple branch points, one at i and the other at $-i$. These branch points obstruct the convergence of the corresponding complex series, limiting it to the unit disc shown in [2.34b].

In greater detail, the notation of [2.34b] enables us to write

$$f(z) = \sqrt{r_1 r_2} e^{i(\theta_1 + \theta_2)/2}. \quad (2.28)$$

Here we must bear in mind that the figure illustrates only one possibility (out of infinitely many) for each of the angles θ_1 and θ_2 . To see that i is indeed a branch



[2.34] [a] The real function $f(x) = (1+x^2)^{\frac{1}{2}}$ can be expressed as a power series, but only for $-1 < x < 1$; the dashed curve shows the 20th degree polynomial obtained by truncating the binomial series at x^{20} . **[b]** In the complex plane, $f(z) = \sqrt{(z-i)(z+i)} = \sqrt{r_1 r_2} e^{i(\theta_1 + \theta_2)/2}$, which makes it clear that $f(z)$ has branch points at $\pm i$, and these obstruct the convergence of the complex power series beyond the unit disc. If z loops around i along L , as illustrated, then the net changes in θ_1 and θ_2 are 2π and 0 , respectively. So $f(z) \rightsquigarrow e^{i\pi} f(z) = -f(z)$.

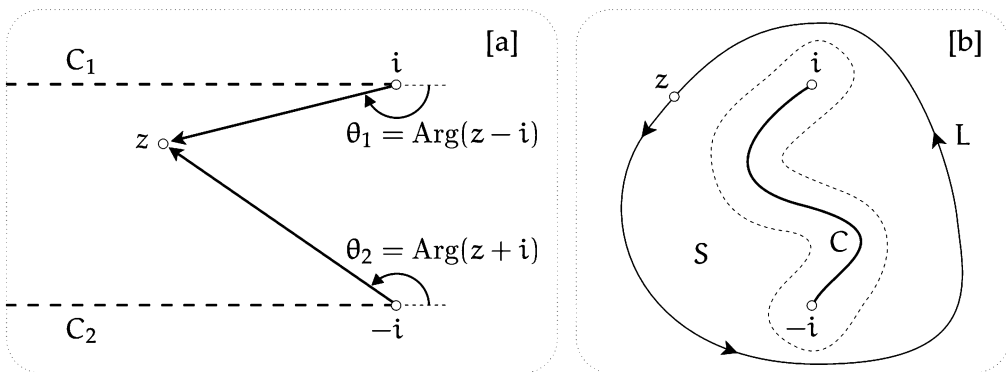
point, suppose we start with the value of $f(z)$ given by the illustrated values of θ_1 and θ_2 . Now let z travel round the illustrated loop L . As it does so, $(z + i)$ rocks back and forth, so θ_2 merely oscillates, finally returning to its original value. But $(z - i)$ undergoes a complete revolution, and so θ_1 increases by 2π . Thus when z returns to its original position, (2.28) shows that $f(z)$ does not return to its original value, but rather to

$$f_{\text{new}}(z) = \sqrt{r_1 r_2} e^{i(\theta_1 + 2\pi + \theta_2)/2} = e^{i\pi} \sqrt{r_1 r_2} e^{i(\theta_1 + \theta_2)/2} = -f_{\text{old}}(z).$$

Of course the same thing happens if z travels along a loop that goes once round $-i$, instead of round $+i$.

In order to dissect $f(z)$ into two single-valued branches, we appear to need two branch cuts: one cut C_1 from i to infinity (to prevent us encircling the branch point at i), and another cut C_2 from $-i$ to infinity, for the same reason. Figure [2.35a] illustrates a particularly common and important choice of these cuts, namely, rays going due west. If we do not allow z to cross the cuts then we may restrict the angle $\theta_1 = \arg(z - i)$ to its principal value, in the range $-\pi < \theta_1 \leq \pi$. For example, the angle in [2.34b] is not the principal value, while the one in [2.35a] is. If θ_2 is likewise restricted to its unique principal value then (2.28) becomes the single-valued principal branch of $f(z)$, say $F(z)$. The other branch of $f(z)$ is simply $-F(z)$.

Let us return to the previous situation in which we allowed θ_1 and θ_2 to take general values rather than their principal values. Figure [2.35b] illustrates the fact that it is possible to define two branches of $f(z)$ using only a single branch cut C that connects the two branch points. If z is restricted to the shaded, multiply-connected region S , then it cannot loop around either branch point singly. It can,



[2.35] [a] We can define two single-valued branches of $f(z) = \sqrt{z^2 + 1} = \sqrt{r_1 r_2} e^{i(\theta_1 + \theta_2)/2}$ by making branch cuts going due west from both branch points, and then taking θ_1 and θ_2 to be their principal values, as illustrated. **[b]** In fact we only need a single branch cut connecting the two branch points, such as C . For then a loop L cannot encircle either branch point singly, and if it encircles both, as illustrated, then $f(z)$ does not change its value.

however, travel along a loop such as L that encircles both branch points together. But in this case both θ_1 and θ_2 increase by 2π , so (2.28) shows that $f(z)$ returns to its original value. Thus we can define two single-valued branches on S . Finally, we may expand S until it borders on C .

2.7 The Logarithm Function

2.7.1 Inverse of the Exponential Function

The complex logarithm function $\log(z)$ may be introduced as the “inverse” of e^z . More precisely, we define $\log(z)$ to be any complex number such that $e^{\log(z)} = z$. It follows [exercise] that

$$\log(z) = \ln|z| + i \arg(z).$$

Since $\arg(z)$ takes infinitely many values, differing from each other by multiples of 2π , we see that $\log(z)$ is a multifunction taking infinitely many values, differing from each other by multiples of $2\pi i$. For example,

$$\log(2 + 2i) = \ln 2\sqrt{2} + i(\pi/4) + 2n\pi i,$$

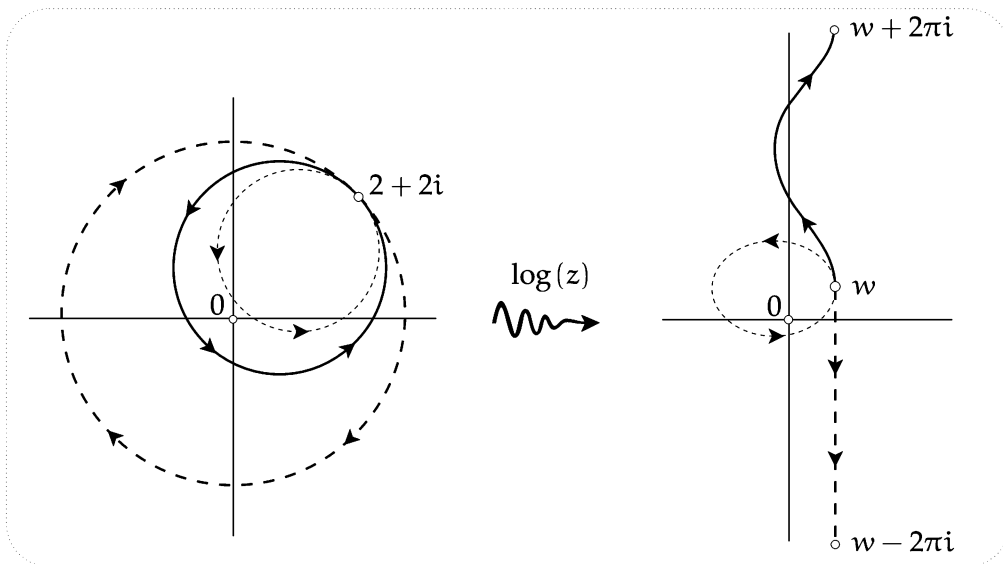
where n is an arbitrary integer.

The reason we get infinitely many values is clear if we go back to the exponential mapping shown in [2.19], p. 90: each time z travels straight upward by $2\pi i$, e^z executes a complete revolution and returns to its original value. Figure [2.36] rephrases this using the above example of $\log(2 + 2i)$. If we arbitrarily choose the initial value $w = \ln 2\sqrt{2} + i(\pi/4)$ for $\log(2 + 2i)$, then as z travels along a loop that encircles the origin v times in the counterclockwise direction, $\log(z)$ moves along a path from w to $w + 2v\pi i$. Check that you understand (roughly) the shapes of the illustrated image paths.

Clearly $\log(z)$ has a branch point at $z = 0$. However, this branch point is quite unlike that of $z^{(1/n)}$, for no matter how many times we loop around the origin (say counterclockwise), $\log(z)$ *never* returns to its original value, rather it continues moving upwards forever. You can now understand the previously introduced term, “logarithmic branch point”.

Here is another difference between the branch points of $z^{(1/n)}$ and $\log(z)$. As z approaches the origin, say along a ray, $|z^{(1/n)}|$ tends to zero, but $|\log(z)|$ tends to infinity, and in this sense the origin is a singularity as well as a branch point. On the other hand, algebraic branch points can also be singularities; consider $(1/\sqrt{z})$.

To define single-valued branches of $\log(z)$ we make a branch cut from 0 out to infinity. The most common choice for this cut is the negative real axis. In this cut plane we may restrict $\arg(z)$ to its principal value $\text{Arg}(z)$; remember, this is



[2.36] Here we see that 0 is a branch point of $\log(z)$. However, this branch point is quite unlike that of $z^{(1/n)}$, for no matter how many times we loop around the origin (say counterclockwise), $\log(z)$ never returns to its original value, rather it continues moving upwards forever. If we make a branch cut along the negative real axis, we can restrict the angle to be its principal value, in which case we obtain the principal value of $\log(z)$, written $\text{Log}(z) \equiv \ln|z| + i \text{Arg}(z)$.

defined by $-\pi < \text{Arg}(z) \leq \pi$. This yields the *principal branch* or *principal value* of the logarithm, written $\text{Log}(z)$, and defined by

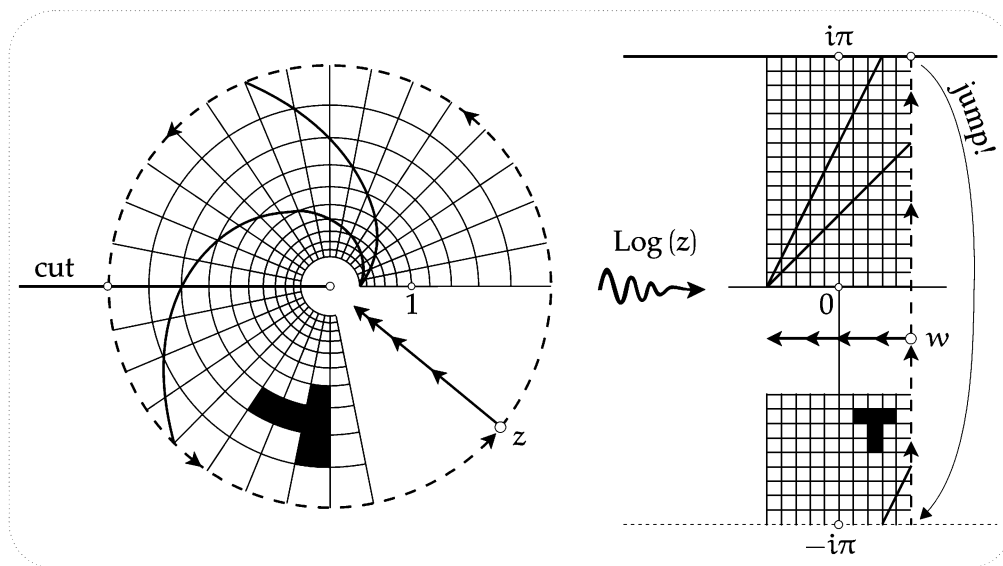
$$\text{Log}(z) \equiv \ln|z| + i \text{Arg}(z).$$

For example, $\text{Log}(-\sqrt{3} - i) = \ln 2 - i(5\pi/6)$, $\text{Log}(i) = i(\pi/2)$, and $\text{Log}(-1) = i\pi$. Note that if $z = x$ is on the positive real axis, $\text{Log}(x) = \ln(x)$.

Figure [2.37] illustrates how the mapping $z \mapsto w = \text{Log}(z)$ sends rays to horizontal lines, and circles to vertical line-segments connecting the horizontal lines at heights $\pm\pi$; the entire z -plane is mapped to the horizontal strip of the w -plane bounded by these lines. Study this figure until you are completely at peace with it. You can see the price we pay for forcing the logarithm to be single-valued: it becomes discontinuous at the cut. As z crosses the cut travelling counterclockwise, the height of w suddenly jumps from π to $-\pi$. If we wish w to instead move continuously, then we must switch to the branch $\text{Log}(z) + 2\pi i$ of the logarithm.

Another problem with restricting ourselves to the principal branch is that the familiar rules for the logarithm break down. For example, $\text{Log}(ab)$ is *not* always equal to $\text{Log}(a) + \text{Log}(b)$; try $a = -1$ and $b = i$, for example. However, if we keep all values of the logarithm in play then it *is* true [exercise] that

$$\log(ab) = \log(a) + \log(b) \quad \text{and} \quad \log(a/b) = \log(a) - \log(b),$$



[2.37] **The geometry of $\text{Log}(z)$.** Note that the price we pay for having a single-valued logarithm is that it is discontinuous: as we cross the branch cut, its value jumps.

in the sense that every value of the LHS is *contained amongst* the values of the RHS, and vice versa.

2.7.2 The Logarithmic Power Series

If we wish to find a power series for the complex logarithm, two problems immediately arise. First, since a power series is single-valued, the best we can hope for is to represent a single branch of $\log(z)$; let's choose the principal branch, $\text{Log}(z)$. Second, the origin is both a singularity and a branch point of $\text{Log}(z)$ so we cannot have a power series centred there (i.e., in powers of z); let us therefore try an expansion centred at $z = 1$, i.e., in powers of $(z - 1)$. [Of course any other non-zero point would be equally suitable.] Writing $Z = (z - 1)$, our problem, then, is to expand $\text{Log}(1 + Z)$ in powers of Z .

Let us use the abbreviation $L(z) = \text{Log}(1 + z)$. Since the branch point of $L(z)$ is $z = -1$, the largest disc of convergence we can have is the unit disc. To find the series we will use the fact that $e^{L(z)} = (1 + z)$. Recall from (2.21) [on p. 91] that by taking n to be a sufficiently large positive integer, we can approximate e^L as precisely as we wish using $[1 + (L/n)]^n$. Thus, as $(1/n)$ vanishes,⁶

$$\left(1 + \frac{L}{n}\right)^n \asymp e^L = (1 + z) \implies 1 + \frac{L}{n} \asymp (1 + z)^{\frac{1}{n}}.$$

⁶ Once again, " $\asymp$ " denotes Newton's concept of *ultimate equality*; see the new Preface.

There are n branches of $(1+z)^{\frac{1}{n}}$ within the unit disc, but since $L(0) = 0$ we need the branch of $(1+z)^{\frac{1}{n}}$ that equals 1 when $z = 0$. Appealing to the Binomial series for this principal branch, we obtain

$$1 + \frac{L}{n} \asymp 1 + \frac{1}{n}z + \frac{\frac{1}{n}(\frac{1}{n}-1)}{2!}z^2 + \frac{\frac{1}{n}(\frac{1}{n}-1)(\frac{1}{n}-2)}{3!}z^3 + \dots,$$

and hence

$$L(z) \asymp z + \frac{(\frac{1}{n}-1)z^2}{2} + \frac{(\frac{1}{n}-1)(\frac{1}{2n}-1)z^3}{3} + \frac{(\frac{1}{n}-1)(\frac{1}{2n}-1)(\frac{1}{3n}-1)z^4}{4} + \dots.$$

Finally, since this ultimate equality becomes equality in the limit that $(1/n)$ tends to zero, we obtain the following *logarithmic power series*:

$$\text{Log}(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \frac{z^5}{5} - \frac{z^6}{6} + \dots. \quad (2.29)$$

For other approaches to this series, see Exs. 31, 32.

Using the ratio test, you can check for yourself that this series does indeed converge inside the unit circle. In fact it can be shown [see Ex. 11] that the series also converges everywhere *on* the unit circle, except obviously at $z = -1$. This yields some very interesting special cases. For example, putting $z = i$ and then equating real and imaginary parts, we get

$$\ln \sqrt{2} = \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \dots$$

$$\text{and} \quad \pi = 4 \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots \right].$$

Try checking the first series by noting that if $z = 1$, then $\ln \sqrt{2} = \frac{1}{2} \ln(1+z)$. The remarkable second series for π was first published by Leibniz in 1676, though it was already known to Newton and Gregory. It is usually obtained from the power series for $\tan^{-1} x$, evaluated at $x = 1$. Astonishingly, *that* power series was discovered *centuries* before Newton, Gregory, and Leibniz roamed the Earth, by the Indian mathematician, Mādhava of Sangamagrāma (c.1340 – c.1425); see (Stillwell, 2010, p. 166).

For other interesting applications of the logarithmic series, see Exs. 36, 37, 38.

2.7.3 General Powers

If x is a real variable then we are accustomed to being able to express x^3 , for example, as $e^{3 \ln x}$. Let's see whether we can do the same thing using the complex exponential and logarithm. That is, let us investigate the possibility of writing

$$z^k = e^{k \log(z)}. \quad (2.30)$$

Let $z = r e^{i\theta}$, where θ is chosen to be the principal value, $\text{Arg}(z)$. Then

$$e^{3 \text{Log}(z)} = e^{3(\ln r + i\theta)} = e^{3 \ln r} e^{i3\theta} = r^3 e^{i3\theta} = z^3.$$

But the most general branch of $\log(z)$ is simply $\text{Log}(z) + 2n\pi i$, where n is an integer, so

$$e^{3 \log(z)} = e^{6n\pi i} e^{3 \text{Log}(z)} = e^{6n\pi i} z^3 = z^3$$

is true irrespective of which branch of the logarithm is chosen. Clearly, by the same argument, (2.30) is true for all integer values of k .

Next, consider the three branches of $z^{\frac{1}{3}}$. Recalling that the principal branch $[z^{\frac{1}{3}}]$ of this function is $\sqrt[3]{r} e^{i(\theta/3)}$, where θ again represents the principal angle, you can easily check that $e^{\frac{1}{3} \text{Log}(z)} = [z^{\frac{1}{3}}]$. Thus the general branch of the logarithm yields

$$e^{\frac{1}{3} \log(z)} = e^{i \frac{2n\pi}{3}} [z^{\frac{1}{3}}].$$

Thus we have again confirmed (2.30), in the sense that the infinitely many branches of $\log(z)$ yield precisely the three branches of the cube root: $[z^{\frac{1}{3}}]$, $e^{i(2\pi/3)} [z^{\frac{1}{3}}]$, and $e^{i(4\pi/3)} [z^{\frac{1}{3}}]$. By the same reasoning, if (p/q) is a fraction reduced to lowest terms then $e^{\frac{p}{q} \log(z)}$ yields precisely the q branches of $z^{\frac{p}{q}}$.

Finally, note that the RHS of (2.30) is still meaningful if $k = (a + ib)$ is a *complex* number. Emboldened by the above successes, we now take (2.30) as the *definition* of a complex power. If we use $\text{Log}(z)$ in (2.30) then we find that the principal branch of $z^{(a+ib)}$ is given by [exercise]

$$[z^{(a+ib)}] = e^{(a+ib) \text{Log}(z)} = r^a e^{-b\theta} e^{i(a\theta + b \ln r)}.$$

If z now travels along a closed loop encircling the origin n times, then $\log(z)$ moves along a path from $\text{Log}(z)$ to $\text{Log}(z) + 2n\pi i$, and $z^{(a+ib)}$ moves along a path from $[z^{(a+ib)}]$ to

$$z^{(a+ib)} = e^{i2\pi n a} e^{-2\pi n b} [z^{(a+ib)}].$$

If $b \neq 0$ then the factor $e^{-2\pi n b}$ makes it obvious that $z^{(a+ib)}$ never returns to its original value, no matter how many times we go round the origin. Thus $z = 0$ is a logarithmic branch point in this case. This is still true even if $b = 0$, provided [exercise] that the real power a is *irrational*. Only when a is a rational number does z^a return to its original value after a finite number of revolutions, and only when a is an integer does z^a become single-valued.

We end with an important observation on the use of “ e^z ” to denote the single-valued exponential mapping. Reversing the roles of the constant and variable in (2.30), we are forced to define $f(z) = k^z$ to be the “multifunction” [see Ex. 29] $f(z) = e^{z \log(k)}$. But if we now put $k = e = 2.718 \dots$ then we are suddenly in hot water: the exponential mapping “ e^z ” is merely one branch [what are the others?] of the newly defined multifunction $(2.718 \dots)^z$. To avoid this confusion, some authors always

write the exponential mapping as $\exp(z)$. However, we shall retain the notation “ e^z ”, which is both convenient and rooted in history, with the understanding that e^z always refers to the single-valued exponential mapping, and never to the multifunction $(2.718\dots)^z$.

2.8 Averaging over Circles*

2.8.1 The Centroid

This entire section is optional because the chief result to which we shall be led (*Gauss’s Mean Value Theorem*) will be derived again later, in fact more than once. It is nevertheless fun and instructive to attempt to understand the result using only the most elementary of methods.

Consider a set of n point particles in $\mathbb{C}$, located at $z_1, z_2, \dots, z_n$. If the mass of the particle at z_j is m_j then the *centroid* Z of the set of particles (also called the “centre of mass”) is defined to be

$$Z \equiv \frac{\sum_{j=1}^n m_j z_j}{\sum_{j=1}^n m_j}.$$

If we imagine the plane to be massless, Z is the point at which we could rest the plane on a pin so as to make it balance.

Throughout this section *we shall take the masses of the particles to be equal*, in which case the centroid becomes the average position of the particles:

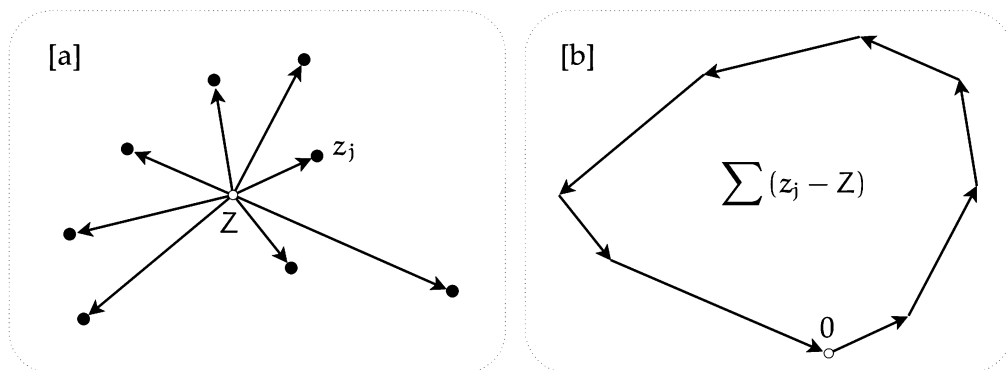
$$Z = \frac{1}{n} \sum_{j=1}^n z_j.$$

This is the case depicted in [2.38a]. An immediate consequence of this definition is that $\sum (z_j - Z) = 0$. In other words, *the complex numbers from Z to the particles cancel*. This vanishing sum is illustrated in [2.38b]. Conversely, if some point Z has the property that the complex numbers connecting it to the particles cancel, then Z must be the centroid.

Another immediate result is that if we translate the set of points by b , then the centroid will translate with them, i.e., the new centroid will be $Z + b$. The same thing happens if we rotate the set of points about the origin—the centroid rotates with them. In general,

$$\text{If } Z \text{ is the centroid of } \{z_j\}, \text{ then the centroid of } \{az_j + b\} \text{ is } aZ + b. \quad (2.31)$$

Given a second set of n points $\{\tilde{z}_j\}$ (with centroid $\tilde{Z}$), we may add pairs from the two sets to obtain the set $\{z_j + \tilde{z}_j\}$, and it is easy to see that the centroid of the latter is $Z + \tilde{Z}$. In particular, the centroid Z of $\{z_j = x_j + iy_j\}$ is the sum $X + iY$ of the centroid X of the points $\{x_j\}$ on the real axis and the centroid iY of the points $\{iy_j\}$ on the imaginary axis.



[2.38] [a] The **centroid** Z of masses m_j located at z_j is $Z \equiv \frac{\sum_{j=1}^n m_j z_j}{\sum_{j=1}^n m_j}$. In the case that the masses are all equal, join Z to the locations of the masses, and sum these connecting complex numbers $(z_j - Z)$... [b] ... and discover that their sum vanishes! Furthermore, this vanishing sum characterizes the location of Z .

Our next result will play a minor role at the end of this section, but later we shall see that it has other interesting consequences. The *convex hull* H of the set of particles $\{z_j\}$ is defined to be the smallest convex polygon such that each particle lies on H or inside it. More intuitively, first imagine pegs sticking out of the plane at each point z_j , then stretch an imaginary rubber band so as to enclose all the pegs. When released, the rubber band will contract into the desired polygon H shown in [2.39a]. We can now state the result:

$$\text{The centroid } Z \text{ must lie in the interior of the convex hull } H. \quad (2.32)$$

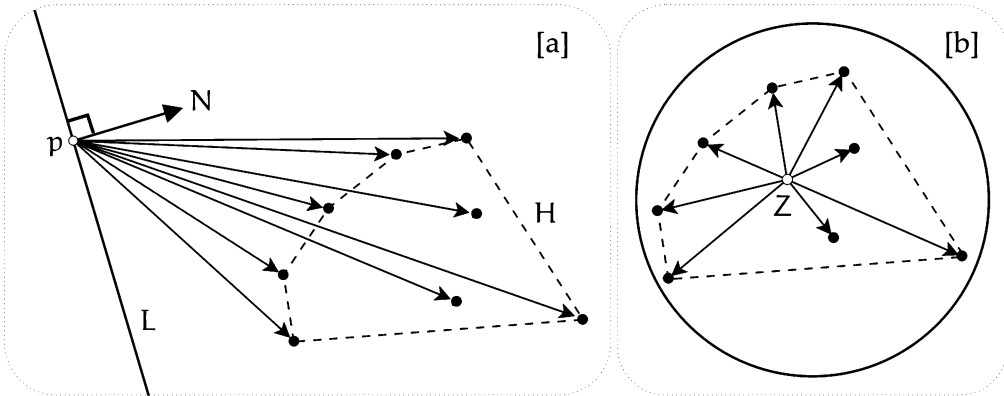
For if p is outside this set, we see that the complex numbers from p to the particles cannot possibly cancel, as they must do for Z . More formally, we take it as visually evident that through any exterior point p we may draw a line L such that H and its shaded interior lie entirely on one side of L . The impossibility of the complex numbers cancelling now follows from their lying entirely on this side of L , for they all must have positive components in the direction of the illustrated complex number N normal to L . Except when the particles are collinear (in which case H collapses to a line-segment), the same reasoning forbids Z from lying *on* H .

As illustrated in [2.39b], an immediate consequence of (2.32) is that

$$\text{If all the particles lie within some circle then their centroid must also lie within that circle.} \quad (2.33)$$

The main result we wish to derive in this section is based on the following fact. Defining the *centre* of a regular n -gon to be the centre of the circumscribing circle,

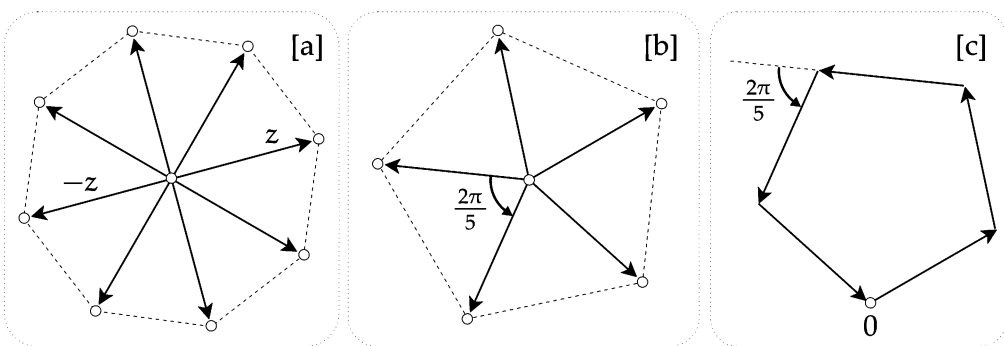
$$\text{The centre of a regular } n\text{-gon is the centroid of its vertices.} \quad (2.34)$$



[2.39] The centroid must lie within the convex hull. [a] If p lies outside the convex hull H of the masses, then the connecting complex numbers from p to the masses cannot vanish, therefore the centroid Z must lie within H . [b] Therefore, if all the particles lie within some circle then their centroid must also lie within that circle.

By virtue of (2.31), we may as well choose the n -gon to be centred at the origin, in which case the claim is that the sum of the vertices vanishes. As illustrated in [2.40a], this is obvious if n is even since the vertices then occur in opposite pairs.

The explanation is not quite so obvious when n is odd; see [2.40b], which illustrates the case $n = 5$. However, if we draw $\sum z_j$ systematically, taking the vertices z_j in counterclockwise order, then we obtain [2.40c], and the answer is suddenly clear: the sum of the vertices of the regular 5-gon forms another regular 5-gon. The figure explains why this happens. Since the angle between successive vertices in [2.40b] is



[2.40] The sum of the vertices of a regular n -gon vanishes. [a] If n is even, it is obvious that the sum of the vertices of any regular, origin-centred n -gon must vanish, for the vertices occur in diametrically opposite pairs. [b] In fact the sum of the vertices must also vanish when n is odd, but this is no longer immediately obvious. Begin by noting that the angle between successive vertices is $(2\pi/n)$. [c] Drawing the sum of the vertices in counterclockwise order, all is suddenly clear: the sum itself forms a regular n -gon, and therefore it vanishes.

$(2\pi/5)$, this is also the angle between successive terms of the sum in [2.40c]. Clearly this argument generalizes to arbitrary n (both odd and even), thereby establishing (2.34). For a different approach, see Ex. 40.

2.8.2 Averaging over Regular Polygons

If a complex mapping $z \mapsto w = f(z)$ maps the set of points $\{z_j\}$ to the set $\{w_j = f(z_j)\}$, then the centroid W of the image points may be described as the *average of $f(z)$ over the set $\{z_j\}$ of n points*. Writing this average as $\langle f(z) \rangle_n$,

$$\langle f(z) \rangle_n = \frac{1}{n} \sum_{j=1}^n f(z_j).$$

Note that if $f(z) = c$ is constant, then its average over any set of points is equal to c .

Henceforth, we shall restrict ourselves to the case where $\{z_j\}$ are the vertices of a regular n -gon; correspondingly, $\langle f(z) \rangle_n$ will be understood as the average of $f(z)$ over the vertices of such a regular n -gon. Note that if we write $f(z) = u(z) + iv(z)$, then

$$\langle f(z) \rangle_n = \langle u(z) \rangle_n + i \langle v(z) \rangle_n. \quad (2.35)$$

Initially, we consider only *origin-centred* polygons.

Consider, then, the average of $f(z) = z^m$ over the vertices of such a regular n -gon. Figure [2.41] illustrates the case $n = 6$. In the centre of the figure is a shaded regular hexagon, and on the periphery are the images of its vertices under the mappings $z, z^2, \dots, z^6$. Study this figure carefully, and see if you can understand what's going on. If we take still higher values of m , then this pattern repeats cyclically: z^7 is like z^1 , z^8 is like z^2 , and so on.

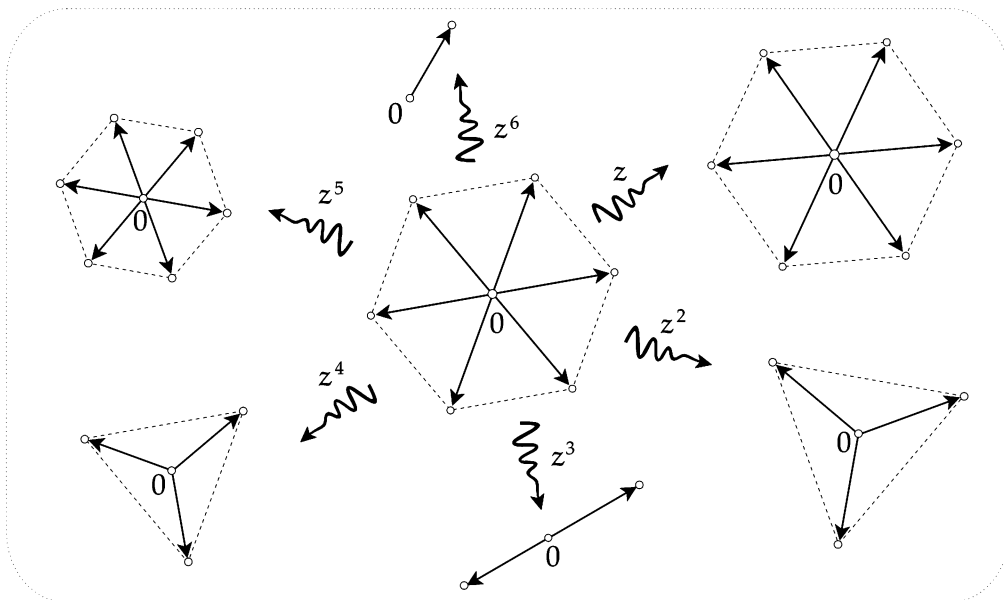
For us the essential feature of this figure is that unless m is a multiple of 6, the image under z^m of the regular 6-gon is another regular polygon. [Note that we count two equal and opposite points as a regular 2-gon, but we do not count a single point as a regular polygon.] More precisely, and in general,

Unless m is a multiple of n , the image under z^m of an origin-centred regular n -gon is an origin-centred regular N -gon, where $N = (n \text{ divided by the highest common factor of } m \text{ and } n)$. If m is a multiple of n , then the image is a single point. (2.36)

Check that this agrees with [2.41]. Try to establish the result on your own, but see Ex. 41 if you get stuck.

Combining this result with (2.34), we obtain the following key fact: *If $n > m$ then $\langle z^m \rangle_n = 0$. This is easy to generalize. If*

$$P_m(z) = c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \dots + c_m z^m$$



[2.41] Unless m is a multiple of n , the image under z^m of an origin-centred regular n -gon is an origin-centred regular N -gon, where $N = (n \text{ divided by the highest common factor of } m \text{ and } n)$. If m is a multiple of n , then the image is a single point.

is a general polynomial of degree m , then its average over the vertices of the n -gon is

$$\langle P_m(z) \rangle_n = \langle c_0 \rangle_n + c_1 \langle z \rangle_n + c_2 \langle z^2 \rangle_n + c_3 \langle z^3 \rangle_n + \cdots + c_m \langle z^m \rangle_n.$$

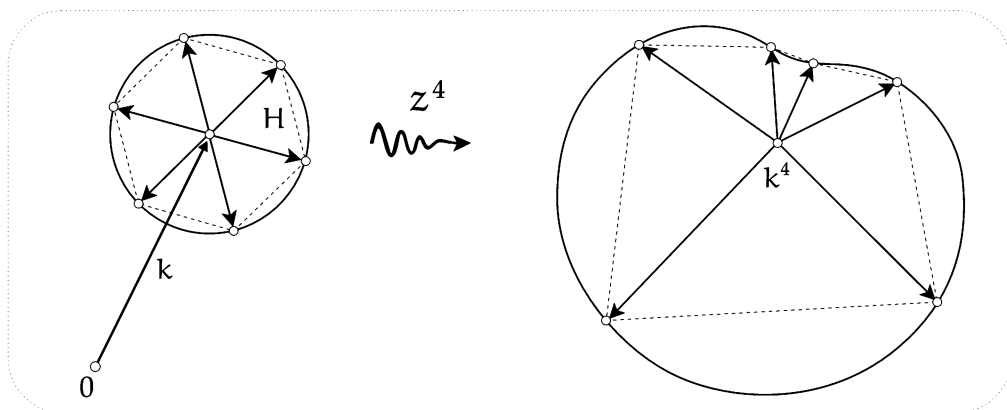
If the number n of vertices is greater than the degree m of the polynomial, we therefore obtain

$$\langle P_m(z) \rangle_n = \langle c_0 \rangle_n = c_0 = P_m(0).$$

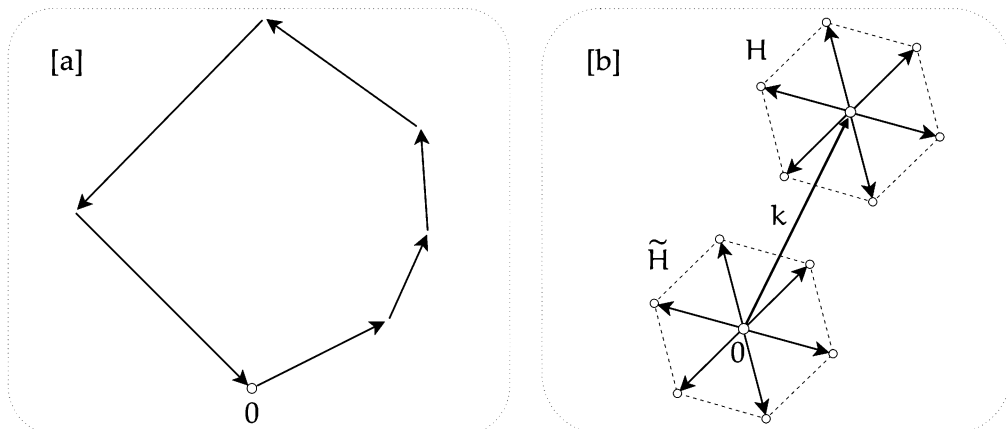
In other words, the centroid of the image points is the image of the centroid. Expressing this result in the language of averages,

If $n > m$ then the average of an m^{th} degree polynomial $P_m(z)$ over the vertices of an origin-centred regular n -gon is its value $P_m(0)$ at the centre of the n -gon. (2.37)

Finally, let us generalize to regular n -gons that are centred at an arbitrary point k , instead of the origin. Of course when we apply z^m to the vertices of such a regular polygon, the image points do *not* form a regular polygon. See [2.42], which shows the effect of z^4 on the vertices of a regular hexagon H centred at k , together with the image of the entire circle on which these vertices lie. Nevertheless, the figure also illustrates the surprising and beautiful fact that, once again, the centroid of the image points is the image of the centroid of H . Figure [2.43a] confirms this empirically



[2.42] It is a surprising and beautiful fact (here illustrated with $n = 6$ and $m = 4$) that **the centroid of the image points of the vertices of a regular n -gon H under $z \mapsto z^m$ is the image of the centroid of H .**



[2.43] [a] Empirical verification of the claim made in the previous figure: the sum of the connecting complex numbers from k^4 to images of the vertices of H vanishes, so k^4 is indeed the centroid of these image points. **[b]** In order to mathematically explain this phenomenon, we view H to be the translation by k of the origin-centred $\tilde{H}$; see main text for the argument.

by showing that the sum of complex numbers connecting k^4 to the image points is indeed zero.

Extending our notation slightly, we may write the average of z^m over the vertices of H as $\langle z^m \rangle_H$, so what we must show is that $\langle z^4 \rangle_H = k^4$. It is no harder to treat the general case of z^m acting on the vertices of a regular n -gon H centred at k . First note that H can be obtained by translating an origin-centred n -gon $\tilde{H}$ by k . See the example in [2.43b]. Since a vertex z_j of $\tilde{H}$ translates to a vertex $z_j + k$ of H , it follows that

$$\langle z^m \rangle_H = \langle (z + k)^m \rangle_{\tilde{H}}.$$

But $(z + k)^m = \sum_{j=0}^m \binom{m}{j} z^j k^{m-j}$ is just an m^{th} degree polynomial which maps 0 to k^m . Using (2.37), we conclude that if $n > m$ then $\langle z^m \rangle_H = k^m$, as was to be shown.

Generalizing the argument that led to (2.37), we see that (2.37) is a special case of the following result:

If $n > m$ then the average of an m^{th} degree polynomial $P_m(z)$ over the vertices of a regular n -gon centred at k is its value $P_m(k)$ at the centre of the n -gon. (2.38)

2.8.3 Averaging over Circles

Since at least the time of Archimedes, mathematicians have found it fruitful to think of a circle as the limit of a regular n -gon as n tends to infinity. We will now use this idea to investigate the average of a complex function over a circle.

Inscribing a regular n -gon in a given circle, and taking the limit as n tends to infinity, (2.38) shows that

The average over a circle C of a polynomial of arbitrarily high degree is equal to the value of the polynomial at the centre of C . (2.39)

By (2.35), the average $\langle f(z) \rangle_C$ of a complex function $f(z) = u(z) + iv(z)$ over a circle C may be expressed as $\langle f(z) \rangle_C = \langle u(z) \rangle_C + i \langle v(z) \rangle_C$. Using a familiar idea from ordinary calculus, the averages of the two real functions u and v may be expressed as integrals. If C has centre k and radius R , then as θ varies between 0 and 2π , $z = k + R e^{i\theta}$ traces out C . Thus,

$$\langle u(z) \rangle_C = \frac{1}{2\pi} \int_0^{2\pi} u(k + R e^{i\theta}) d\theta \quad \text{and} \quad \langle v(z) \rangle_C = \frac{1}{2\pi} \int_0^{2\pi} v(k + R e^{i\theta}) d\theta.$$

More compactly, we may write

$$\langle f(z) \rangle_C = \frac{1}{2\pi} \int_0^{2\pi} f(k + R e^{i\theta}) d\theta,$$

in which it is understood that the complex integral may be evaluated in terms of the real integrals above.

Once again denoting a general m^{th} degree polynomial by $P_m(z)$, (2.39) can therefore be expressed as an integral formula:

$$\frac{1}{2\pi} \int_0^{2\pi} P_m(k + R e^{i\theta}) d\theta = \langle P_m(z) \rangle_C = P_m(k). \quad (2.40)$$

For example, if C is centred at the origin and $P_m(z) = z^m$, then

$$\langle z^m \rangle_C = \frac{1}{2\pi} \int_0^{2\pi} R^m e^{im\theta} d\theta = \frac{R^m}{2\pi} \int_0^{2\pi} [\cos m\theta + i \sin m\theta] d\theta = 0,$$

in agreement with (2.40).

The fact that (2.40) holds for polynomials of arbitrarily high degree immediately suggests that it might also hold for *power series*. We shall show that it does.

As usual we will only give the details for origin-centred power series, the generalization to arbitrary centres being straightforward. Let $P(z) = \sum_{j=0}^{\infty} c_j z^j$ be the power series, so that $P_m(z) = \sum_{j=0}^m c_j z^j$ are its approximating polynomials. If the circle C lies inside the disc of convergence of $P(z)$, then (2.12) implies the following. No matter how small we choose a real number ϵ , we can find a sufficiently large m such that $P_m(z)$ approximates $P(z)$ with accuracy ϵ throughout C and its interior. If we write $\mathcal{E}(z)$ for the complex number from the approximation $P_m(z)$ to the exact answer $P(z)$, then

$$P(z) = P_m(z) + \mathcal{E}(z), \quad \text{where} \quad |\mathcal{E}(z)| < \epsilon$$

for all z on and inside C , and in particular at the centre k of C .

At this point we could immediately study $\langle P(z) \rangle_C$ in terms of its integral representation, but it is more instructive to first consider the average $\langle P(z) \rangle_n$ of $P(z)$ over a regular n -gon inscribed in C . Once this is done, we may let n tend to infinity to obtain $\langle P(z) \rangle_C$.

First note that $\mathcal{E}(z)$ maps the vertices of the n -gon to points lying inside an origin-centred disc of radius ϵ . By (2.33), or directly from the generalized triangle inequality (1.8) on p. 8, the centroid $\langle \mathcal{E}(z) \rangle_n$ of these points must also lie in this disc. Choosing n greater than m , say $n = (m+1)$, (2.38) yields

$$\begin{aligned} \langle P(z) \rangle_{m+1} &= \langle P_m(z) \rangle_{m+1} + \langle \mathcal{E}(z) \rangle_{m+1} \\ &= P_m(k) + \langle \mathcal{E}(z) \rangle_{m+1} \\ &= P(k) + [\langle \mathcal{E}(z) \rangle_{m+1} - \mathcal{E}(k)]. \end{aligned}$$

The term in square brackets is the connecting complex number from $\mathcal{E}(k)$ to $\langle \mathcal{E}(z) \rangle_{m+1}$, and since both these points lie within a disc of radius ϵ , their connecting complex number must be shorter than 2ϵ . Finally, since the term in square brackets may also be interpreted as the connecting complex number from $P(k)$ to $\langle P(z) \rangle_{m+1}$, we have the following result:

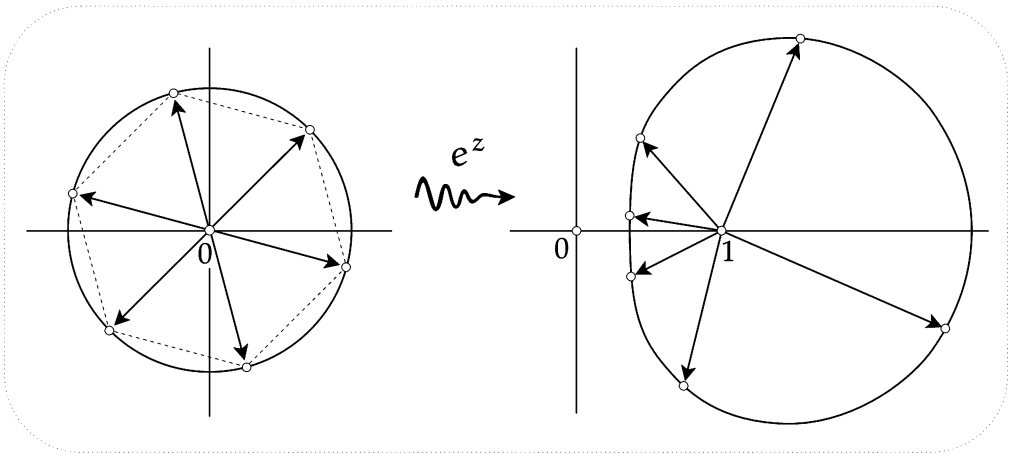
Let m be chosen so that $P_m(z)$ approximates the power series $P(z)$ with accuracy ϵ on and within a circle C centred at k . If a regular $(m+1)$ -gon is inscribed in C , then the average $\langle P(z) \rangle_{m+1}$ of $P(z)$ over its vertices will approximate $P(k)$ with accuracy 2ϵ . (2.41)

We have thus transformed an exact result concerning the approximation $P_m(z)$ into an approximation result concerning the exact mapping $P(z)$.

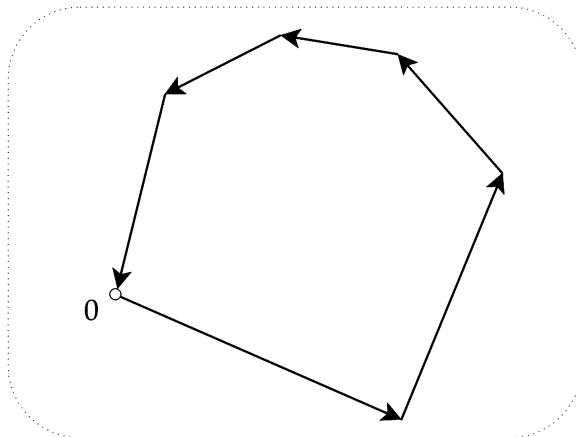
For example, let C be the unit circle, and let $P(z) = e^z$. If we desire an accuracy of $\epsilon = 0.004$ everywhere on the unit disc then it turns out that $m = 5$ is sufficient, i.e., the approximating polynomial of lowest degree that has this accuracy is

$$P_5(z) = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \frac{1}{4!}z^4 + \frac{1}{5!}z^5.$$

Figure [2.44] shows the image under $z \mapsto e^z$ of C , and in particular it shows the images of the vertices of a regular hexagon inscribed in C . According to the result, the centroid of these image points should differ from $e^0 = 1$ by no more than 0.008—an indiscernible discrepancy in a drawing done to this scale. This prediction is convincingly borne out in [2.45], which shows the sum of the complex numbers connecting 1 to the images of the vertices of the hexagon. To within the accuracy of the drawing, the sum is indeed zero!



[2.44] Applying $z \mapsto e^z$ to the unit circle and to the vertices of the inscribed regular hexagon on the left yields the figure on the right. **The centroid of the six image points is almost the image of the centre of the original hexagon, namely, $e^0 = 1$.** As the number of vertices goes to infinity, the result is ultimately exact.



[2.45] **Empirical confirmation of the claim made in the previous figure:** the connecting complex numbers from 1 to the images of the vertices of the hexagon do indeed sum to zero, at least to within the accuracy of this drawing.

In the limit that ϵ tends to zero and m tends to infinity, (2.41) yields a form of Gauss's Mean Value Theorem:

If a complex function $f(z)$ can be expressed as a power series, and a circle C (radius R and centre k) lies within the disc of convergence of that power series, then

$$\langle f(z) \rangle_C = \frac{1}{2\pi} \int_0^{2\pi} f(k + R e^{i\theta}) d\theta = f(k).$$

In addition to its theoretical importance, this formula can sometimes be used to evaluate difficult real integrals. For example, the exact version of [2.44] is $\langle e^z \rangle_C = e^0 = 1$, and this implies [exercise] that $\int_0^{2\pi} e^{\cos \theta} \cos[\sin \theta] d\theta = 2\pi$. See Ex. 43 for another example of this idea.

2.9 Exercises

- 1 Sketch the circle $|z-1| = 1$. Find (geometrically) the polar equation of the image of this circle under the mapping $z \mapsto z^2$. Sketch this image curve, which is called a *cardioid*.
- 2 Consider the complex mapping $z \mapsto w = (z-a)/(z-b)$. Show geometrically that if we apply this mapping to the perpendicular bisector of the line-segment joining a and b , then the image is the unit circle. In greater detail, describe the motion of w round this circle as z travels along the line at constant speed.

- 3 Consider the family of complex mappings

$$z \mapsto M_a(z) = \frac{z-a}{\bar{a}z-1} \quad (a \text{ constant}).$$

[These mappings will turn out to be fundamental to non-Euclidean geometry.] Do the following problems algebraically; in the next chapter we will provide geometric explanations.

- (i) Show that $M_a[M_a(z)] = z$. In other words, M_a is self-inverse.
- (ii) Show that $M_a(z)$ maps the unit circle to itself.
- (iii) Show that if a lies inside the unit disc then $M_a(z)$ maps the unit disc to itself.

Hint: Use $|q|^2 = q\bar{q}$ to verify that

$$|\bar{a}z-1|^2 - |z-a|^2 = (1-|a|^2)(1-|z|^2).$$

- 4 In figure [2.7] we saw that if $q^2 \leq p^3$ then the solutions of $x^3 = 3px + 2q$ are all real. Draw the corresponding picture in the case $q^2 > p^3$, and deduce that one solution is real, while the other two form a complex conjugate pair.
- 5 Show that the mapping $z \mapsto z^2$ doubles the angle between two rays coming out of the origin. Use this to deduce that the lemniscate (see [2.9] on p. 69) must self-intersect at right angles.
- 6 This question refers to the Cassinian curves in [2.9] on p. 69.
 - (i) On a copy of this figure, sketch the curves that intersect each Cassinian curve at right angles; these are called the *orthogonal trajectories* of the original family of curves.
 - (ii) Give an argument to show that each orthogonal trajectory hits one of the foci at ± 1 .
 - (iii) If the Cassinian curves are thought of as a geographical contour map of the modular surface (cf. [2.10]) of $(z^2 - 1)$, then what is the interpretation of the orthogonal trajectories in terms of the surface?

- (iv) In Chapter 4 we will show that if two curves intersect at some point $p \neq 0$, and if the angle between them at p is ϕ , then the image curves under $z \mapsto w = z^2$ will *also* intersect at angle ϕ , at the point $w = p^2$. Use this to deduce that as z travels out from one of the foci along an orthogonal trajectory, $w = z^2$ travels along a ray out of $w = 1$.
 - (v) Check the result of the previous part by using a computer to draw the images under $w \mapsto \sqrt{w}$ of (A) circles centred at $w = 1$; (B) the radii of such circles.
 - (vi) Writing $z = x + iy$ and $w = u + iv$, find u and v as functions of x and y . By writing down the equation of a line in the w -plane through $w = 1$, show that the orthogonal trajectories of the Cassinian curves are actually segments of *hyperbolas*.
- 7 Sketch the modular surface of $C(z) = (z+1)(z-1)(z+1+i)$. Hence sketch the Cassinian curves $|C(z)| = \text{const.}$, then check your answer using a computer. To answer the following questions, recall that if $R(z)$ is a real function of position in the plane, then $R(p)$ is a *local minimum* of R if $R(p) < R(z)$ for all z in the immediate neighbourhood of p . A *local maximum* is defined similarly.
- (i) Referring to the previous exercise, what is the significance of the orthogonal trajectories of the Cassinian curves you have just drawn?
 - (ii) Does $|C(z)|$ have any local maxima?
 - (iii) Does $|C(z)|$ have any non-zero local minima?
 - (iv) If D is a disc (or indeed a more arbitrary shape), can the maximum of $|C(z)|$ on D occur at a point inside D , or must the maximum occur at a boundary point of D ? What about the minimum of $|C(z)|$ on D ?
 - (v) Do you get the same answers to these questions if $C(z)$ is replaced by an arbitrary polynomial? What about a complex function that is merely known to be expressible as a power series?
- 8 On page 69 we saw that the polar equation of the lemniscate with foci at ± 1 is $r^2 = 2 \cos 2\theta$. In fact James Bernoulli and his successors worked with a slightly different lemniscate having equation $r^2 = \cos 2\theta$. Let us call this the *standard lemniscate*.
- (i) Where are the foci of the standard lemniscate?
 - (ii) What is the value of the product of the distances from the foci to a point on the standard lemniscate?
 - (iii) Show that the Cartesian equation of the standard lemniscate is

$$(x^2 + y^2)^2 = x^2 - y^2.$$

9 Here is an attempt [ultimately doomed] at using real methods to expand $H(x) = 1/(1+x^2)$ into a power series centred at $x = k$, i.e., into a series of the form $H(x) = \sum_{j=0}^{\infty} c_j X^j$, where $X = (x - k)$. According to Taylor's Theorem, $c_j = H^{(j)}(k)/j!$, where $H^{(j)}(k)$ is the j^{th} derivative of H .

- (i) Show that $c_0 = 1/(1+k^2)$ and $c_1 = -2k/(1+k^2)^2$, and find c_2 . Note how it becomes increasingly difficult to calculate the successive derivatives.
- (ii) Recall (or prove) that the n^{th} derivative of a product AB of two functions $A(x)$ and $B(x)$ is given by *Leibniz's rule*:

$$(AB)^{(n)} = \sum_{j=0}^n \binom{n}{j} A^{(j)} B^{(n-j)}.$$

By applying this result to the product $(1+x^2)H(x)$, deduce that

$$(1+k^2)H^{(n)}(k) + 2nkH^{(n-1)}(k) + n(n-1)H^{(n-2)}(k) = 0.$$

Because the coefficients in this recurrence relation depend on n , we cannot solve it using the technique of Ex. 30 on p. 55.

- (iii) Deduce from the previous part that the recurrence relation for the c_j 's is

$$(1+k^2)c_n + 2kc_{n-1} + c_{n-2} = 0,$$

which *does* have constant coefficients.

- (iv) Solve this recurrence relation, and hence recover the result (2.17) on p. 85.

10 Reconsider the series (2.18) on p. 85.

- (i) Show that we recover the correct series (missing the odd powers of x) when the centre k of the series is at the origin.
- (ii) Find a value of k such that the series is missing all the powers X^n , where $n = 2, 5, 8, 11, 14, \dots$. Check your answer using a computer.

11 Show that each of the following series has the unit circle as its circle of convergence, then investigate the convergence *on* the unit circle. You can guess the correct answers by "drawing the series" in the manner of [2.18] on p. 89.

$$(i) \sum_{n=0}^{\infty} z^n, \quad (ii) \sum_{n=1}^{\infty} \frac{z^n}{n}, \quad (iii) \sum_{n=1}^{\infty} \frac{z^n}{n^2}.$$

[By virtue of (2.29), note that the second series is $-\text{Log}(1-z)$.]

12 Consider the geometric series $P(z) = \sum_{j=0}^{\infty} z^j$, which converges to $1/(1-z)$ inside the unit disc. The approximating polynomials in this case are $P_m(z) = \sum_{j=0}^m z^j$.

- (i) Show that the error $E_m(z) \equiv |P(z) - P_m(z)|$ is given by

$$E_m(z) = \frac{|z|^{m+1}}{|1-z|}.$$

- (ii) If z is any fixed point in the disc of convergence, what happens to the error as m tends to infinity?
 - (iii) If we fix m , what happens to the error as z approaches the boundary point $z = 1$?
 - (iv) Suppose we want to approximate this series in the disc $|z| \leq 0.9$, and further suppose that the maximum error we will tolerate is $\epsilon = 0.01$. Find the lowest degree polynomial $P_m(z)$ that approximates $P(z)$ with the desired accuracy throughout the disc.
- 13 We have seen that if we set $P_n(z) = z^n$, then the representation of a complex function $f(z)$ as an infinite series $\sum_{n=0}^{\infty} c_n P_n(z)$ (i.e., a power series) is unique. This is *not* true, however, if $P_n(z)$ is just any old set of polynomials. The following example is taken from [Boas and Boas \(2010\)](#). Defining

$$P_0(z) = -1, \quad \text{and} \quad P_n(z) = \frac{z^{n-1}}{(n-1)!} - \frac{z^n}{n!} \quad (n = 1, 2, 3, \dots),$$

show that

$$-2P_0 - P_1 + P_3 + 2P_4 + 3P_5 + \dots = e^z = P_1 + 2P_2 + 3P_3 + 4P_4 + \dots$$

- 14 Consider two power series, $P(z) = \sum_{j=0}^{\infty} p_j z^j$ and $Q(z) = \sum_{j=0}^{\infty} q_j z^j$, which have approximating polynomials $P_n(z) = \sum_{j=0}^n p_j z^j$ and $Q_m(z) = \sum_{j=0}^m q_j z^j$. If the radii of convergence of $P(z)$ and $Q(z)$ are R_1 and R_2 then both series are uniformly convergent in the disc $|z| \leq r$, where $r < \min\{R_1, R_2\}$. Thus if ϵ is the maximum error we will tolerate in this disc, we can find a sufficiently large n such that

$$P_n(z) = P(z) + \mathcal{E}_1(z) \quad \text{and} \quad Q_n(z) = Q(z) + \mathcal{E}_2(z),$$

where the (complex) errors $\mathcal{E}_{1,2}(z)$ both have lengths less than ϵ . Use this to show that by taking a sufficiently high value of n we can approximate $[P(z) + Q(z)]$ and $P(z)Q(z)$ with arbitrarily high precision using $[P_n(z) + Q_n(z)]$ and $P_n(z)Q_n(z)$, respectively.

- 15 Give an example of a pair of origin-centred power series, say $P(z)$ and $Q(z)$, such that the disc of convergence for the product $P(z)Q(z)$ is *larger* than either of the two discs of convergence for $P(z)$ and $Q(z)$. [*Hint*: think in terms of rational functions, such as $[z^2/(5 - z^3)]$, which are known to be expressible as power series.]
- 16 Our aim is to give a *combinatorial* explanation of the Binomial Theorem (2.14) for all negative integer values of n . The simple yet crucial first step is to write $n = -m$ and to change z to $-z$. Check that the desired result (2.14) now takes

the form $(1 - z)^{-m} = \sum_{r=0}^{\infty} c_r z^r$, where c_r is the binomial coefficient

$$c_r = \binom{m+r-1}{r}. \quad (2.42)$$

[Note that this says that the coefficients c_r are obtained by reading Pascal's triangle *diagonally*, instead of horizontally.] To begin to understand this, consider the special case $m = 3$. Using the geometric series for $(1 - z)^{-1}$, we may express $(1 - z)^{-3}$ as

$$[1 + z + z^2 + z^3 + \cdots] \bullet [1 + z + z^2 + z^3 + \cdots] \bullet [1 + z + z^2 + z^3 + \cdots],$$

where $\bullet$ simply denotes multiplication. Suppose we want the coefficient c_9 of z^9 . One way to get z^9 is to take z^3 from the first bracket, z^4 from the second, and z^2 from the third.

(i) Write this way of obtaining z^9 as the sequence $zzz \bullet zzzz \bullet zz$ of 9 z 's and 2 $\bullet$'s, where the latter keep track of which power of z came from which bracket. [I got this nice idea from my friend Paul Zeitz.] Explain why c_9 is the number of distinguishable rearrangements of this sequence of 11 symbols. Be sure to address the meaning of sequences in which a $\bullet$ comes first, last, or is adjacent to the other $\bullet$.

(ii) Deduce that $c_9 = \binom{11}{9}$, in agreement with (2.42).

(iii) Generalize this argument and thereby deduce (2.42).

17 Here is an inductive approach to the result of the previous exercise.

(i) Write down the first few rows of Pascal's triangle and circle the numbers $\binom{5}{3}, \binom{4}{2}, \binom{3}{1}, \binom{2}{0}$. Check that the sum of these numbers is $\binom{6}{3}$. Explain this.

(ii) Generalize your argument to show that

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-2}{r-1} + \binom{n-3}{r-2} + \cdots + 1.$$

(iii) Assume that $(1 - z)^{-M} = \sum_{r=0}^{\infty} \binom{M+r-1}{r} z^r$ holds for some positive integer M . Now multiply this series by the geometric series for $(1 - z)^{-1}$ to find $(1 - z)^{-(M+1)}$. Deduce that the binomial series is valid for all negative integer powers.

18 The basic idea of the following argument is due to Euler. Initially, let n be any real (possibly irrational) number, and define

$$B(z, n) \equiv \sum_{r=0}^{\infty} \binom{n}{r} z^r \quad \text{where} \quad \binom{n}{r} \equiv \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!},$$

and $\binom{n}{0} \equiv 1$. We know from elementary algebra that if n is a positive integer then $B(z, n) = (1 + z)^n$. To establish the Binomial Theorem (2.14) for rational

powers, we must show that if p and q are integers then $B(z, \frac{p}{q})$ is the principal branch of $(1+z)^{\frac{p}{q}}$.

(i) With a fixed value of n , use the ratio test to show that $B(z, n)$ converges in the unit disc, $|z| < 1$.

(ii) By multiplying the two power series, deduce that

$$B(z, n) B(z, m) = \sum_{r=0}^{\infty} C_r(n, m) z^r \quad \text{where} \quad C_r(n, m) = \sum_{j=0}^r \binom{n}{j} \binom{m}{r-j}.$$

(iii) If m and n are positive integers, then show that

$$B(z, n) B(z, m) = B(z, n+m), \quad (2.43)$$

and deduce that $C_r(n, m) = \binom{n+m}{r}$. But $C_r(n, m)$ and $\binom{n+m}{r}$ are simply *polynomials* in n and m , and so the fact that they agree at infinitely many values of m and n [positive integers] implies that *they must be equal for all real values of m and n* . Thus the key formula (2.43) is valid for all real values of m and n .

(iv) By substituting $n = -m$ in (2.43), deduce the Binomial Theorem for negative integer values of n .

(v) Use (2.43) to show that if q is an integer then $[B(z, \frac{1}{q})]^q = (1+z)$. Deduce that $B(z, \frac{1}{q})$ is the principal branch of $(1+z)^{\frac{1}{q}}$.

(vi) Finally, show that if p and q are integers, then $B(z, \frac{p}{q})$ is indeed the principal branch of $(1+z)^{\frac{p}{q}}$.

19 Show that the ratio test cannot be used to find the radius of convergence of the power series (2.18) on p. 85. Use the root test to confirm that $R = \sqrt{1+k^2}$.

20 Show that if m and n are integers, then $\int_0^{2\pi} \cos m\theta \cos n\theta d\theta$ vanishes unless $m = n$, in which case it equals π . Likewise, establish a similar result for $\int_0^{2\pi} \sin m\theta \sin n\theta d\theta$. Use these facts to verify (2.19), at least formally.

21 Do the following problems by first substituting $z = r e^{i\theta}$ into the power series for e^z , then equating real and imaginary parts.

(i) Show that the Fourier series for $[\cos(\sin \theta)] e^{\cos \theta}$ is $\sum_{n=0}^{\infty} \frac{\cos n\theta}{n!}$, and write down the Fourier series for $[\sin(\sin \theta)] e^{\cos \theta}$.

(ii) Deduce that $\int_0^{2\pi} e^{\cos \theta} [\cos(\sin \theta)] \cos m\theta d\theta = (\pi/m!)$, where m is a positive integer.

(iii) By writing $x = (r/\sqrt{2})$, find the power series for $f(x) = e^x \sin x$.

(iv) Check the first few terms of the series for $f(x)$ by multiplying the series for e^x and $\sin x$.

- (v) Calculate the n^{th} derivative $f^{(n)}(0)$ using (1.14) on p. 25 of Chapter 1. By using these derivatives in Taylor's Theorem, verify your answer to part (iii).

22 Reconsider the formula,

$$e^z = \lim_{n \rightarrow \infty} P_n(z), \quad \text{where} \quad P_n(z) = \left(1 + \frac{z}{n}\right)^n.$$

- (i) Check that $P_n(z)$ is the composition of a translation by n , followed by a contraction by $(1/n)$, followed by the power mapping $z \mapsto z^n$.
 - (ii) Referring to figure [2.4] on p. 64, use the previous part to sketch the images under $P_n(z)$ of circular arcs centred at $-n$, and of rays emanating from $-n$.
 - (iii) Let S be an origin-centred square (say of unit side) in the z -plane. With a large value of n , sketch just those portions of the arcs and rays (considered in the previous part) that lie within S .
 - (iv) Use the previous two parts to quantitatively explain figure [2.19] on p. 90.
- 23 If you did not do so earlier, sketch the image of a vertical line $x = k$ under $z \mapsto w = \cos z$ by drawing the analogue of [2.26]. Deduce that the asymptotes of this hyperbola are $\arg w = \pm k$. Check this using the equation of the hyperbola.
- 24 Consider the multifunction $f(z) = \sqrt{z-1} \sqrt[3]{z-i}$.
- (i) Where are the branch points and what are their orders?
 - (ii) Why is it *not* possible to define branches using a single branch cut of the type shown in [2.35b]?
 - (iii) How many values does $f(z)$ have at a typical point z ? Find and then plot all the values of $f(0)$.
 - (iv) Choose one of the values of $f(0)$ which you have just plotted, and label it p . Sketch a loop L that starts and ends at the origin such that if $f(0)$ is initially chosen to be -1 , then as z travels along L and returns to the origin, $f(z)$ travels along a path from -1 to p . Do the same for each of the other possible values of $f(0)$.
- 25 Describe the branch points of the function $f(z) = 1/\sqrt{1-z^4}$. What is the smallest number of branch cuts that may be used to obtain single-valued branches of $f(z)$? Sketch an example of such cuts. [Remark: This function is historically important, owing to the fact (Ex. 20, p. 242) that $\int f(x) dx$ represents the arc length of the lemniscate. This integral (the *lemniscatic integral*) cannot be evaluated in terms of elementary functions—it is an example of a new kind of function called an *elliptic integral*. See Stillwell (2010), for more background and detail.]

- 26 For each function $f(z)$ below, find and then plot all the branch points and singularities. Assuming that these functions may be expressed as power series centred at k [in fact they can be], use the result (2.27) on p. 107 to verify the stated value of the radius of convergence R .
- (i) If $f(z) = 1/(e^{\pi z} - 1)$ and $k = (1 + 2i)$, then $R = 1$.
 - (ii) If $f(z)$ is a branch of $\sqrt[5]{z^4 - 1}$ and $k = 3i$, then $R = 2$.
 - (iii) If $f(z)$ is a branch of $\sqrt{z - i}/(z - 1)$ and $k = -1$, then $R = \sqrt{2}$.
- 27 Until Euler cleared up the whole mess, the complex logarithm was a source of tremendous confusion. For example, show that $\log(z)$ and $\log(-z)$ have no common values, then consider the following argument of John Bernoulli:

$$\begin{aligned}\log[(-z)^2] &= \log[z^2] \implies \log(-z) + \log(-z) = \log(z) + \log(z) \\ &\implies 2\log(-z) = 2\log(z) \\ &\implies \log(-z) = \log(z).\end{aligned}$$

What is wrong with this argument?!

- 28 What value does z^i take at $z = -1$ if we start with the principal value at $z = 1$ (i.e., $1^i = 1$), and then let z travel one and a half revolutions clockwise round the origin?
- 29 In this exercise you will see that the “multifunction” k^z is quite different in character from all the other multifunctions we have discussed. For integer values of n , define $l_n \equiv [\text{Log}(k) + 2n\pi i]$.
- (i) Show that the “branches” of k^z are $e^{l_n z}$.
 - (ii) Suppose that z travels along an arbitrary loop, beginning and ending at $z = p$. If we initially choose the value $e^{l_2 p}$ for k^z , then what value of k^z do we arrive at when z returns to p ? Deduce that k^z has no branch points.
- Since we cannot change one value of k^z into another by travelling round a loop, we should view its “branches” $\{\dots, e^{l_{-1} z}, e^{l_0 z}, e^{l_1 z}, \dots\}$ as an infinite set of *completely unrelated* single-valued functions.
- 30 Show that all the values of i^i are real! Are there any other points z such that z^i is real?
- 31 In the case of a real variable, the logarithmic power series was originally discovered [see next exercise] as follows. First check that $\ln(1+X)$ can be written as $\int_0^X [1/(1+x)] dx$, and then expand $[1/(1+x)]$ as a power series in x . Finally, integrate your series term by term. [Later in the book we will be able to generalize this argument to the complex plane.]

- 32 Here is another approach to the logarithmic power series. As before, let $L(z) = \text{Log}(1+z)$. Since $L(0) = 0$, the power series for $L(z)$ must be of the form $L(z) = az + bz^2 + cz^3 + dz^4 + \dots$. Substitute this into the equation

$$1+z = e^L = 1 + L + \frac{1}{2!}L^2 + \frac{1}{3!}L^3 + \frac{1}{4!}L^4 + \dots,$$

then find a , b , c , and d by equating powers of z . [Historically the logarithmic series came first—both Mercator and Newton discovered it using the method in the previous exercise—then Newton reversed the reasoning of the present exercise to obtain the series for e^x . See [Stillwell \(2010\)](#).]

- 33 (i) Use [2.26] to discuss the branch points of the multifunction $\cos^{-1}(z)$.
 (ii) Rewrite the equation $w = \cos z$ as a quadratic in e^{iz} . By solving this equation, deduce that $\cos^{-1}(z) = -i \log[z + \sqrt{z^2 - 1}]$. [Why do we not need to bother to write $\pm$ in front of the square root?]
 (iii) Show that as z travels along a loop that goes once round either 1 or -1 (but not both), the value of $[z + \sqrt{z^2 - 1}]$ changes to $1/[z + \sqrt{z^2 - 1}]$.
 (iv) Use the previous part to show that the formula in part (ii) is in accord with the discussion in part (i).
- 34 Write down the origin-centred power series for $(1 - \cos z)$. Use the Binomial Theorem to write down the power series (centred at $Z = 0$) for the principal branch of $\sqrt{1 - Z}$, then substitute $Z = (1 - \cos z)$. Hence show that if we choose the branch of $\sqrt{\cos z}$ that maps 0 to 1, then

$$\sqrt{\cos z} = 1 - \frac{z^2}{4} - \frac{z^4}{96} - \frac{19z^6}{5760} - \dots$$

Verify this using a computer. Where does this series converge?

- 35 What value does $(z/\sin z)$ approach as z approaches the origin? Use the series for $\sin z$ to find the first few terms of the origin-centred power series for $(z/\sin z)$. Check your answer using a computer. Where does this series converge?
- 36 By considering $\text{Log}(1+ix)$, where x is a real number lying between ± 1 , deduce that

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \frac{x^{11}}{11} + \dots$$

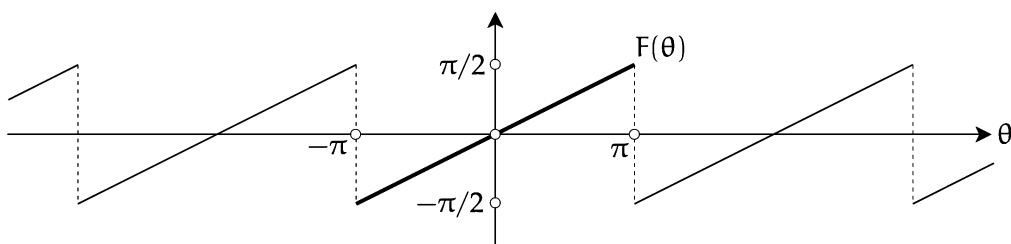
In what range does this value of $\tan^{-1}(x)$ lie? Give another derivation of the series using the idea in Ex. 31.

- 37 (i) Show geometrically that as $z = e^{i\theta}$ goes round and round the unit circle (with ever increasing θ), $\text{Im}[\text{Log}(1+z)] = (\Theta/2)$, where Θ is the principal value of θ , i.e., $-\pi < \Theta \leq \pi$.

- (ii) Consider the periodic “saw tooth” function $F(\theta)$ whose graph is shown below. By substituting $z = e^{i\theta}$ in the logarithmic series (2.29), use the previous part to deduce the following Fourier series:

$$F(\theta) = \sin \theta - \frac{\sin 2\theta}{2} + \frac{\sin 3\theta}{3} - \frac{\sin 4\theta}{4} + \dots$$

- (iii) Check this Fourier series by directly evaluating the integrals (2.19).
 (iv) Use a computer to draw graphs of the partial sums of the Fourier series. As you increase the number of terms, observe the magical convergence of this sum of smooth waves to the jagged graph below. If only Fourier could have seen this on the screen, not just in his mind’s eye!



38 As in the previous exercise, let $\Theta = \text{Arg}(z)$.

- (i) Use (2.29) to show that

$$\frac{1}{2} \text{Log} \left[\frac{1+z}{1-z} \right] = z + \frac{z^3}{3} + \frac{z^5}{5} + \frac{z^7}{7} + \dots$$

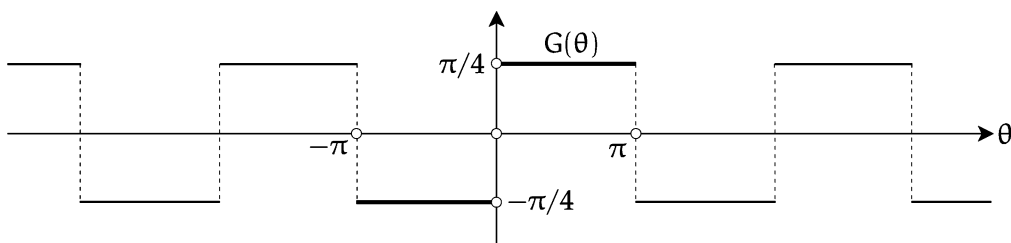
- (ii) Show geometrically that as $z = e^{i\theta}$ goes round and round the unit circle,

$$\text{Im} \left\{ \frac{1}{2} \text{Log} \left[\frac{1+z}{1-z} \right] \right\} = (\text{sign of } \Theta) \left[\frac{\pi}{4} \right].$$

- (iii) Consider the periodic “square wave” function $G(\theta)$ whose graph is shown below. Use the previous two parts to deduce that its Fourier series is

$$G(\theta) = \sin \theta + \frac{\sin 3\theta}{3} + \frac{\sin 5\theta}{5} + \frac{\sin 7\theta}{7} + \dots$$

Finally, repeat parts (iii) and (iv) of the previous exercise.



- 39 Show that (2.32) is still true even if the (positive) masses of the particles are not all equal.

- 40 Here is another simple way of deriving (2.34). If the vertices of the origin-centred regular n -gon are rotated by ϕ , then their centroid Z rotates with them to $e^{i\phi}Z$. By choosing $\phi = (2\pi/n)$, deduce that $Z = 0$.
- 41 To establish (2.36), let $z_0, z_1, z_2, \dots, z_{n-1}$ be the vertices (labelled counterclockwise) of the regular n -gon, and let C be the circumscribing circle. Also, let $w_j = z_j^m$ be the image of vertex z_j under the mapping $z \mapsto w = z^m$. Think of z as a particle that starts at z_0 and orbits counterclockwise round C , so that the image particle $w = z^m$ starts at w_0 and orbits round another circle *with m times the angular speed of z* .
- (i) Show that each time z travels from one vertex to the next, w executes (m/n) of a revolution. Thus as z travels from z_0 to z_k , w executes $k(m/n)$ revolutions as it travels from w_0 to w_k .
 - (ii) Let w_k be the first point in the sequence w_1, w_2 , etc., such that $w_k = w_0$. Deduce that if (M/N) is (m/n) reduced to lowest terms, then $k = N$. Note that $N = (n \text{ divided by the highest common factor of } m \text{ and } n)$.
 - (iii) Explain why $w_{N+1} = w_1, w_{N+2} = w_2$, etc.
 - (iv) Show that $w_0, w_1, \dots, w_{N-1}$ are distinct.
 - (v) Show that $w_0, w_1, \dots, w_{N-1}$ are the vertices of a regular N -gon.
- 42 Consider the mapping $z \mapsto w = P_n(z)$, where $P_n(z)$ is a general polynomial of degree $n \geq 2$. Let S_q be the set of points in the z -plane that are mapped to a particular point q in the w -plane. Show that *the centroid of S_q is independent of the choice of q* , and is therefore a property of the polynomial itself. [Hint: This is another way of looking at a familiar fact about the sum of the roots of a polynomial.]
- 43 Use Gauss's Mean Value Theorem [p. 78] to find the average of $\cos z$ over the circle $|z| = r$. Deduce (and check with a computer) that for all real values of r ,

$$\int_0^{2\pi} \cos[r \cos \theta] \cosh[r \sin \theta] d\theta = 2\pi.$$

